Call: HORIZON-CL4-2021-TWIN-TRANSITION-01
(TWIN GREEN AND DIGITAL TRANSITION 2021)

Topic: HORIZON-CL4-2021-TWIN-TRANSITION-01-05

Type of Action: HORIZON-RIA

Proposal number: 101057049

Proposal acronym: BIO-UPTAKE

Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

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1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-CL4-2021-TWIN-TRANSITION-01-05	Type of Action	HORIZON-RIA
Call	HORIZON-CL4-2021-TWIN-TRANSITION-01	Type of Model Grant Agreement	HORIZON-AG

Acronym: BIO-UPTAKE

Proposal title: BIOcomposites in smart plastic transformation processes to pave the way for the large-scale UPTAKE of sustainable bio-based products

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months: 42

Fixed keyword 1: Materials engineering

Fixed keyword 2: Technology development

Fixed keyword 3: Automation and control systems

Free keywords: *Conformal Manufacturing Technologies, intermediate formats, biobased end-products, digitalization, bioplastic composites manufacturing as a Service, thermosets, thermoplastics, fibres, circularity*

Abstract *

The general objective of Bio-Uptake project is to ensure a sustainable uptake (increase the use in a 39%) of bioplastic composites through boosting a twin green and digital transformation in the European manufacturing industry. In particular, Bio-Uptake solution will focus scientific and technology efforts on developing flexible manufacturing processes to produce biobased end-products for the construction, medical and packaging sectors based on the combination of intermediate formats made of natural and/or biobased synthetic fibres reinforced with biopolymers, which are easily adaptable to new market demands. The novel approach on which Bio-Uptake project relies is based on modularity or pre-fabrication: a smart combination of intermediate formats (organosheets, tapes and pellets) into a final end-product which allows to overcome the current technical and environmental limitations to meet the demanding requirements of a specific sector/application where a single biobased material doesn?t. Thus, the synergistic potential of composite materials will pave the way for the integration and uptake of bio-based materials in mass customised manufacturing (manufacturing as a service). Three disruptive manufacturing processes (based on conformal technologies) will be developed within Bio-Uptake focused on the plastics manufacturing sector, which is on the centrality of a variety of value chains, and being demonstrated in 3 demo cases (bathroom ceiling cabinet, feet orthosis and garbage container lid). Sustainability criteria will be applied since the design phase to reach circularity by design, obtaining products with more than 75% biobased content and decreasing GHG up to 33%. Bio-Uptake solution does not request large investment in complex equipment too. Bio-Uptake consortium is formed by 13 interdisciplinar and complementary partners (6 industries, 4 RTOs, 1 academia and 2 Other organizations). The overall budget is 5,994,886 Euros.

Remaining characters 42

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? ○ Yes ◉ No

Please give the proposal reference or contract number.

Administrative forms

Proposal ID **101057049**

Acronym **BIO-UPTAKE**

Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. *	☒
2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions).	☒
3) We declare: - to be fully compliant with the eligibility criteria set out in the call - not to be subject to any exclusion grounds under the EU Financial Regulation 2018/1046 - to have the financial and operational capacity to carry out the proposed project.	☒
4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions.	☒
5) We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits).	☒
6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct.	☒
7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 428/2009, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used).	☒
8) We confirm that the activities proposed do not - aim at human cloning for reproductive purposes; - intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or - intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer. - lead to the destruction of human embryos (for example, for obtaining stem cells) These activities are excluded from funding.	☒
9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State.	☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Action
1	FUNDACION AITIIP	Spain	
2	CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL	Belgium	
3	FUNDACION CIDETEC	Spain	
4	SPECIFIC POLYMERS	France	
5	ASOCIACION DE INVESTIGACION METALURGICA DEL NOROESTE	Spain	
6	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	Spain	
7	SIMCON France	France	
8	PODCOMP AB	Sweden	
9	COMFIL APS	Denmark	
10	MOSES PRODUCTOS SL	Spain	
11	UNIVERSITY OF LIMERICK	Ireland	
12	POLYMERIS	France	
13	ASOCIACION ESPANOLA DE NORMALIZACION	Spain	

Organisation data

PIC	Legal name
997301427	*FUNDACION AITIIP*

Short name: AIT

Address

Street	POLIGONO EMPRESARIUM, CALLE ROMERO N 12
Town	ZARAGOZA
Postcode	50720
Country	Spain
Webpage	http://www.aitiip.com/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	06/06/2003 - no
SME self-assessment	unknown
SME validation	06/06/2003 - no

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant
Controls	MOSES PRODUCTOS SL

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **David**

Last name* **PONCE**

E-Mail* **david.ponce@aitiip.com**

Position in org. Head of European projects

Department FUNDACION AITIIP

☒ Same as organisation name

☒ Same as proposing organisation's address

Street POLIGONO EMPRESARIUM, CALLE ROMERO N 12-14

Town ZARAGOZA

Post code 50720

Country Spain

Website www.aitiip.com

Phone +34 976 46 45 44

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Julio	Vidal	julio.vidal@aitiip.com	+34 684 46 87 01
Lidia	Garcia	lidia.garcia@aitiip.com	+34 976 46 45 44
Pere	Castell	pere.castell@aitiip.com	+34 976 46 45 44

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	David	Ponce	Man	Spain	david.ponce@atiiip.com	Category B Senior re	Team member	0000-0001-6044-3848	Orcid ID
Dr	Lidia	Garcia	Woman	Spain	lidia.garcia@aitiip.com	Category B Senior re	Team member	0000-0003-1558-5526	Orcid ID
Dr	Pere	Castell	Man	Spain	pere.castell@aitiip.com	Category A Top grad	Leading	0000-0002-3611-9957	Orcid ID
Mr	Julio	Vidal	Man	Spain	julio.vidal@aitiip.com	Category D First stag	Team member	0000-0002-1094-0720	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*G. Quiles, L.; Vidal, J.; Luzi, F.; Dominici, F.; Fernandez Cuello, A.; Castell, P. Color Fixation Strategies on Sustainable Poly-Butylene Succinate Using Biobased Itaconic Acid. Polymers 2021,13, 79.*
Publication	*Julio Vidal, David Ponce, Antonio Miravete, Jesus Cuartero & Pere Castell (2019): Bio-binders for the improvement of the performance of natural fibers as reinforcements in composites to increase the sustainability in the transport sector, Mechanics of Advanced Materials and Structures, DOI: 10.1080/15376494.2019.1633447.*
Publication	*Garcia-Quiles, L.; Valdes, A.; Fernandez Cuello, A.; Jimenez, A.; Garrigos, M.C.; Castell, P. Reducing off-Flavour in Commercially Available Polyhydroxyalkanoate Materials by Autooxidation through Compounding with Organoclays. Polymers 2019, 11, 945. https://doi.org/10.3390/polym11060945*
Publication	*Lidia Garcia-Quiles, Angel Fernendez Cuello and Pere Castell. Sustainable Materials with Enhanced Mechanical Properties Based on Industrial Polyhydroxyalkanoates Reinforced with Organomodified Sepiolite and Montmorillonite. Polymers 2019, 11(4), 696; https://doi.org/10.3390/polym11040696*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
VIBES	*BBI-2020-RIA. 101023190,* *IMPROVING RECYCLABILITY OF THERMOSET COMPOSITE MATERIALS THROUGH A GREENER RECYCLING TECHNOLOGY BASED ON REVERSIBLE BIOBASED BONDING MATERIALS*
HELACS	*H2020-CS2-CFP11-2020-101007871,* *Holistic processes for the cost-effective and sustainable management of End of Life of Aircarft Composite Structures (Role: Coordinator)*
INN-PAEK	*H2020-CS2-CFP11-2020-101007868, www.aitiip.com/innpaek.html* *INNovative thermoplastic transformation processes to produce a flange wheel in PAEK reinforced with short Fibres*
BIZENTE	*H2020-BBI-JTI-2019- 886567, www.bizente.eu/* *Apply ligninases to resolve end-of-life issues of thermoset composite plastics targeting the recycling of thermoset composites (Role: Coordinator).*
MANDALA	*H2020-BBI-JTI-2018_837715, www.mandalaproject.eu/* *Boosting the transition of MultilAyer/multipolymer packagiNg into more sustainable multilayer/single polymer products for the fooD and phArma sectors through the deveLopment of innovative functional Adhesives? (Role: Coordinator).*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Recycling and valorisation facilities	*Extensive facilities (4000 m2) for the recycling and valorisation of thermoplastic and composite materials. In addition, there is also a plant for processing large parts/components including: dismantling, separation, sorting and recycling strategies (enzymatic, chemical and mechanical) of composites*
Pilot processing plant	*Industrial plant with over 6.000 m2 for the pre-processing (sorting, dismantling and cutting) and mechanical recycling of large composite parts together with large tooling equipment.*

Polymer Laboratory	*Mechanical recycling systems: grinder, cutting* *Chemical reactors for materials modification: additive modification, functionalization, surface treatment.* *Chemical/Enzymatical laboratory: 1L sizevolumetric chemical reactors for the enzymatic optimization of composite materials at lab scale*
Characterization Laboratory	*Mechanical characterization: rheology and universal testing system* *Thermal characterization: thermal stability and controlled pyrolysis* *Ceast CR20 Capillary rheometer* *HDT heat deflection temperature* *Charpy impact* *MFI analyzer* *Optical microscopy ZEISS* *IR thermal camera* *Densimeter* *Moisture analyzer*
Digitalization and process monitoring	*Automatization and robotic systems for the monitoring and digitalisation of the recycling processes.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998868171	*CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL*

Short name: CTB

Address

Street	RUE MONTOYER 24/2
Town	BRUXELLES
Postcode	1000
Country	Belgium
Webpage	www.centexbel.eu

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	31/12/2011	- yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Guy**

Last name* **Buyle**

E-Mail* **gbu@centexbel.be**

Position in org. Manager EU research

Department CENTRE SCIENTIFIQUE & TECHNIQUE DEL'INDUSTRIE TEXTILE BELGE ASBL

☒ Same as organisation name

☒ Same as proposing organisation's address

Street RUE MONTOYER 24/2

Town BRUXELLES

Post code 1000

Country Belgium

Website https://www.centexbel.be/en

Phone +32 92 438 253

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Isabel	De Schrijver	ids@centexbel.be	+32 488 999 226

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Guy	Buyle	Man	Belgium	gbu@centexbel.be	Category B Senior re	Team member	0000-0002-1060-9851	Orcid ID
Ms	Isabel	De Schrijver	Woman	Belgium	ids@centexbel.be	Category B Senior re	Leading	0000-0002-1890-322X	Orcid ID
Ms	Elke	Demeyer	Woman	Belgium	edm@vkc.be	Category D First stag	Team member	0000-0002-8335-3914	Orcid ID
Mr	Robbe	De Bisschop	Man	Belgium	rdi@vkc.be	Category D First stag	Team member	0000-0002-5585-6397	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*L. Van der Schueren et al. - BIO4SELF: Biobased self-functionalised self-reinforced composite materials based on high-performance nanofibrillar PLA fibres. Bioplastics magazine [04/16] Vol. 11, p. 36-37, 2016*
Publication	*L. Van der Schueren, Guy Buyle, Luc Ruys. Biobased self-reinforced polymer composites from polylactic acid, Plastics are Future congress, Valencia, 2016*
Publication	*I De Schrijver - Impact of regulation on processing and recycling in the textile and plastic industry, EFRA, general assembly meeting, Belgium, 2016*
Publication	*D De Smet, M. Vanneste - Development and implementation of biomaterials in textile coating and finishing, ICTF 2015: 17th International Conference on Textiles and Fashion, Paris 2015*
Publication	*Book chapter: Biobased textile coatings and composites; book title: Biobased products and industries, ISBN 978-0-12-818493-6 (2020, Elsevier), David De Smet, Frederik Goethals, Brecht Demedts, Willem Uyttendaele and Myriam Vanneste*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BIO4SELF	*Bio-based self-functionalised self-reinforced composite materials based on high performance nanofibrillar PLA fibres (H2020 2016-2019) ? Coordinated by CTB.*
SEABIOCOMP	*Development and demonstrators of durable biobased composites for a marine environment (Interreg 2 Seas 2019-2023)*
PRESERVE	*High-performance sustainable bio-based packaging with tailored end-of-life and upcycled secondary use (H2020 2021-2024)*
Fibriltex	*Microfibrillar reinforced textiles and derived composites (CORNET)*
Ecoxy	*Bio-based recyclable, reshapable and repairable (3R) fibre-reinforced EpOXY composites for automotive and construction sectors. (H2020 - BBI 2017-2021) - WP-leader: high tenacity PLA fibre reinforcements*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Compounder	*twin-screw compounder under water pelletizer for formulating biopolymers*
Pilot line for mono & multifilament production	*Mono- and bicomponent structures; FDY, POY, BCF (throughput 2-14kg/hour, production speed up to 3500 m/min)*
Semi-industrial extrusion line	*Injection moulding machines, profile and sheet extrusion line (width 50 cm), blown film extrusion line*
Textile production	*Knitting, weaving, braiding equipment*
Analytical instruments	*Characterization of polymer properties including, rheological behavior, stability, thermal properties and morphology, mechanical properties of textiles and plastics*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999461229	*FUNDACION CIDETEC*

Short name: CID

Address

Street	PASEO MIRAMON 196 PARQUE TECNOLOGICO D
Town	SAN SEBASTIAN
Postcode	20014
Country	Spain
Webpage	www.cidetec.es

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	06/07/2018	- no
SME self-assessment	07/04/1998	- no
SME validation	07/04/1998	- no

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Aratz**

Last name* **Genua**

E-Mail* **agenua@cidetec.es**

Position in org. Senior Researcher. Project manager

Department Polymers & Composites Unit. Surface Engineering Institute

☐ Same as organisation name

☒ Same as proposing organisation's address

Street PASEO MIRAMON 196 PARQUE TECNOLOGICO DE MIRAMON

Town SAN SEBASTIAN

Post code 20014

Country Spain

Website https://www.cidetec.es/en/home

Phone +34 943318212

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Alaitz	Rekondo	arekondo@cidetec.es	+34 943318212
Elena	Jubete	ejubete@cidetec.es	+34 943318212
Jon	Lacunza	jlacunza@cidetec.es	+34 943309022

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Aratz	Genua	Woman	Spain	agenua@cidetec.es	Category B Senior re	Leading	0000-0003-0437-1119	Orcid ID
Dr	Alaitz	Rekondo	Woman	Spain	arekondo@cidetec.es	Category A Top grad	Team member	0000-0003-2195-5167	Orcid ID
Dr	Elena	Jubete	Woman	Spain	ejubete@cidetec.es	Category A Top grad	Team member	0000-0003-0598-2325	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Genua, A. et al. Build-To-Specification Vanillin and Phloroglucinol Derived Biobased Epoxy-Amine Vitrimers. Polymers 2020, 12, 2645; doi:10.3390/polym12112645*
Publication	*Ruiz de Luzuriaga, A. et al. Epoxy resin with exchangeable disulfide crosslinks to obtain reprocessable, repairable and recyclable fiber-reinforced thermoset composites. Materials Horizons 2016, 3, 241; doi:10.1039/C6MH00029K*
Publication	*Odriozola I. et al. Thermomechanically reprocessable epoxy composites and processes for their manufacturing. Patent Priority number: EP 14382184.1.*
Publication	*Ruiz de Luzuriaga, A. et al. Transient mechanochromism in epoxy vitrimer composites containing aromatic disulfide crosslinks. Journal of Materials Chemistry C 2016, 4, 6220; doi: 10.1039/C6TC02383E.*
Publication	*Aranberri, I. et al. Thermoformable and recyclable CFRP pultruded profile manufactured from an epoxy vitrimer . Polymer testing 2021, 93, 106931 ; doi:10.1016/ j.polymertesting.2020.106931*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ECOXY	*Bio-based recyclable, reshapable and repairable (3R) fibre-reinforced epoxy composites for automotive and construction sectors. Coordinated by CIDETEC. (H2020-BBI, GA 744311; 2017-2020, www.ecoxy.eu)*
RECYSITE	*Production of fully recyclable and reusable green composites based on bioresins and natural fibres. (LIFE15,ENV/BE/000204; 2016-2019, www.recysite.eu)*
CARBO4POWER	*High performance multifunctional and innovative 3R composites for wind energy sector. Demonstrators (tidal and wind turbine blades) will be manufactured with the developed 3R resin formulation(s) via infusion, RTM or ATL (3R prepregs) processes (H2020-EU; GA 953192; 2020-2024, www.carbo4power.net)*
AIRPOXY	*High performance 3R thermosets for the aeronautics sector. High performance 3R thermoset, epoxy based composites. Coordinated by CIDETEC. (H2020-MG; GA 769274 2018-2022, www.airpoxy.eu)*
HARVEST	*Development of nano-modified 3R thermosets for the aeronautics sector. (H2020-MG; GA 769140; 2018-2021, www.harvest-project.eu)*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
PREPREG MACHINE	*Coatema BASECOATER BC 54 prepregging machine for epoxy systems to produce prepregs with a maximum width of 300 mm, equipped with knife and foulard impregnation systems.*
RESIN TRANSFER MOULDING (RTM)	*Isojet DPE COMPACT injection machine for epoxy and polyurethane resins, 1K or 2K, tanks of 7,6 litres heatables up to 120 C, flow between 50 to 300 cc/min and ratio between 100/100 to 100/10 for 2K, PLC control.*
PRESS	*Press for injection mould equipped with 2 heatable platens of 600x600 mm, with a clamping pressure of 36 Tons.*
REACTOR FOR RESIN FORMULATION	*1-5 L volume Netzsch Laboratory Dissolver SHEARMASTER.*
MECHANICAL TESTING MACHINE	*Universal extensometer Instron 5985 equipped with 250 kN load cell.*

DSC	*DSC (Differencial Scanning Calorimeter) TA INSTRUMENTS Discovery DSC 25 auto.*
TGA	*TA Instruments Thermogravimetric Analyzer Q500.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998808807	*SPECIFIC POLYMERS*

Short name: SP

Address

Street	AVENUE DES COCARDIERES 150 ZAC VIA DOMIT
Town	CASTRIES
Postcode	34160
Country	France
Webpage	www.specificpolymers.fr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	23/01/2003	- yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Alain**

Last name* **Graillot**

E-Mail* **alain.graillot@specificpolymers.fr**

Position in org. COO

Department R&D

☐ Same as organisation name

☒ Same as proposing organisation's address

Street AVENUE DES COCARDIERES 150 ZAC VIA DOMITIA

Town CASTRIES

Post code 34160

Country France

Website www.specificpolymers.fr

Phone +33499749135

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Marion	Combre	marion.combre@specificpolymers.fr	+33499749135
Samuel	Malburet	samuel.malburet@specificpolymers.fr	+xxx xxxxxxxxx
Agathe	Bouvet-Marchand	agathe.bouvet-marchand@specificpolymers.fr	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Alain	Graillot	Man	France	alain.graillot@specificpolymers.fr	Category B Senior re	Leading	0000-0001-5970-5687	Orcid ID
Mr	Samuel	Malburet	Man	France	samuel.malburet@specificpolymers.fr	Category C Recognis	Team member	0000-0002-4224-9520	Orcid ID
Dr	Agathe	Bouvet-Marchand	Woman	France	agathe.bouvet-marchand@specificpolymers.fr	Category C Recognis	Team member	0000-0003-0371-1126	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Good	*Biobased expoxy resins, cyclocarbonates, epoxidized oils*
Publication	*Enhancing the Recyclability of a Vegetable Oil-Based Epoxy Thermoset through Initiator Influence, Chiara di Mauro, Thi-Nguyet Tran, Alain Graillot, and Alice Mija, ACS Sustainable Chemistry & Engineering 2020 8 (20), 7690-7700 https://doi.org/10.1021/acssuschemeng.0c01419*
Publication	*Sustainable access to fully biobased epoxidized vegetable oil thermoset materials prepared by thermal or UV-cationic processes, S. Malburet, C. Di Mauro, C. No, A. Mija, M. Sangermano and A. Graillot. RSC Advances, 2020. DOI: 10.1039/d0ra07682a*
Publication	*Dual Cross-linking of Epoxidized Linseed Oil with Combined Aliphatic/Aromatic Diacids Containing Dynamic S-S Bonds Generating Recyclable Thermosets, T. Thi-Nguyet, C. Di Mauro, S. Malburet, A. Graillot and A. Mija., ACS Applied Biomaterials, 2020. https://doi.org/10.1021/acsabm.0c00788*
Publication	*Build-To-Specification Vanillin and Phloroglucinol Derived Biobased Epoxy-Amine Vitrimers, A. Genua, S. Montes, I. Azcune, A. Rekondo, S. Malburet, B. Dayde-Cazals, A. Graillot, Polymers 2020 https://doi.org/10.3390/polym12112645*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
VIBES	*Improving recyclability of thermoset composite materials through a greener recycling technology based on reversible biobased bonding materials. European Project funded under H2020-BBI-JTI-2020 (grant agreement 101023190 - 2021-2025). The project presents an innovative solution to rsolve end-of-life issues of thermoset composites based on the development of a new green technology focused on the controlled separation and recovery of composite material components by means of developing customised*
BIZENTE	*Apply ligninases to resolve end-of-life issues of thermoset composite plastic. European Project funded under H2020-BBI-JTI-2019 (grant agreement 886567 - 2020-2023). The project will determine directed enzyme evolution strategies to improve ligninolytic oxidoreductases which have been selected as the most suited biocatalysts to degrade the three types of thermoset composites addressed in the project: epoxy, polyester and vinyl ester.*
ECOXY	*Bio-based recyclable, reshapable and repairable (3R) fibre-reinforced EPOXY composites for automotive and construction sectors. European project funded under H2020-BBI-JTI-2016 (grant agreement 744311 - 2016-2019). The project will develop innovative bio-based epoxy resins and fibre reinforcements to produce new sustainable and techno-economically competitive fibre reinforced thermoset composites by targeting advanced functionalities: reparability, reprocessability and recyclability (3R).*
BAMCO	*BAmboo long fibre reinforced biobased Matrix Composites. 2018-2021, funded by the French national fund Fonds Unique Interminesteriel. The objective of the project is to manufacture new bio-composites created using long bamboo fibres that could be used in the aerospace industry for cabin furnishings and in cover panels and fuselage cladding panels.*
BIOVANILLINE	*Bio-sourced materials based on vanillin synthetized from bio-lignin. 2016-2019, funded by the French Agency of Environment.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Organic synthesis equipment	*In our 600m2 lab allow us to produce up to few kg products and resins (from classical round bottom flasks to high capacity synthesis reactors)*
Polymer synthesis equipment	*In our 600m2 lab allow us to produce up to few kg product products and resins (from classical round bottom flasks to high capacity synthesis reactors)*
Materials formulation and processing equipment	*spin-coating, bare-coating, UV-conveyor, spin-mixer, mechanical disperser, mechanical specimen moulds, vacuum ovens, etc.*
Physico-chemical analysis equipment	*Nuclear Magnetic Resonance-Bruker 300 MHz, COSY, 2D DOSY, Size Exclusion Chromatography-Agilent technologies, Differential Scanning Calorimetry-TA Instrument Q2000, InfraRed Analysis-Perkin Elmer ATR Module, Brookfield viscosity, Optical contact angle measuring system*
Material numerical modeling	*Relationship with an experienced academic partner and 2 employees working full time on the subject (including PhD)*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998801532	*ASOCIACION DE INVESTIGACION METALURGICA DEL NOROESTE*

Short name: AIM

Address

Street	CALLE RELVA TORNEIROS 27A
Town	PORRINO
Postcode	36410
Country	Spain
Webpage	www.aimen.es

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	29/09/2008	- no
SME self-assessment	unknown	
SME validation	29/09/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Smart Systems and Smart Manufacturing	☐ not applicable
	☒ Same as proposing organisation's address	
Street	CALLE RELVA TORNEIROS 27A	
Town	PORRINO	
Postcode	36410	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mrs

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Maria**

Last name*: **Cervino**

E-Mail*: **gpfint@aimen.es**

Position in org.: Funding Programmes Manager

Department: Funding Programmes Unit

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: CALLE RELVA TORNEIROS 27A

Town: PORRINO

Post code: 36410

Country: Spain

Website: www.aimen.es

Phone: 003498634400

Phone 2: *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Lucia	Alonso Ferreira	Woman	Spain	lucia.alonso@aimen.es	Category C Recognis	Leading	0000-0001-8608-2438	Orcid ID
Mr	Baltasar	Lodeiro Senaris	Man	Spain	baltasar.lodeiro@aimen.es	Category C Recognis	Team member	0000-0002-0531-7432	Orcid ID
Ms	Lara	Barja Besada	Woman	Spain	lara.barja@aimen.es	Category D First stag	Team member	0000-0001-8740-2720	Orcid ID
Mr	Carlos	Gonzalez Val	Man	Spain	carlos.gonzalez@aimen.es	Category C Recognis	Team member	0000-0003-2760-1650	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Babcinschi, M., Freire, B., Neto, P., Ferreira, L. A., Senaris, B. L., & Vidal, F. (2019, September). AutomationML for Data Exchange in the Robotic Process of Metal Additive Manufacturing. In 2019 24th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA) (pp. 65-70). IEEE.*
Publication	*Babcinschi, M., Freire, B., Ferreira, L., Senaris, B., Vidal, F., Vaz, P., & Neto, P. (2020). An integrated and interoperable AutomationML-based platform for the robotic process of metal additive manufacturing. Procedia Manufacturing, 51, 26-31.*
Publication	*Freire, B., Babcinschi, M., Ferreira, L., Senaris, B., Vidal, F., & Neto, P. (2020). Direct Energy Deposition: a complete workflow for the additive manufacturing of complex shape parts. Procedia Manufacturing, 51, 671-677.*
Publication	*Christian Eike Precker, Andrea Gregores-Coto, and Santiago Muinos-Landin,Generative Adversarial Networks for Generation of Synthetic High Entropy Alloys. To appear on Procc. International Conference on Material Science and Artificial Intelligence (ICSCML2021)*
Publication	*Ferreira, L. A., Souto, M., Chappuis, C., & El Khaldi, F. (2020). Off-line programming of a flexible and adaptive production line for composite-metal multi-material manufacturing based on OPC-UA communication. Procedia Manufacturing, 51, 520-526.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
INTEGRADDE (H2020-NMBP-FOF-2018 DT-FOF-04-2018)	*The goal is to develop a new manufacturing methodology capable of ensuring the manufacturability and quality of a metal component from initial product design via Direct Energy Deposition technologies, implementing a zero-defect manufacturing approach. AIMEN role within this project consists in developing a pilot- involving knowledge in laser-based processing together with robotics, control and data analytics.* *H2020-NMBP-FOF-2018 DT-FOF-04-2018-820776*
PENELOPE (DT-FOF-10-2020)	*The main goal is to develop a novel closed-loop digital pipeline based on the development of modular and reconfigurable production approach for the manufacturing of high-precision large-scale parts. AIMEN role within this project consists in a) development of interoperable hardware for monitoring and process control, b) development of artificial intelligence algorithms for cross-correlation between data gathering from process monitoring and quality control.* *DT-FOF-10-2020-958303*
DOMMINIO (MG-3-5-2020)	*The project will develop an innovative methodology to ensure cost-effective, efficient and sustainable manufacturing of high quality multifunctional and intelligent airframe parts. Lead by AIMEN deals with the development of a multidisciplinary optimization framework for ATL and FFF processes.* *MG-3-5-2020-101007022*
DIMOFAC (DT-FOF-08-2019)	*The project partnered by AIMEN dealing with the digitalization in the multimaterial components enabling modularity by integrating Plug-and-Produce modules.* *DT-FOF-08-2019*
ACHIEF (LC-SPIRE-08-2020)	*AIMEN role within this project deals with the development of artificial intelligence algorithms for virtually design new materials.* *LC-SPIRE-08-2020*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Digital infrastructure	*AIMEN counts with multiple GPU servers (CPU: 16 cores, Memory: 768 GB, GPU: 2x RTX 8000 48GB) and a cluster for the development of AI and distributed computing solutions (2 servers Dell PowerEdge 740, 1 storage system DellEMC de 80 TB, Software VMWare vSphere Essentials Plus).*
Embedded systems	*A laboratory of CPS and embedded systems for electronics design (FPGA, GPU, ARM) and software development (VHDL, CUDA, C++) applied to condition monitoring, real-time signal processing and pattern recognition.*
Robotics laboratory	*A Laboratory for Industrial Robotics with several manipulators (from ABB, KUKA, MOTOMAN) that could be easily integrated with different systems and tools to test different pilot applications into laboratory environment.*
Machine vision laboratory	*Machine vision laboratory including different vision systems and vision software.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998993786	*IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA*

Short name: IRI

Address

Street	CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA
Town	MADRID
Postcode	28006
Country	Spain
Webpage	www.iris.cat

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	unknown
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	24/02/2009	- yes

Departments carrying out the proposed work

Department 1

Department name	Project Management Office	☐ not applicable
	☐ Same as proposing organisation's address	
Street	Carretera d'Esplugues 39-41	
Town	Cornell de Llobregat (Barcelona)	
Postcode	08940	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Ana Maria**

Last name* **Lopez**

E-Mail* **alopez@iris-eng.com**

Position in org. Project and Innovation Manager

Department Project Management Office

☐ Same as organisation name

☐ Same as proposing organisation's address

Street Carretera d'Esplugues 39-41

Town Cornell de Llobregat (Barcelona)

Post code 08940

Country Spain

Website https://www.iris-eng.com/

Phone +34 93 554 25 00

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Tomeu	Coll	tcoll@iris-eng.com	+34 93 554 25 00
Albert	Torres	albert.torres@iris-eng.com	+xxx xxxxxxxxx
Alexandra	Poch	apoch@iris-eng.com	+xxx xxxxxxxxx
Colm	Digby	cdigby@iris.cat	+xxx xxxxxxxxx
Joan	Puig	jpuig@iris-eng.com	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Ana Maria	Lopez Sabiron	Woman	Spain	alopez@iriseng.com	Category B Senior re	Team member	0000-0002-8509-7109	Orcid ID
Dr	Laura	Rodriguez Turienzo	Woman	Spain	lrodriguez@iriseng.com	Category B Senior re	Team member	0000-0003-4656-1674	Orcid ID
Dr	David	Nettleton	Man	United Kingdom	david.nettleton@iris-eng.com	Category B Senior re	Leading	0000-0002-5852-7716	Orcid ID
Dr	Sara	Sanchez	Woman	Spain	ssanchez@iriseng.com	Category B Senior re	Team member	0000-0002-1154-165X	Orcid ID
Dr	Victor	Olmos	Man	Spain	volmos@iriseng.com	Category B Senior re	Team member	0000-0003-0132-6849	Orcid ID
Dr	Nicola	Palomo Blascetta	Man	Italy	npalombo@iriseng.com	Category B Senior re	Team member	0000-0003-3827-5360	Orcid ID
Dr	Johannes	Ziegler	Man	Germany	jziegler@iriseng.com	Category B Senior re	Leading	0000-0002-2022-4314	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	*An integrated optical system for the examination of sample materials, EU Patent Application EP18198458, 03/10/2018, Rosales Alejandro, Kehoe Timothy, Garrick Meabh. https://worldwide.espacenet.com/patent/search/family/060270416/publication/EP3467478A1?q=pn%3DES2817438T3*
Other achievement	*Device for detecting foreign matter in an edible substance or in a container containing the edible substance and method employing the device, EP19382101 Patent, Alejandro Rosales, Adrian Manotas, Joan Puig*
Good	*VisumNIR: In-Line NIR monitoring for product standardization and increase of production efficiency. https://www.iris-eng.com/*
Publication	*Araujo-Andrade, C., Bugnicourt, E., Philippet, L., Rodriguez-Turienzo, L., Nettleton, D., Hoffmann, L., & Schlummer, M.. Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes in view of their posterior recycling. Waste Management & Research, 2021, 39(5), 631?651. https://doi.org/10.1177/0734242X21997908*
Publication	*A. Obisesan K, Neri S, Bugnicourt E, Campos I, Rodriguez-Turienzo L. Determination and Quantification of the Distribution of CN-NL Nanoparticles Encapsulating Glycyrrhetic Acid on Novel Textile Surfaces with Hyperspectral Imaging. Journal of Functional Biomaterials. 2020; 11(2):32. https://doi.org/10.3390/jfb11020032*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
PRESERVE	*H2020-NMBP-BIOTEC, GA 952983. (2021-2025) IRIS: Advanced monitoring solutions for new microfibril formation*
BIONANOPOLYS	*H2020-NMBP, GA 953206, (2021-2024) IRIS: Inline monitoring systems for bio nanocomposites dispersion and composition leveraging photonic technologies*
BIOMAC	*H2020-NMBP, GA 952941. (2021-2024) IRIS: Monitoring solutions including AI for measuring the composition and the reaction evolution*
MULTIPLE	*H2020 ICT, GA 871783. (2019 ? 2022) IRIS: Photonic solutions and advanced chemometrics for process optimization*
REMADYL	*H2020 SC5, GA 821136. (2019-2023) IRIS: Advanced monitoring systems for sorting relevant materials for the recycling process*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spectroscopic Laboratory	*Equipped with several instruments: FTIR-ATR, 2 UV-VIS, Extended-SWIR and a NIR spectrophotometers as well as a SWIR Hyperspectral imager. The associated staff are properly trained and skilled in chemometrics modelling by using a commercial suite as well as ad hoc noncommercial chemometric tools*
Electronics mechanical & optical workshops & labs	*Equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others*

Food and wet chemistry lab	*Equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter, Ford cup viscometers, water vapour permeation cells, turbidimeter, climatic chambers, manual and automatic coaters.*
Optical photonics-based technologies & PAT system	*HSI and NIR based cameras suitable for the surface and composition analysis of different materials and for the optimization and monitoring of numerous processes. They can be applied for the digitalisation of biowaste valorisation processes, adding value to the circular and biobased economy.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
889684971	*SIMCON France*

Short name: SIM

Address

Street	19, Rue Louis Gu?rin
Town	VILLEURBANNE
Postcode	69100
Country	France
Webpage	www.simcon.fr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	23/08/2021 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Raphael**

Last name* **Mosnier**

E-Mail* **raphael.mosnier@simcon.fr**

Position in org. Director

Department SIMCON France

☒ Same as organisation name

☒ Same as proposing organisation's address

Street 19, Rue Louis Gu?rin

Town VILLEURBANNE

Post code 69100

Country France

Website https://www.simcon.fr/

Phone +33 6 07 18 42 71

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Romain	Chariere	romain.chariere@simcon.fr	+33 6 34 48 28 36

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Romain	Chariere	Man	France	romain.chariere@simcon.fr	Category C Recognis	Leading		
Mr	Ludovic	Potier	Man	France	ludovic.potier@simcon.fr	Category C Recognis	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Artificial intelligence is applied to injection moulding to optimize the process through a DoE. Learning is done by automatically launching simulations in Cadmould. The user defined the variations on the process, topological parameters and geometrical tolerances.* *Varimos Virtual establish a response law for each physical quantity, then optimizes all parameters to reach the defined quality criteria, and then suggests the optimal process settings.* *Plastiques & Caoutchoucs Magazine n?957*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
CONPROMI	*The overall objective of the Con-Pro-Mi project is to develop new convergence processes in micro-fabrication in order to develop hybrid solutions for high value-added applications. The project aims to create new manufacturing lines adapted to hybrid replication processes with several materials to reduce the cost of and integrating adaptable modules for greater flexibility and efficiency at each phase of the process.*
PACKPLAST	*Development of multi-material packaging technologies for automotive Li-ion batteries and various uses*
SILICONE	*The ambition of the SILICONE LSR project was to position itself more specifically on the market of LSR overmoulding on thermoplastic materials, and to provide optimised and very competitive solutions.* *- To improve productivity (cycle time reduction of at least 20%), quality and competitiveness* *- To develop industrial tools and equipment* *- To develop innovative numerical simulation tools* *- Demonstrate the validity and performance of the solutions developed*
ACAPULCO	*The objective of the project is to develop an innovative solution for the production of tools necessary for the manufacture of injected plastic parts.* *These new tools made of polymer by additive manufacturing make possible to inject parts with a quality similar to an industrial product injected into a metal production mould. This solution will replace multiple long and costly stages of modeling, part prototyping and prototype tooling testing.* *https://www.acapulco-fabrication-additive.com/*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
CADMOULD	*Cadmould simulates the injection process of thermoplastic, resin (RIM), elastomer and silicone. It also allows to simulate the overmoulding of a plastic or metal insert (for example electrical connectors) and to calculate the deformations and stresses suffered by these inserts during the process.*
VARIMOS	*Design of experiments software based on simulations made with CADMOULD. This software allows you to identify the optimal pocess parameters for plastic injection.*
T-SIM	*T-SIM allows to simulate the thermoforming of thin and thick thermoplastic plates.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
916404591	*PODCOMP AB*

Short name: POD

Address

Street	SKYLVAGEN 1
Town	OJEBYN
Postcode	943 33
Country	Sweden
Webpage	www.podcomp.se

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	01/11/2016 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Ms

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Birgitha**

Last name*: **Nystrom**

E-Mail*: **birgitha.nystrom@podcomp.se**

Position in org.: Sustainability coordinator

Department: PODCOMP AB

☒ Same as organisation name

☐ Same as proposing organisation's address

Street: Skylv?gen 1

Town: OJEBYN

Post code: 94333

Country: Sweden

Website: www.podcomp.se

Phone: +46 70 312 66 40

Phone 2: *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Birgitha	Nystrom	Woman	Sweden	birgitha.nystrom@podcomp.se	Category B Senior re	Leading		
Dr	Peter	Lundmark	Man	Sweden	peter.lundmark@podcomp.se	Category B Senior re	Team member		
Ms	Asa	Lindstrom	Woman	Sweden	asa.lindstrom@podcomp.se	Category D First stag	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*JEC Composites ISSUE #138 - JANUARY/FEBRUARY 2021, RAMSSES: Lightweighting ships for a cleaner environment*
Publication	*Aitomaki Y., Nystrom B., Biobased composite in a green office walls: from ecodesign to manufacture, Biobased materials - 9th WPC, Natural Fibre and other innovative Composites 2012*
Service	*Development and serial production of prefabricated composite bathroom unit TOTALTAT for construction industry.*
Service	*Development and serial production of composite floorings TOTALTAT for construction industry.*
Publication	*PhD thesis P. Lundmark LTU and Lic Thesis B. Nystrom LTU*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
WOODY	*Development of innovative biobased composites*
NCC FOAM	*Development of biobased foam core*
GreenLight	*Development of biobased carbon fibres*
ThroughLife	*Composites for marine applications focus on sustainability*
RAMSSES	*Composites for marine applications focus on fire*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Factory	*7 000 m2 production facility using vacuum infusion to produce composite sandwich panels and bathroom modules, 5 robots*
Space dedicated to development work	*30 m2*
Space available for development of pilot line	*100 m2*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998394617	*COMFIL APS*

Short name: COM

Address

Street	KAROLINELUNDSVEJ 2
Town	GJERN
Postcode	8883
Country	Denmark
Webpage	www.comfil.biz

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	29/07/2008	- yes
SME self-assessment	unknown	
SME validation	29/07/2008	- yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Hans**

Last name* **Knudsen**

E-Mail* **hans.knudsen@comfil.biz**

Position in org. Technical director

Department COMFIL APS

☒ Same as organisation name

☒ Same as proposing organisation's address

Street KAROLINELUNDSVEJ 2

Town GJERN

Post code 8883

Country Denmark

Website www.comfil.biz

Phone +45 20 13 94 37

Phone 2 *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	*1. JEC Innovation Award In the category "Sustainablity" 2. Techtextil Innovation Award in the category "Sustainable Solution" 3. Global Bioplastics Award*
Publication	*Influence of fiber-matrix adhesion on mechanical properties of unidirectional glass/ polybutylene terephthalate composites. 18th International Conference on Composites Materials (ICCM-18), August 21-27, 2011, Jeju, South Korea.*
Publication	*Wind Blades Using Cost-Effective Advanced Lightweight Design - Innovative Solutions for Wind Blades Made of Advanced Thermoplastic Materials, ITHEC 2014 - 2nd International Conference and Exhibition on Thermoplastic Composites,27-28 October 2014, Bremen, Germany*
Publication	*EP 717 133 Production and application of a shrinkable, permanently deformable textile material made out of hybrid yarn How to make a yarn and/or a fabric deformable to use in production of 3D composite parts.*
Good	*Brand for SR-PLA system*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Bio4Self H2020	*BIO4SELF aims at fully biobased self-reinforced polymer composites (SRPC).*
PassEnger Eureka	*Project about SR-PA materials for automotive*
Esprit FP7	*Resource-efficient self-reinforced plastic materials and processing, 2008-2012*
Walid FP7	*Development of thermoplastic hybrid yarn materials for off-shore wind blades, 2013-2017*
Genvind, DK	*The purpose of GENVIND is to identify and develop new and existing strategies for sustainable recycling of composites to provide Denmark with an environmental and financial solution. 2013-2017*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Drying	*Equipment to dry polymers works with dry heated air*
Spinning	*Small melt spin equipment for spinning continues fibres for research and development. It can go up to 400C. Comfil also has production machine for spinning*
Tensile/flex & impact testing	*A Tinius Olsen 50 k testing machine. Can test 3and 4 point flexural, tensile, ILSS and yarn testing. An Avery Charpy and Izod test machine*
Vacuum consolidation	*Comfil has 2 vacuum consolidation machines, which can go to 350C and 400C (upto 120*80*20cm laminates) in house development*
Pultrusion line	*Pultrusion line for development. It can go up to 400C*
Hybrid yarn machines	*Many different machines for texturizing and mingling of fibres*
Steamtex yarn machins	*Machine which can texturize with steam. Especially good for bringing bulk into yarn without loosing tension properties*

Weaving looms	*Comfil has 2 m rapier looms which works from creel*
Interval press	*Comfil has a continues interval press which manufactures continues sheets with a maximum width of 650 mm. thickness from ,3mm to 5mm*
Automatic pressing line for composites	*Inhouse developed pressline, where the material is heated up in a isothermal system and transferred to a forming press which is having a tempered tool. Parts up to 1200*650*150 mm is possible.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
900200547	*MOSES PRODUCTOS SL*

Short name: MOS

Address

Street	CALLE ROMERO PL EMPRESARIUM 12
Town	ZARAGOZA
Postcode	50720
Country	Spain
Webpage	https://mosesproductos.com/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2018 - yes
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	R&D department	☐ not applicable
	☒ Same as proposing organisation's address	
Street	CALLE ROMERO PL EMPRESARIUM 12	
Town	ZARAGOZA	
Postcode	50720	
Country	Spain	

Links with other participants

Type of link	Participant
Is controlled by	FUNDACION AITIIP

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mrs

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Berta**

Last name*: **Gonzalvo**

E-Mail*: **berta.gonzalvo@mosesproductos.com**

Position in org.: CEO & Founder

Department: R&D department

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: CALLE ROMERO PL EMPRESARIUM 12

Town: ZARAGOZA

Post code: 50720

Country: Spain

Website: https://mosesproductos.com/

Phone: +34 976 46 45 44

Phone 2: *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mrs	Berta	Gonzalvo	Woman	Spain	berta.gonzalvo@mosesproductos.com	Category A Top grad	Leading	0000-0003-3421-4528	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
UpPE-T	*Title: Upcycling of PE and PET wastes to generate biodegradable bioplastics for food and drink packaging* *Grant agreement ID: 953214* *Funded under Programme: H2020* *Topic: CE-BIOTEC-09-2020.* *Budget: 7.826.685 euros*
ACROBA	*Title: AI-Driven Cognitive Robotic Platform for Agile Production environments* *Grant agreement ID: 101017284* *Funded under Programme: H2020* *Topic: Industrial Leadership- ICT- 2020-2* *Budget: 8.052.261,88 euros*
LIA	*Title: Implementation of a novel, advanced rotomoulding line* *Grant agreement ID: 20200037* *Funded under Programme: CDTI-IDITopic:* *Innovation Line. Support to Innovative Technologies* *Budget: 652.014,00 euros*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Industrial processing systems of polymers	*Injection moulding machine of 400 tons and 220 Tm clamp force* *Leonardo Smart V 2.8 rotomoulding machine* *Dryer system/dehumidifier for polymeric materials* *Extrusion-compounding system (Coperion ZSK 26 mm) adapted to biobased and recycled materials including recovered fibers from composites*
Bioenzymatic treatment of polymeric materials sys	*Pre-industrial mill system to facilitate the size reduction of the PUR-based polymeric materials to be latter recycled by enzymatic treatments for the obtention of monomers and secondary raw materials* *Ad hoc designed recycling biotechnological reactor-100 litre size*
Pilot processing plants	*Two plants (shared with AITIIP), one of 1.200 m2 and other of 1.500 m2, one destined to produce packages and the second plant for the production of blends and nanocomposites for the materials industry.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999809071	*UNIVERSITY OF LIMERICK*

Short name: LIM

Address

Street	NATIONAL TECHNOLOGICAL PARK, PLASSEY
Town	LIMERICK
Postcode	-
Country	Ireland
Webpage	www.ul.ie

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	29/10/2008	- no
SME self-assessment	unknown	
SME validation	29/10/2008	- no

Departments carrying out the proposed work

Department 1

Department name	School of engineering	☐ not applicable
	☒ Same as proposing organisation's address	
Street	NATIONAL TECHNOLOGICAL PARK, PLASSEY	
Town	LIMERICK	
Postcode	-	
Country	Ireland	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Maurice**

Last name* **Collins**

E-Mail* **maurice.collins@ul.ie**

Position in org. Senior Lecturer

Department UNIVERSITY OF LIMERICK

☒ Same as organisation name

☒ Same as proposing organisation's address

Street NATIONAL TECHNOLOGICAL PARK, PLASSEY

Town LIMERICK

Post code -

Country Ireland

Website www.ul.ie

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Mario	Culebras	mario.culebrasrubio@ul.ie	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Maurice	Collins	Man	Ireland	Maurice.collins@ul.ie	Category B Senior re	Leading	0000-0003-2536-4508	Orcid ID
Dr	Mario	Culebras	Man	Spain	Mario.culebrasrubio@ul.ie	Category C Recognis	Team member	0000-0001-9244-9347	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Collins, M.N., Nechifor, M., Tanasa, F., Zanoaga, M., McLoughlin, A., Strozyk, M.A., Culebras, M. and Teaca, C.A., 2019. Valorization of lignin in polymer and composite systems for advanced engineering applications-a review. International journal of biological macromolecules*
Publication	*Culebras, M., Ren, G., O'Connell, S., Vilatela, J. J., & Collins, M. N. (2020). Lignin doped carbon nanotube yarns for improved thermoelectric efficiency. Advanced Sustainable Systems, 4(11), 2000147*
Publication	*Culebras, M., Beaucamp, A., Wang, Y., Clauss, M.M., Frank, E. and Collins, M.N., 2018. Biobased structurally compatible polymer blends based on lignin and thermoplastic elastomer polyurethane as carbon fiber precursors. ACS Sustainable Chemistry & Engineering, 6(7), pp.8816-8825*
Publication	*Culebras, M., Sanchis, M.J., Beaucamp, A., Carsi, M., Kandola, B.K., Horrocks, A.R., Panzetti, G., Birkinshaw, C. and Collins, M.N., 2018. Understanding the thermal and dielectric response of organosolv and modified kraft lignin as a carbon fibre precursor. Green chemistry, 20(19), pp.4461-4472.*
Publication	*Ortiz-Serna, P., Carsi, M., Culebras, M., Collins, M. N., & Sanchis, M. J. (2020). Exploring the role of lignin structure in molecular dynamics of lignin/bio-derived thermoplastic elastomer polyurethane blends. International journal of biological macromolecules, 158, 1369-1379*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
VIBES	*Improving recyclability of thermoset composite materials through a greener recycling technology based in reversible biobased bonding materials* *2021-2024 : H2020 - BioBased Industries call*
NXTGENWOODS	*2020-2025 : Science foundation Ireland. Sustainable materials*
Lignin based carbon fiber for composite (LIBRE)	*2016-2021 : H2020 - BioBased Industries call: Lead Principal investigator (PI) and coordinator*
Controlled photo polymerisation of methacrylate c	*2016-2018: Enterprise Ireland and J&J: PI;*
Biobased Composites	*2013 University of Limerick Seed Funding: PI;*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
SEM SU-70	*Scanning electron microscopes SU-70 and Focused Ion Beam SEM (FIB-SEM)*
Composite Lab	*Composite lab equipped with a High temperature/pressure autoclave, hot drape former, resin transfer moulding injection system, clean room: Composite test panel manufacture;* *Automated tape laying machine: Preform and composite test panel manufacture*
FTIR	*Perkin Elmer Fourier Transform Infrared (FTIR),*
DLS	*Scattering Particle Size Distribution Analyser Horiba LA-920*
UV-VIs	*Double-beam UV-Visible spectrometer, Perkin-Elmer Lambda 35*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998394229	*POLYMERIS*

Short name: POL

Address

Street	180 RUE PIERRE ET MARIE CURIE MAISON DES EN
Town	BELLIGNAT
Postcode	01115
Country	France
Webpage	www.plastipolis.fr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	02/03/2009	- no
SME self-assessment	unknown	
SME validation	02/03/2009	- no

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mrs

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Annabelle**

Last name*: **Sion**

E-Mail*: **annabelle.sion@polymeris.fr**

Position in org.: Head of EU developments

Department: POLYMERIS

☒ Same as organisation name

☒ Same as proposing organisation's address

Street: 180 RUE PIERRE ET MARIE CURIE MAISON DES ENTREPRISES

Town: BELLIGNAT

Post code: 01115

Country: France

Website: www.polymeris.fr

Phone: +33 6 33 70 06 74

Phone 2: *+xxx xxxxxxxxx*

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Annabelle	Sion	Woman	France	annabelle.sion@polymeris.fr				

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☒

Stakeholder involvement

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	*Support for innovation projects: Polymeris supports its members in their innovation activities, using its large network of companies and RTOs to create collaborative project*
Service	*Scientific committee: group of scientific experts to evaluate and give insights on innovation potential*
Service	*Industrial committee: group of industrial experts to evaluate and give insights on innovation potential*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
CIRC-PACK, H2020	*The CIRC-PACK project will produce breakthrough biodegradable plastics using alternative bio-based raw materials, which will have an important role to play throughout the plastic value chain. CIRC-PACK will also develop smart eco-designs for these plastics. They'll be adapted to the new bio-based materials we develop to allow for better collection and recycling of multilayer and multi-material packaging*
DigiPrime, H2020	*Develop a new concept of Circular Economy digital platform overcoming current information asymmetry among value-chain stakeholders, in order to unlock new circular business models based on the data-enhanced recovery and re-use of functions and materials from high valueadded post-use products with a cross-sectorial approach*
Citrus Pack, Life+	*The project aims to demonstrate that sustainability and efficiency can be applied to agricultural and industrial practices by researching and boosting the potential of byproducts through their valorisation in a number of new value chains.*
AlpLink BioEco, Interreg Alpine Space	*The project's main objective is precisely to develop a cross-regional circular bio-based economic strategy that includes a roadmap and demonstrators for intelligently assessing, selecting and creating innovative VCs for the AS wood, agriculture, food/pharma and chemistry sectors.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
984232326	*ASOCIACION ESPANOLA DE NORMALIZACION*

Short name: UNE

Address

Street	CALLE GENOVA 6
Town	MADRID
Postcode	28004
Country	Spain
Webpage	www.une.org

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	24/03/1997 - no
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Fernando**

Last name* **Utrilla**

E-Mail* **futrilla@une.org**

Position in org. Head of research and Innovation Unit

Department ASOCIACION ESPANOLA DE NORMALIZACION

☒ Same as organisation name

☒ Same as proposing organisation's address

Street CALLE GENOVA 6

Town MADRID

Post code 28004

Country Spain

Website www.une.org

Phone (+34) 914 326 003

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Steffen	Jenkel	sjenkel@une.org	(+34) 914 326 003

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Fernando	Utrilla	Man	Spain	futrilla@une.org		Leading		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☒
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Other achievement	*UNE has at least 12 national standardization committees and subcommittees related with bio-based products, circular economy, packaging, sustainable manufacturing and information technology, among them CTN 49 Packaging. Horizontal aspects and environmental management, CTN 53 Plastics and rubber, CTN 71 Enabling digital technologies, CTN 168 Machinery for plastic and rubber, UNE/CTN 323 Circular economy etc.* *More than 500 published standards are directly related with these committe*
Other achievement	*The total number of standardization committees in UNE is over 200, involving more than 10.000 experts and having published more than 32.000 standards.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BRDGIT/BRIDGIT 2	*BRIDGIT - Bridging the gap between research and standardization FPA 2013-2014, and BRIDGIT 2 Bridge the gap between standardization, innovation and research communities FPA 2017-2019 http://www.cencenelec.eu/research/BRIDGIT2/Pages/default.aspx*
BIOMAT	*An Open Innovation Test Bed for Nano-Enabled Bio-Based PUR Foams and Composites - H2020 (2020 Tomorrow's Industry) 2021-2024 https://cordis.europa.eu/project/id/953270*
UPPE-T	*Upcycling of PE and PET wastes to generate biodegradable bioplastics for food and drink packaging - H2020 (2020 Transformation of Industry) 2020-2024 https://uppet.eu/*
INN-PRESSME	*Open innovation ecosystem for sustainable Plant-based nano-enabled biomaterials deployment for packaging, transport and consumer goods -* *H2020 (2020 Tomorrow's Industry) 2021-2025 - https://www.inn-pressme.eu/*
CIRCPACK	*Towards circular economy in the plastic packaging value chain - Horizon 2020 (2016-Circular economy) 2017-2020 https://circpack.eu/*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Database	*UNE, as a member of the European/International standardization system, has full access to its entire internal information infrastructure which will be required to fulfil the relevant tasks in the project.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
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    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

Proposal ID **101057049**

Acronym **BIO-UPTAKE**

3 - Budget

?

No.	Name of beneficiary	Country	Role	Personnel costs/€	Subcontracting costs/€	Purchase costs - Travel and substistence /€	Purchase costs - Equipment/€	Purchase costs - Other goods, works and services/€	Internally invoiced goods and services/€ (Unit costs-usual accounting practices)	Indirect costs/€	Total eligible costs	Funding rate	Maximum EU contribution to eligible costs	Requested EU contribution to eligible costs/€	Max grant amount	Income generated by the action	Financial contributions	Own resources	Total estimated income
1	Fundacion Aitiip	ES	Coordinator	560,250	0	26,000	9,216	135,500	0	182741.00	913707.00	100	913707.00	913,707	913707.00				913707.00
2	Centre Scientifique & Technique Del'industrie Textile Belge Asbl	BE	Partner	305,500	0	15,000	12,000	18,000	0	87625.00	438125.00	100	438125.00	438,125	438125.00				438125.00
3	Fundacion Cidetec	ES	Partner	333,600	0	5,950	0	30,000	0	92387.00	461937.00	100	461937.00	461,937	461937.00				461937.00
4	Specific Polymers	FR	Partner	206,010	30,000	10,500	42,587	80,500	0	84899.00	454496.00	100	454496.00	454,496	454496.00				454496.00
5	Asociacion De Investigacion Metalurgica Del Noroeste	ES	Partner	311,500	0	44,800	15,000	39,500	0	102700.00	513500.00	100	513500.00	513,500	513500.00				513500.00
6	Iris Technology Solutions, Sociedad Limitada	ES	Partner	361,200	0	25,000	0	144,000	0	132550.00	662750.00	100	662750.00	662,750	662750.00				662750.00
7	Simcon France	FR	Partner	174,300	44,000	8,000	13,000	500	0	48950.00	288750.00	100	288750.00	288,750	288750.00				288750.00
8	Podcomp Ab	SE	Partner	296,700	0	12,600	0	19,000	0	82075.00	410375.00	100	410375.00	410,375	410375.00				410375.00

9	Comfil Aps	DK	Partner	413,000	0	18,000	0	39,400	0	117600.00	588000.00	100	588000.00	588,000	588000.00				588000.00
10	Moses Productos Sl	ES	Partner	178,500	0	8,000	7,163	101,000	0	73665.00	368328.00	100	368328.00	368,328	368328.00				368328.00
11	University Of Limerick	IE	Partner	312,900	0	17,500	27,000	43,000		100100.00	500500.00	100	500500.00	500,500	500500.00				500500.00
12	Polymeris	FR	Partner	200,000	0	21,000	0	19,000	0	60000.00	300000.00	100	300000.00	300,000	300000.00				300000.00
13	Asociacion Espanola De Normalizacion	ES	Partner	64,235	0	7,000	0	4,300	0	18883.00	94418.00	100	94418.00	94,418	94418.00				94418.00
			TOTAL	3,717,695	74,000	219,350	125,966	673,700	0	1184175.00	5994886.00		5994886.00	5,994,886	5994886.00	0	0	0	5994886.00

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	◯ Yes	◉ No	
Does this activity involve the use of human embryos?	◯ Yes	◉ No	

2. Humans			Page
Does this activity involve human participants?	◯ Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	◯ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	◯ Yes	◉ No	

3. Human Cells / Tissues (not covered by section 1)			Page
Does this activity involve the use of human cells or tissues?	◯ Yes	◉ No	

4. Personal Data			Page
Does this activity involve processing of personal data?	◯ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	◯ Yes	◉ No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	◯ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	◯ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	◯ Yes	◉ No	

5. Animals			Page
Does this activity involve animals?	◯ Yes	◉ No	

6. Non-EU Countries			Page
Will some of the activities be carried out in non-EU countries?	◯ Yes	◉ No	
In case non-UE countries are involved, do the activities undertaken in these countries raise potential ethics issues?	◯ Yes	◉ No	
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	◯ Yes	◉ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	◯ Yes	◉ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	◯ Yes	◉ No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	◯ Yes	◉ No	
Could the situation in the country put the individuals taking part in the activity at risk?	◯ Yes	◉ No	

7. Environment, Health and Safety	Page

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Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	◯ Yes	◉ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	◯ Yes	◉ No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	◯ Yes	◉ No	

8. Artificial Intelligence			Page
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	◯ Yes	◉ No	

9. Other Ethics Issues			Page
Are there any other ethics issues that should be taken into consideration?	◯ Yes	◉ No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

Explain in detail the identified issues in relation to:
- objectives of the activities (e.g. study of vulnerable populations, etc.)
- methodology (e.g. clinical trials, involvement of children, protection of personal data, etc.)
- the potential impact of the activities (e.g. environmental damage, stigmatisation of particular social groups, political or financial adverse consequences, misuse, etc.)

Remaining characters 4594

Compliance with ethical principles and relevant legislations

Describe how the issue(s) identified in the ethics issues table above will be addressed in order to adhere to the ethical principles and what will be done to ensure that the activities are compliant with the EU/national legal and ethical requirements of the country or countries where the tasks are to be carried out. It is reminded that for activities performed in a non-EU countries, they should also be allowed in at least one EU Member State.

Remaining characters 5000

Proposal ID **101057049**

Acronym **BIO-UPTAKE**

Security issues table

1. EU Classified Information (EUCI)[2]			Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	○ Yes	◉ No	
Does this activity involve non-EU countries?	○ Yes	◉ No	
2. Misuse			**Page**
Does this activity have the potential for misuse of results?	○ Yes	◉ No	
3. Other Security Issues			**Page**
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	○ Yes	◉ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	○ Yes	◉ No	

[2]According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, "European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States".

[3]Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

[4]EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

BIOcomposites in smart plastic transformation processes to pave the way for the large-scale **UPTAKE** of sustainable bio-based products

Coordinator: PhD. Pere Castell Muixí (AITIIP) – pere.castell@AITIIP.com +34 976 464 544, @AITIIP

No.	ACR.	Participant organisation name – *Role in the project*	Country	*Type*
1	AIT	FUNDACIÓN AITIIP – *Project coordinator, improvement of plastic transformation processes, exploitation and dissemination of bio-based results & training activities.*	ES	RTO
2	CTB	CENTRE SCIENTIFIQUE & TECHNIQUE DE L'INDUSTRIE TEXTILE BELGE – *Self reinforcing technologies for thermoplastic bio-fibres & improvement of extrusion processes*	BE	RTO
3	CID	FUNDACION CIDETEC – *Enduring technologies for thermoset bio-resins & improvement of pre-impregnation processes*	ES	RTO
4	SP	SPECIFIC POLYMERS – *Novel covalent adaptable networks technologies for recyclability, remanufacturing and reusing of bio-based formats*	FR	SME
5	AIM	ASOCIACION DE INVESTIGACION METALURGICA DEL NOROESTE – *Data-driven technologies to support the design and production of reusable and recyclable bio-based products*	ES	RTO
6	IRI	IRIS TECHNOLOGY SOLUTIONS – *Sensor and monitoring technologies adapting current controlling systems to bio-based materials*	ES	SME
7	SIM	SIMCON – *Modelling and Simulation technologies for bio-based formats to enhance current available applications*	FR	IND
8	POD	PODCOMP – *Optimisation of pre-impregnated process & demonstration in construction sector*	SE	IND
9	COM	COMFIL – *Optimisation of thermoplastic sheets/tapes production and thermoforming processes & demonstration in orthopaedic sector*	DK	SME
10	MOS	MOSES PRODUCTS– *Optimisation of over-injection moulding process & demonstration in tertiary packaging and urban furniture sectors*	ES	SME
11	LIM	LIMERICK UNIVERSITY –M*aterial characterisation support and LCA-LCC & HSS*	IE	UNI
12	POL	POLIMERIS – *Stakeholder engagement, results communication, support to the exploitation and dissemination activities, transferability and replicability activities*	FR	OTH
13	UNE	ASOCIACIÓN ESPAÑOLA DE NORMALIZACIÓN – *Standardization activities*	ES	OTH

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1. EXCELLENCE

1.1. Objectives and ambition

- Objectives

The general objective of Bio-Uptake project is to ensure a sustainable uptake (increase the use in a **39%)** of bioplastic composites through boosting a twin green and digital transformation in the European manufacturing industry. In particular, Bio-Uptake solution will focus scientific and technology efforts on developing flexible manufacturing processes to produce biobased end-products for the construction, medical and packaging sectors based on the combination of intermediate formats made of natural and/or biobased synthetic fibres reinforced with biopolymers, which are easily adaptable to new market demands.

The Specific Objectives of Bio-Uptake project are summarized below:

SO1. To develop **3 disruptive innovative manufacturing processes for bio-based products with a better overall performance of the current fossil-based alternatives:** one for thermoset composites, one for highly customised thermoplastics and one for one-shot large volume thermoplastics

SO2. To **adapt and upgrade 6 conformal manufacturing technologies** (filament extrusion, extrusion-compounding, resin impregnation, press consolidation, thermoforming and over-injection) that constitutes the manufacturing processes for deploying biobased materials to substitute traditional materials.

SO3. To implement a smart **digital end-to-end platform for manufacturing biobased materials** (design, modelling, process simulation, in-line monitoring and quality diagnosis) to ensure product quality and process interoperability required to optimize and upgrade each manufacturing process being demonstrated for each user case and its scalability to any other manufacturing process.

SO4. To develop and validate **4 new biobased intermediate formats** or semi-finished products: 1) endured thermoset organosheets, 2) monocomponent & bicomponent thermoplastic organosheets and 3) unidirectional tapes, 4) thermoplastic pellets or granulates.

SO5. To **demonstrate** the Bio-Uptake solution **in 3 manufacturing value chains (VC)**: construction, medical & packaging

SO6. To develop **3 end-products** (bathroom ceiling cabinet, feet orthosis and garbage container lid) that are made of **more than 75% biobased** materials and to validate that satisfy the technical and environmental requirements

SO7. To demonstrate **product circularity** (reusability, remanufacturability and/or recyclability) by validating their de-manufacturing process (due to reversible dynamic chemicals (CANs) and aided by eco-design principles).

SO8. To create **2 training programmes** for upskilling professionals in circular economy and smart & green manufacturing practices: one for young generations of scientists & engineers and one for industrial workforce

SO9. To contribute to their **standardization** applied to the **3 manufacturing value chains** driving biomaterials uptake

Table 1: Objective's contribution

OBJ	Bio-Uptake Contributions	Means of Verification	When
GO	Upgrading TRL of the activities related to functionality, digitalization & circularity plus achieving the project KPIs (comm., diss., training, outcomes and wider impact)	Achievement of 8 Ms	M42
SO1	TRL3→5 (fibre yarns) + TRL4→6 (intermediates) + TRL5→6 (conformal products) + TRL5→6 (in-line monitoring & simulation) + TRL4→5 (digital – interop. algorithm)	Ms1 + Ms2 + Ms3 + Ms4 + Ms5	M38
SO2	TRL3→5 (fibre yarns) + TRL4→6 (intermediates) + TRL5→6 (conformal products)	Ms1+Ms2+Ms3+Ms4	M30
SO3	TRL5→6 (in-line monitoring & simulation) + TRL4→5 (digital – interop. algorithm)	Ms5	M38
SO4	TRL3→5 (fibre yarns) + TRL4→6 (intermediates)	Ms3	M30
SO5	TRL3→5 (fibre yarns) + TRL4→6 (intermediates) + TRL5→6 (conformal products)	Ms4	M38
SO6	TRL5→6 (conformal products) + TRL4→TRL5 (CANs integration)	Ms6	M41
SO7	TRL4→TRL5 (CANs integration) + TRL4→TRL5 (de-manufacturing, reuse, remanufacture, recycling)	Ms7	M41
SO8	Achieving KPIs for training as described in *Table 10* and task 1.4	Ms8	M42
SO9	Achieving KPIs for standardization described in *Table 10* and task 1.3	Ms8	M42

- Beyond the state-of-the-art

Biobased materials, due to their intrinsic nature, have demonstrated that they can be adequate candidates with a performance good enough to replace petroleum derivates´ in various application areas[1]. However, due to differences in physical and thermochemical properties, a direct replacement of the current material with newly developed biobased is often a mistake. To reach the full potential of biobased materials product eco-design is essential (part drawing, materials to be used, processability, end-of-life etc)[2]. The novel approach on which Bio-Uptake project relies is based on modularity or pre-fabrication: the development of high-tech biobased intermediate formats (*organosheets, tapes and pellets*). The smart combination of these intermediates into

[1] S. Thomas et al (2020). Biopolymers & their industrial applications: From Plant, Animal & Marine Sources to Functional Products. Elsevier ISBN0128192593.
[2] ETC-Eionet Report (ETC/WMGE 2021/2): Business Models in a Circular Economy

a final end-product allows to overcome the current technical and environmental limitations to meet the demanding requirements of a specific sector/application where a single biobased material doesn´t. Thus, the synergistic potential of composite materials will be multiplied by the possibility of combining different intermediate formats to pave the way for the integration and uptake of bio-based materials in mass customised manufacturing (*manufacturing as a service*)[3]. Moreover, Bio-Uptake solution does not involve major machine modifications and large investment in complex equipment, but rather it involves the development of peripherals and the adaptation of simulation models to the intrinsic characteristics of biomaterials.

Three different manufacturing processes will be developed within Bio-Uptake (from the twin green and digital perspective) focused on the plastics manufacturing sector, which is on the centrality of a variety of value chains. These processes are designed to address diverse kinds of bio-based composite materials, including thermoset or thermoplastic polymers reinforced with natural fibres (including carbon positive materials). Sustainability criteria will be applied since the design phase to reach circularity by design, considering de-manufacturing routes for the separation and recovery of materials to ensure their reuse, remanufacture and/or recycling, boosting regenerated new products. To demonstrate the full potential of the manufacturing processes developed within Bio-Uptake project, three manufacturing sectors have been selected: construction, medical and packaging. One end-product has been selected to represent the full potential of biobased materials in each sector:

Table 2: Comparison of Bio-Uptake new products respect to State of the Art (SoA) products

Current Product and Manufacturing Process	Bio-Uptake Innovations in the Product and the Process
Demo Case for Construction: Sandwich Panels for the Ceiling for pre-fabricated bathroom pods	
The current manufacturing process used is **vacuum infusion** with conventional Epoxy resin and Glass Fibre (fully Oil-based) plus a balsa core (in a sandwich structure). Handmade & slow Process, Not Recyclable. Large scrap (needs of refrigeration/freezer for resin storage)	Semi-automated process to produce endured pre-impregnated organosheets and **thermoforming** of the sandwich structure with the balsa core. Flax fabric pre-impregnated with Biobased 3R-CANs-Epoxy (from Vanillyl alcohol – DGEVA, coming from lignin) + wood core with bonding-debonding reversible adhesive 30% faster process. Reprocesable and Recyclable. Storable at room temperature → No scraps (no value lost in the VC).
Demo Case for Medical: Feet Insoles - Orthopaedics	
Glued layers and assembly of cut-outs from different materials anatomically shaped. Each time a layer has been glued has to be **finished** afterwards: manual, multi-stage, time consuming, costly and not made of biobased materials. Epoxy resin reinforced with Carbon/Glass Fibres and PEEK + Carbon Fibre for AFO high-tech adaptable insoles	Consolidation of layers (organosheets and UDTapes) anatomically structured in a one-shot free forming **(press consolidation + thermoforming)**. Only a final trimming is required. PLA and PCL (mono and multi component), which has the particularity of once the final shape is ready, it can be warmed up if necessary, for an anatomical adaptation (which cannot be done with epoxy, GF or CF). Reusability and Recyclability. 34% more efficient, 12% lower costs, 50% shorter process and allows to put biobased materials that otherwise is unfeasible....)
Demo Case for Packaging: Lid for a garbage container (tertiary packaging – big-sized packaging)	
Injection moulding of oilbased PE (polyelythelene) + EPDM (ethylene propylene diene rubber). Average weight of a lid for a 10.000L container is 22Kg. Requirements for outdoors service life is high-demanding (mechanically, waterproof, UV resistance…)	Conforming of self-reinforced PLA organosheet & **overinjection** of reinforced bioPolyamide (short lignin-Carbon and wood fibres). This hybridization will enable to reduce by 35% the final weight of the lid and by 22% the plastic material used. BioPA will ensure the lid to withstand outdoors environment & the application of a reversible adhesive on the organosheet´s surface will ensure materials´ separation for remanufacture and recycling.

Bio-Uptake ground-breaking concept tackles 3 pillars: 1) achieving material FUNCTIONALITY through the composite lay-up, in which semi-finished structures called intermediate formats permits to better influence in the eco-design and customization of properties, allowing to meet technical quality criteria and requirements; 2) introducing process DIGITALISATION by building up a digital hybrid twin, that allows the interoperability of manufacturing technologies and enables the training of the workforce to adapt and normalize the use of biobased materials in the European manufacturing industry; and 3) ensuring a SUSTAINABLE end-of-life by introducing scientific novelties in the materials that permits remanufacturing and a controlled de-manufacturing:

MAIN BREAK-TRHOUGH NOVELTIES that brings Bio-Uptake beyond the State of the Art:
FUNCTIONALITY: The combination of biobased intermediate formats to meet industry requirements: sheets (unidirectional tapes + organosheets) based on self-reinforced biobased thermoplastic polymers and endured thermosets, plus thermoplastic pellets formulations, will ensure to achieve products with enhanced mechanical, physical and chemical properties *(technical disruption).* These are conceived as **high-performance semi-finished materials** that are not only considerably lighter, but that can also include many **functions into end-product**. In Bio-Uptake project 4 formats can be envisaged: **endured prepegs** using thermoset bioepoxy and flax natural fabrics, ad-hoc **biobased organosheets** and **unidirectional tapes** using self-

[3] EFFRA – European Commission – MADE IN EUROPE: The manufacturing partnership in Horizon Europe

reinforce biobased thermoplastic synthetic fibres (monocomponent PLA & bicomponent PLA/PCL) and biobased **thermoplastic pellets** (bioPA) reinforced with short biobased lignin fibres: carbon fibre and recycled wood.

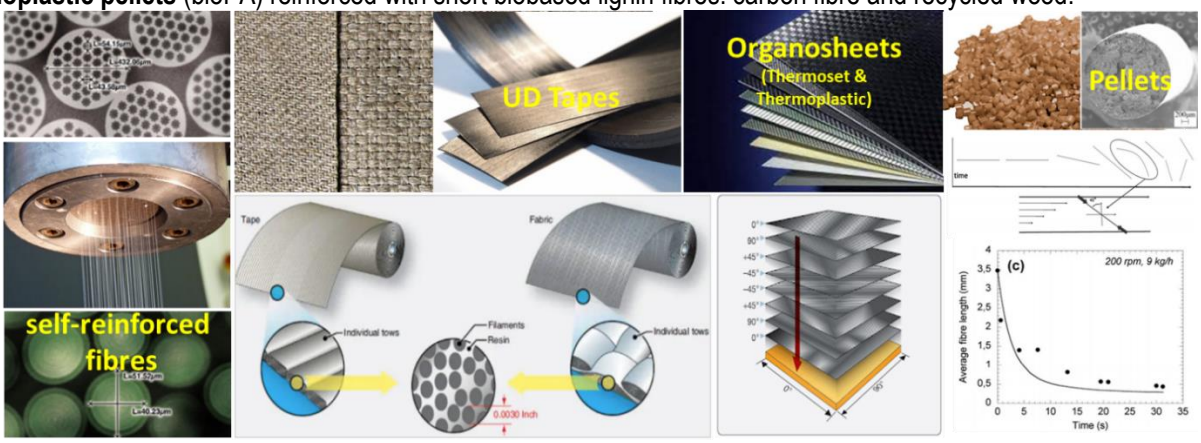


Fig. 1: Bio-Uptake Conceptual Intermediate Formats

The strength and stiffness of a composite build-up depends on its **design and in the micro- and macro- structures** conferred. Regarding micro-structure of polymeric fibres and matrices, it is essential to have a precise controlled on the **crystallinity of the polymer used** (X_c%, size and polymorphisms). In addition, the **orientation of the fibres** and of the polymer chains, length and molecular weight are crucial to obtain intermediate formats with enhanced properties respect to the current SoA. Moreover, at macro-structure level, when combining natural and synthetic fabrics and polymeric resin matrices in an organosheet concept, the **orientation sequence of the plies** (layers) and of the **fibres within each ply** play a crucial role to control the porosity and final properties of those composite formats and to provide a structurally efficient design.

The use of intermediate formats **provides flexibility on the design** and allows product designers to use them either as **main structure** of the end-product (insoles) or as **selective reinforcement** (lid container) according to product needs. In addition, the accurate characterization and in-line monitoring control of their structure permits to customise the materials obtaining end-products with increased and customised quality and properties. Intermediates permit to **boost manufacturing** technologies for end-products based on **smart conformal technologies**, like press consolidation, thermoforming, and injection-moulding as well as adapting the manufacturing industry of biobased products for a *de-centralised manufacturing-as-a-service* model (highly customised, decreasing production cost & times, more accessible biomaterials to manufacturers & consumers).

DIGITALISATION: Building a digital twin to allow a smart and trustful production and interaction across the value chain. Digital tools and technologies in every stage of manufacturing are opening opportunities to achieve a whole new level of productivity[4]. However, the current tools and systems have been developed to work with conventional oil-based materials and needs to be adapted to the peculiarities of biobased materials to ensure their effectiveness and trustfulness, allowing decision making when introducing a new biobased product in the market. This requires a holistic system approach that integrates and connects a series of complex data-driven events along the entire value chain (Digital Thread) enabling control of manufacturing systems[5]. The novel approach underpinning the digital pillar lies in implementing a **digital end-to-end engineering concept embedded in the full manufacturing chain**, combining design and modelling, process simulation, fabrication, monitoring systems and quality diagnosis of products.

The combination of information technology (IT) -coming from engineering phase- and operational technology (OT) -coming from manufacturing phase- enables a seamless data connection between physical assets and information processing technologies[6] and the creation of an evolved and updated Digital Twin of the product and/or manufacturing processes[7].

In this sense there are two key elements that will allow to build this architecture system:

a. *Digital Thread* that allows a seamless dataflow integration of a data-driven pipeline linking the information during the lifecycle of the product from the design, to simulation (CADMOULD and TSIM) to manufacturing, to process in-line monitoring (sensors, photonic & visual techniques) and control, to quality assurance (non-destructive tests) of novel bio-based products (a unified paradigm).

b. *Digital manufacturing* as key driver for the implementation of a flexible production based on a virtual commissioning strategy following a holistic simulation-optimization to ensure optimised system performance in order to minimize the ramp-up phase and to confirm system's manufacturing behaviour of the new biobased product.

The innovation going beyond the State of the Art is the development of this digital communication architecture and of the algorithm that will enable to match, correlate and communicate data. A digital infrastructure capable to manage and analyse different types of data, and to interoperate across different machines and processes, using single/multiple bio-based materials combinations, is crucial to successfully build and operate this approach *(technical disruption).*

[4] ROSEN, Roland, et al. About the importance of autonomy and digital twins for the future of manufacturing. IFAC-PapersOnLine, 2015, 48, 3, p. 567-572.
[5] Hozdić, Elvis, et al. A Cyber-Physical Approach to the Management and Control of Manufacturing Systems. J. Mech. Eng. 66(2020)1, 61-70
[6] https://atos.net/wp-content/uploads/2016/07/atos-smart-factory-ascent-thought-leadership-whitepaper.pdf
[7] Uhlemann, et al. The digital twin: Realizing the cyber-physical production system for industry 4.0. Procedia Cirp,2017,61, 335-340

SUSTAINABILITY: Covalent adaptable networks (CANs) are materials (chemical moieties) that can undergo modifications in their inner structure based on efficient associative and dissociative exchange mechanisms that enable topology reconfiguration without loss of network integrity by thermally activated bond-exchange processes. They structurally resemble thermosets with permanent covalent crosslinks but are able to flow in a manner that resembles thermoplastic behaviour only when a dynamic chemical reaction is active. In Bio-Uptake project innovations material science related to biobased CANs will allow:

1) To make thermosets reshapable (thus reusable and remanufacturable) conferring to thermoset the thermoplastic properties. The use of prepregs is one of the most extended manufacturing methods consisting of a reinforcement fabric impregnated with a thermoset epoxy resin matrix. The epoxy matrix is partially cured (preventing from leaking while keeping enough functional groups) and tacky, and in this form is supplied to the composite manufacturer who needs to store it refrigerated (typically at -18ºC) or frozen. Afterwards an autoclave processing is required to perform the final curing and consolidation; a process which is labour-intensive and high-cost, thought for relatively small production runs. The here described bio-based 3R-CANs bioepoxy resin would allow manufacturing a new class of enduring prepregs[8], as cured prepreg sheets that could still be valid for the fabrication of multilayered composite parts. These makes them processable at room temperature and via a thermoforming process, making thermosets more accessible for high volume production manufacturing technologies *(technical disruption).*
2) CANs have been demonstrated to be excellent bonding-debonding materials[9,10]. By inducing a controlled trigger external stimulus, the change in the polymer topology allows the separation of the structural formats, which combined with chemical and biological catalyst processing permits the recycling of all the components. A second type of vitrimerlike-CANs biobased moieties will be designed to be conceptually developed as an adhesive with the aim of taking advantage of their debonding selectivity and allow the separation of the intermediate formats conforming the final products. In this sense the separation between thermoplastics (PLA and bioPA) and thermoset/wood surfaces occurred under a controlled conditions permitting its circularity, recyclability and reusability of components *(technical disruption).*

The project methodology is designed in a way it is not significantly harming any of the six environmental objectives of the EU Taxonomy Regulation. More details are elaborated in *Table 6: Bio-Uptake contribution to DNSH Objectives*.

- R&I maturity

Table 3: Bio-Uptake R&I Maturity

R&I activity	Maturity: Start TRL→Final TRL
Transforming Commercial Biobased Pellets into Self-Reinforced Filaments (SRF) and microfibrillar bi-component structures *SO2, SO4*	**TRL 3 →TRL 5:** This concept will start developing filaments based on different grades of PLA, combining high melting and low melting grades. Then the Knowledge developed will be transferred to bi-component biopolymer structures (using structures of PLA in a PCL matrix) using in-line monitoring during extrusion to have a precise control of the micro-structure (with the heat deflection temperature of the fibre above the melting temperature of the matrix). Tests will span from lab (CTB, hundreds of grams for characterisation) to pilot scale demonstration (COM, kg scale for thermoplastic sheet formats demo-case validations).
Research & development of biobased intermediate structural formats *SO1, SO2, SO4*	**TRL 4 →TRL 6:** By using forming technology, COM will selectively order the developed hybrid fibres processed in a press-moulding process to obtain two types of thermoplastic biobased formats: unidirectional tapes and thermoplastic organosheets without losing mechanical properties (optimization of prototypes will be carried out using PLA and PLA/PCL to obtain from 5-10 units/each for fully characterization to 100 units/each for validation purposes). AIT will optimise and upscale the production of thermoplastic pellets based on bioPA short lignin-based fibres via extrusion-compounding (1-5 kg for characterization) prior selecting the best formulation (at least 1 per type of matrix) for upscaling demonstration purposes (about 100-200 kg for demo case validation). CID will develop and manufacture pre-impregnate biobased thermoset organosheets based on dynamic bioepoxy and flax reinforcemnts. Endured pre-preg sheet samples (1-2 m^2) for characterisation purposes, and the best combination will be upscaled for demonstration purposes (about 100 m^2).
Research on conformal manufacturing technologies to develop final bio-composite products *SO1, SO2, SO5, SO6*	**TRL5 →TRL6:** Ceilings for bathroom pods based on smart endured pre-preg composites (bioepoxy-flax) will be demonstrated through thermoset-thermoforming by POD. Feet orthopaedics will be demonstrated by COM via press-forming, with freedom operability by using combinations of UDT and thermoplastic organosheets. Finally, MOS will manufacture container lids, using as reinforcing structure the thermoplastic organosheets in combination of the biobased pellets via over-injection moulding (press-forming with the mould and injection of molten bioplastic in a one-shot process). Optimization of the conformal manufacturing processes (machine level) and quality control will be carried out At least 25 final parts of each demonstrator will be produced for validation.

[8] 3R concept: CID EP2949679A1 patent and building on top of ECOXY project. In addition, precursors for bioepoxy thermoset resins coming from DGEVA developed by SP under ECOXY project will be used as SoA biopoexy resin for 3R modifications.

[9] VIBES project 'Improving recyclability of thermoset Composite materials through a greener recycling technology Based on reversible biobased bonding materials' — Grant 101023190 — VIBES — H2020-BBI-JTI-2020

[10] NJ.Van Zee et al. Vitrimers: Permanently crosslinked polymers with dynamic network topology. Prog. Polym. Sci. 2020;104:101233

Sensorization of critical manufacturing processes to design new in-line characterization methods. *SO1, SO3*	**TRL5 → TRL6:** Sensorization based on *Lab-on-Plant* concept (bringing laboratory characterization technologies into manufacturing processes via adapted sensors for in-line monitoring. Hyperspectral imaging (HSI) and Spectral reflectance in probe mode will be tested by IRI to monitor the curing process of thermoset organosheets manufacturing. For the thermoplastic fibres and sheets, spectroscopic techniques such as NIR, RAMAN, LIBS and XRF to monitor the polymeric structures will be used to conceptualise a hardware inline sensor system that will be implemented at CTB facilities. The crystallinity of the polymer extruded will be monitored applying using the Visum Palm monitoring system (infrared spectroscopy characterization & chemical composition analysis)
Simulation FESEM adapted to biobased composites *SO1, SO3*	**TRL5 → TRL6:** simulation from an existing predictive model (CADMOULD for injection moulding and TSIM for thermoforming) and adapt it to the requirements of the biobased materials used in Bio-Uptake. Currently there is no model as such for biomaterials.
Development of a digital communication architecture for interoperability between manufacturing systems *SO1, SO3*	**TRL4 → TRL5:** Bio-Uptake will focus in connecting product and process lifecycles for a flexible and more accurate solution for the manufacturing of novel recirculated bio-based products with increased serviceability by reusing and remanufacturing. Interoperability and connectivity for data exchange will be based on the Reference Architecture Model for Industry 4.0 (RAMI), by relying on OPC-UA protocol for machine-to-machine connectivity, and process monitoring, and AutomationML for modelling and data exchange definition.
Research related to the development and integration of CANs *SO6, SO7*	**TRL 4 →TRL 5:** SP & CID will develop biobased 3R-CAN moieties for enduring pre-pegs by replicating the concept developed under ECOXY project but substituting non-biobased moieties for biobased ones and enhancing the thermo-mechanical stability. Pilot manufacturing of the 3R-CANs will be produced and characterized (WP2) to cover enough quantities for demo-case validations and circularity evaluation (from 50metres up to 500metres of prepreg). **TRL3→ TRL5:** ad-hoc CAN for bonding-debonding adhesive. SP will design and manufacture the reversible bonding agents (based on their own potfolio moieties). SP will apply CAN technology to favour the separation plastic formats made of different plastic nature (combinations of PLA/PCL, bioepoxy/wood) from lab scale (10-20 g) to pilot scale 500g-1Kg of the best candidate/s.
Demonstrate de-manufacturing for debonding components to ease the regeneration of products *SO7*	**TRL4 → TRL5:** AIT with the support of SP, will collaborate to define and test the best activation stimuli to motivate the debonding of CANs and test the easiness to dismantle of the composite components for further reuse and regeneration of products. Thermal technologies (IR; UV light, convection heating in oven) will be tested at pilot level to evaluate process efficiency, environmental and cost up scalability. Industrial end-users in cooperation with SIM will work on the eco-design of the final products and look for industrial symbiosis circularity.

1.2. Methodology

- Overall methodology

1. CONCEPTS, MODELS AND ASSUMPTIONS underpinning the work:

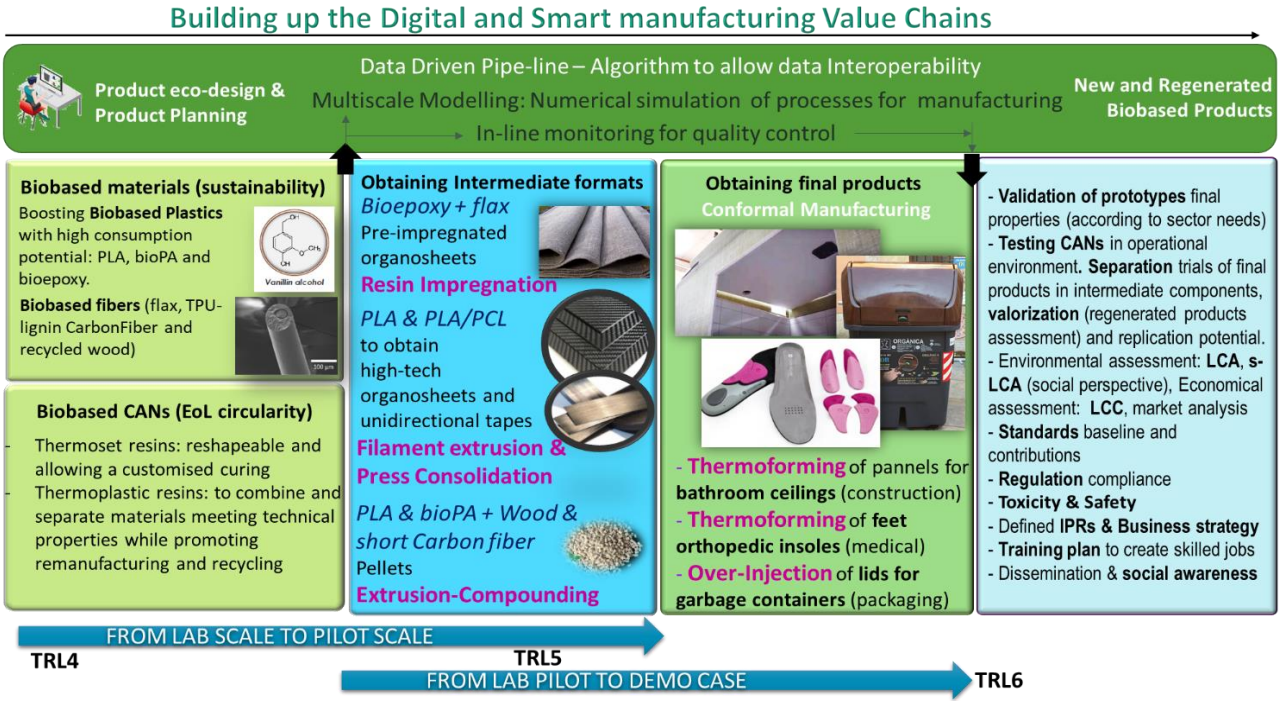


Fig. 2: Bio-Uptake Concept

Bio-Uptake project presents a holistic solution to resolve a sustainable scale production of green products based on bio-based materials and that are circular by design (promoting reuse, remanufacturing and recycling of components), thus, favouring a green transition to build a truly sustainable manufacturing industry. The project aims at establishing new manufacturing processes for biobased materials and the overall concept is based on the **most relevant key points**:

- ✓ Design and develop intermediate formats based on biobased polymers (PLA, bioPA and bioepoxy) and vegetal fibres (flax, lignin-based Carbon Fibre and recycled wood). Modify those polymeric resins with dynamic chemicals (CANs) to make thermosets composites reshapeable (3R disulphide vitrimers) and the interphase between combination of different thermoplastic components easy to split (reversible adhesive) to recover clean flows of materials and formats to be reused or recycle.
- ✓ In depth characterisation of biobased materials to be used in the manufacturing processing will be applied. Comparison of TDS (theoretical material behaviour) with in-line monitoring data (real material behaviour during manufacturing process) of the most critical manufacturing steps will be carried out to avoid structural defects that may affect the final quality of the end-product.
- ✓ To match in-line sensor data with ad-hoc process modelling to evaluate the impact of a variable in the quality criteria in the manufacturing processes. The software tool is able to assess which is for a pre-selected material: the best solution based on the product design, the specifications of the manufacturing process, (Decision Support System) and guide the product designers for example in the wt% load of fibres, length or orientation, as well as potential combinations between different plastic resins (e.g. during the over-injection process).
- ✓ Design of final products (demo cases) considering remanufacturing by applying circular principles by design. Bio-Uptake final products are conceived as the combination of intermediate formats, which are the core structure of those final products. The eco-design of the sandwich panels for bathroom pods (construction value chain), feet insole orthopaedics (medical value chain) and lids for containers (tertiary packaging) will be based on a product design that will include final product requirements as well as guidelines for product dismantling/de-manufacture for its reuse into the same or new regenerated products (industrial symbiosis principles will be applied among the three selected sectors in the project. However, other potential products in the same sector and even other sectors of application will not be excluded).
- ✓ To make the data of the whole manufacturing pilot line interoperable and to create data-based models (virtual) of the manufacturing process, so that the data obtained from the different stages of the overall process -engineering and process data- can be correlated by using AI to improve the design a final product with a biobased material and to provide the required quality aiming at a zero defects process.

2. OVERALL PROJECT METHODOLOGY

The overall methodology followed in the project is divided in 4 main blocks aiming at accelerating the twin green and digital transitions to succeed in building a sustainable manufacturing process for biobased materials. The same methodology will be applied to each value chain, adapted to its particularities as described in *Table 4*. Key efforts and activities are dedicated to:

1. **Materials** (develop and validate biobased materials alternatives from different nature, understand and contribute to their standards, taking into account safety and health issues for workers and end-users)
2. **Manufacturing Processes** (overcoming technical and digitalization challenges to pave the way for a sustainable and smart manufacturing value chain per selected sector and its process replication)
3. **Product** (demonstrate the functional validation in comparison with current benchmark while ensuring circularity by design: including eco-design principles and sustainable end-of-life practices for remanufacturing and recycling). The synergistic potential of **composite materials** will be multiplied by the possibility of **combining different composite formats** to pave the way for the integration of this kind of bio-based materials in large-scale production sectors.
4. Contributions to the **European industry competitiveness and social welfare**: by developing sustainable business models that looks over the environment and the creation of new jobs (alignment to the Green Deal).

1 – Materials: In-depth understanding and control of the biobased materials including material sourcing and recycling
Full characterisation of all the materials to be used in the project. It is important to remark that there are four types of biobased materials feedstocks/sources: a. **Commercial grades and fibres**: PLA (Total Corbion), PCL (Ingevity/BASF), bioPA (Arkema/Natureplast), Flax (Flips&Doobels) used with a double objective: i. To define a **protocol** for the adequate reception, storage and use of commercial materials as well as a consolidated **characterisation** protocol that enables to introduce all the input data required for the manufacturing stages. Currently, there exits lack of material performance and variations in the manufacturing conditions from batch to batch of a same material grade that deviate the real operational conditions respect from their technical data-sheets. ii. To **increase the biobased content** of current petro-based products highly accepted in the market due to their functional behaviour (such as the case of PCL in medicine applications) - CTB b. **Innovative materials** designed and produced by own Bio-Uptake partners as result and in collaboration of other EU projects: BioEpoxy resin (SP/CID - ECOXY), Biobased Carbon Fibre (LIM-VIBES/LIBRE).

Batches are smaller and reproducibility has to be controlled. A characterisation protocol will be defined and followed. In this way Bio-Uptake aims at boosting to a pre-commercial stage SMEs and results coming from EU funded programmes.

c. **Recyclates:** materials that are biobased but are recycled flows coming from other industrial by-processes. These are currently considered contaminants in their own value chain but are considered new feedstock for other value chains instead (boosting industrial symbiosis with local resources): wood dust coming from the manufacturing of furniture made of recycled wood (kindly provided by FINSA).

d. **Novel multifunctional materials based on combinations** of the previous three categories **modified with CANs** (**dual functionality**). This will modify the rheology and thermal behaviour of the plastics containing them and therefore it is a must to ensure a proper characterisation method and identifying critical properties which will have to be monitored and controlled during the manufacturing process (SP-AIT in MANDALA/VIBES).

Identification and alignment to current **standards:** In case of identifying a lack of coverture in the materials´ **characterization** protocols, define correction measures and bring them to the appropriate standardization group (UNE).

2- Manufacturing Processes: Setting up the manufacturing pilot lines.

A detailed description of the unit processes and identification of critical process parameters will be defined. The manufacturing processes have been divided into the technologies required to produce the intermediate formats and the final products. In addition, the monitoring status and main part defects will be analysed in advance and **depending on the mature of each selected technology**, specific needs will have to be addressed regarding technical and digitisation challenges. Some are dedicated to adapting the tooling and hardware stages of the manufacturing line itself and others to control the structure of the material in-line by introducing monitoring sensors that provide real data regarding the transformation process in order to take the most of the biomaterial used:

a. **Solving operational technology challenges (**IRI**): In-line monitoring** (machine vision and spectroscopic techniques) will be implemented in **low mature** technologies *(extrusion of fibrils and self-reinforced fibres, obtaining intermediate biobased thermoplastic organosheets using the self-reinforced fibres, thermoset enduring pre-pregs)* in order to reach high control of the critical parameters that will provide the high-tech performance quality to the intermediate formats above described. This monitoring will allow to establish correlations with already lab-test characterised materials and correct or give indications to the operator to correct the variables of the process, making the process more robust.
For those technologies considered as **high mature** technologies *(extrusion-compounding, injection-moulding and thermoforming)* it will be necessary to slightly adapt and optimise the manufacturing process, peripherals and process parameters for biobased materials. However, there is no need to implement further in-line monitoring systems as the machines used are highly monitored (termocouples, pressure probes, torque control, easy-going user-machine interface…and it is more important to ensure data interoperability and provide staff skills qualification (different challenges should be overcome).

b. **Solving information technology challenges** to boost the use of biobased materials in the selected manufacturing processes: **Collection of data** from the different characterization equipment's and manufacturing machines to make the data "open" allowing **interoperability - *end-to-end smart algorithm* – (**AIM**)** and adapt them to feed the "virtual line" so called digital thread. Real data will serve to feed **simulation** libraries for over-injection moulding and thermoforming technologies in terms of materials behaviour and quality criteria in order to let them extrapolate data and simulate the **design of new biobased products** (validated in the demonstration selected products: feet orthopaedics, ceilings for modular bathrooms and lids for trash containers). Bio-Uptake will support the component design through the numerical simulation of the manufacturing process (collaborative data management framework).

3 – Product: challenges associated to product design, technical validation & environmental performance

a. **Eco-design of the product**, considering the type, format and materials used (biobased content) that conforms the final product, the assembling and disassembling/de-manufacturing of the components (AIT). Once the monitoring and digitalisation of the processes is finished, the algorithm designed to establish the digital twin will be able to match experimental data and extrapolated computer data (coming from process simulation that uses mathematical models describing the physics behind each process– SIM) finding and correcting deviations. In this way the initial design will be adapted to the new biobased and regenerated materials (using remanufactured parts). Therefore, a supporting tool for screening different designs and materials based on the new libraries created with Bio-Uptake biobased plastic components will be created. This tool will aid industry in next-generation product design.

b. **Validation** of the **functional performance and end-of-life** of the demo-product according to each sector and alignment to applicable **standards** (construction - POD, medicine - COM and packaging - MOS). Concerning the product end-of-life, the innovations implemented regarding CANs to ensure a proper decommissioning, separation and remanufacturing or recycling of the components will be carried out (task 5.4). In case of identifying a lack of coverture in the quality and validation protocols, Bio-Uptake partners will define correction measures and bring them to the appropriate standardization group (UNE). Cooperation with other EU initiatives will be seek too to share and enrich results (task 6.5).

c. **Measure the impact** regarding the environmental and cost perspective for the manufacturing of the new and regenerated products: **LCA** and **LCC** assessment (LIM, task 1.2)

d. Activities aimed at **paving the way for the commercialization.** Developing **sustainable business models** for the new biobased products by developing an **exploitation strategy** covering the following steps (AIT):
   i. Definition of Bio-Uptake project results and IPRs identification.
   ii. Identification of Key Exploitable Results
   iii. Definition of the consortium exploitation strategy
   iv. Study about the Replication potential of the manufacturing processes into new sectors.
   v. Industry engagement activities dedicated to attracting potential customers (dedicated workshops and customised meetings to listen their needs and match Bio-Uptake solutions to them).

4 – Industry & Society: transversal actions that have an impact in the human dimension and the uptake of biobased products

a. **Training programme**: aiming at boosting human capital by teaching at academic and vocational levels within the Bio-Uptake training plan (WP1 – task1.4, LIM).
   i. Industry workforce: to create and transfer knowledge to the European industry about circular economy and sustainable manufacturing, generating skilled workforce capable of using and implementing biomaterial-based and digital manufacturing activities.
   ii. Academic students (scientists and engineers): the goal is the development of a training programme to provide knowledge to European students in the main disciplines involved in the project, namely: materials science, engineering, and chemical fields…(STEM) teaching them new skills for new arising demand in technical jobs (to boost the breeding ground of new talents).

b) The implementation of a **Communication and Dissemination** strategy (WP6 - POL): the former one mainly focused on the Social Perspective and Policy maker Perspective and the second one dealing to an open transfer or scientific knowledge to peers and strategic industry (exploitation and replication) than can multiply the impact uptake of the project (boosting resilience: open-mind and quick adaptation to biomaterials).

c) **Measure the impact** of the project in the European industry and society (task 1.2): **s-LCA** (ISO14040), assessing the social and sociological aspects of products, their actual and potential positive as well as negative impacts along the life cycle, as well as health and safety issues. The elaborated indicators (e.g. number of jobs created) will be linked to impact categories (e.g. local employment) which are related to five main stakeholder groups (e.g., [i] worker, [ii] consumer, [iii] local community, [iv] society and [v] value chain actors).

The above-described methodology is applied to create three manufacturing process that can be applied to demonstrate its feasibility and sustainability in each of the selected value chains. If it is applied to the peculiarities of each type of material, we can describe a specific method to define and set up each process (described and summarized in the following tables):

Table 4: Tables explaining the particularities of each demo case or business case

Construction Value Chain: Demo-Case → Modular ceiling´s for prefabricated bathroom *(more than 87% biobased)*

Partners involved in the value chain: SP (bioepoxy provider, adhesive and lab support for characterization), CID (integration of 3R-CANs and set up of flax fabric pre-impregnation pilot line, end-of-life), POD (construction end-user, thermoforming of pre-preg organosheet into ceiling structures for bathrooms), IRI (in-line monitoring precise control of curing stage), SIM (thermoforming modelling of final product).

Materials used for the demo-case: Biobased Epoxy Resin (doped with 3R-CANs) + Flax Fibre + Wood basal core

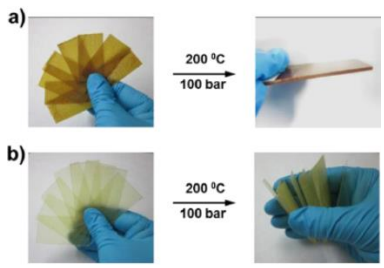


Fig. 3: a) Prepreg sheets of fully cured 3R epoxy resin with 50 v% of GF prior (left) and after being hot-pressed (right).
b) Prepreg sheets of fully cured reference epoxy resin with 50 v% of GF prior (left) and after being hot-pressed (right).

CID has developed an innovative and ground-breaking based epoxy resin system (patent application number EP14382184[8].1, 2014), by using a dynamic and reversible cross-linking system based on a diamine hardener containing a reversible aromatic disulphide bond in substitution to traditional diamines. The use of this resin in the manufacture of composite materials allows the development of reprocessable, recyclable and repairable (3R) thermoset composite materials. DGEBA epoxy matrix was used by CID to manufacture carbon fibre-reinforced 3R composites [11,12] and, in a more recent work[13], CID used fully bio-based epoxy resins developed by SP (diglycidyl ether of vanillyl alcohol (DGEVA) and phloroglucinol tris epoxy (PHTE) for the preparation of taylor-made epoxy resin matrices to be used in fibre reinforced polymer composites (FRPCs). Initial biobased content of the resin will be of 65% (ECOXY SoA). SP will also work on a biobased alternative to 4-AFD hardener to increase the biocontent value over 85%.

[11] De Luzuriaga, A.R., et al. Epoxy resin with exchangeable disulfide crosslinks to obtain reprocessable, repairable and recyclable fiber-reinforced thermoset composites. Mater. Horiz. **2016**, 3, 241.
[12] De Luzuriaga, A.R.,et al.. Correction: Epoxy resin with exchangeable disulfide crosslinks to obtain reprocessable, repairable and recyclable fiber-reinforced thermoset composites. Mater. Horiz. **2020**, 7, 2460.
[13] Genua, A.; et al.. Build-To-Specification Vanillin and Phloroglucinol Derived Biobased Epoxy-Amine Vitrimers. Polymers **2020**, 12, 2645.

Manufacturing technologies: intermediate formats Thermoset Organosheet → Flax Fabric pre-impregnated in Biobased Epoxy – 3R CANs resin: new class of enduring prepregs that can be prepared as totally cured prepregs (not being on a B-stage) and are able to be stored at room temperature (either in rolls, sheets or tapes) and still be appropriate to produce multilayered composite parts. Set-Up of a semi-automated pre-impregnation line + **the final product** thermoforming of endured-organosheets for obtaining a multi-layered biobased composite sandwich panel, with a wood balsa core to provide certain flexibility and insulation. The wood core will be sprayed with bonding-debonding adhesive.

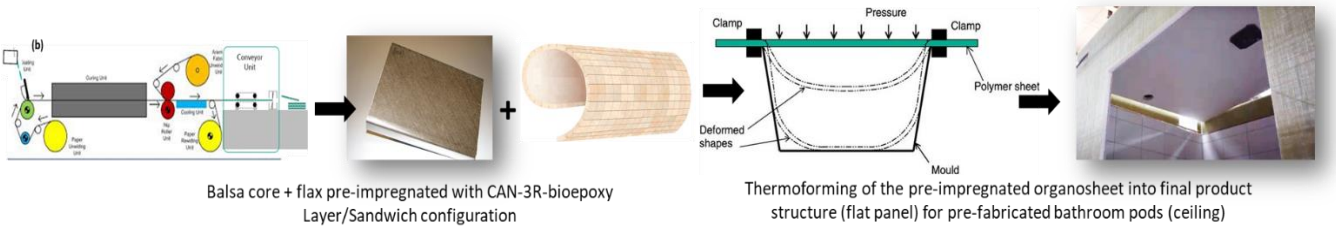


Fig. 4: Technologies that conforms the thermoset production process, including its intermediate and end products

Main Validation tests of demo product according to identified Standards: ISO 9001:2008 processes for enduring line. For panel: ISO 178 three point bending tests; fire properties D-s2,d0, ISO 5660-1 Cone Calorimeter and EN ISO 11925-2 small flame test and EN 13823 Single Burning Item; emission tested ISO 16000-3:2011(Indoor air - Part 3).

End-of-life scenario pursued: Separation of basal core from organosheet to facilitate their refurbishment and remanufacturing thanks to debonding properties of the adhesive (concept based on MANDALA project results). The thermoset based organosheet contains Associative CANs (3R), by applying heat (over the commutation point) and pressure (via thermoforming) the organosheets can be reshaped into a new intermediate to be used as component in a biobased regenerated product.

Digitalization pursued: In-line monitoring to **determine the curing level** of the developed pre-preg through **Hyperspectral imaging (HSI) and Spectral Reflectance (SR)** in probe mode of the intermediate formats *(low mature technology)*. These technologies have been recently used to monitor changes in polymer chemical structures due to temperature stimulation[14,15]. Simulation support: setting up the thermoset pilot line and interoperability among units (digital threat - AIM). **Pre-impregnated unit:** A chemometric model will be developed (mathematical modelling) and predictions about the degree of curing will be obtained. A software intended to process the data and deliver the results by applying machine learning algorithms to improve the accuracy of the results will be developed. **Thermoforming unit:** Results from pre-impregnated unit will feed new material libraries required to run simulations for the final thermoforming unit (TSIM software). **Remanufacture unit:** material modelling and set-up of the innovative digital platform for the production of reusable & recyclable thermoset products.

Medical Value Chain: Demo Case → Feet Orthopaedics – medical insoles *(more than 75% biobased)*

Partners involved in the value chain: LIM (characterisation laboratory support), CTB (fibres and fabrics, end-of-life), COM (upscaling fibres, manufacture of thermoplastic organosheets and free forming of ad-hoc feet orthopaedics), IRI (in-line monitoring), SIM (thermoplastic modelling of final product via thermoforming technology).

Materials used for the demo-case: This value chain is conceived to increase the use of biobased content (PLA) in applications in which a non-biobased material is highly installed (PCL). Polycaprolactone is considered one of the most competitive polymers in the medical sector due to its high mechanical properties (toughness) and biocompatibility. More recently PLA has been considered for the fabrication of devices that are under constant tensile stress and high elongation for the biomedical sector although its major disadvantage is brittleness and low thermal stability[16]. PCL (ductile and semi-crystalline hydrophobic) is compatible to PLA[17] and has been demonstrated to complement it by increasing the mechanical performance[18] in medical applications such as orthopaedics, scaffolds, implants, fixation…

Manufacturing technologies: intermediate formats: needs two stages: 1) melt-spinning to produce the filament yarns (100% biobased PLA **monocomponent or self-reinforced**[19] **fibres** & 50%-80% biobased **bicomponent PLA/PCL reinforced fibres** based on microfibrillar structures) and 2) continuous pressing to consolidate the desired organosheets (a few layers with different direction fibre orientation) and unidirectional tapes (same fibre orientation).

[14] Bleszynski, Monika; et al. Visualizing Polymer Damage Using Hyperspectral Imaging. Polymers, **2020**, 12(9), 2071
[15] Inga Niedermaier, et al. Quantification and classification in process analytics using hyperspectral imaging,"Proc. SPIE 11144, Photonics and Education in Measurement Science **2019**, 1114409
[16] Su S, Kopitzky R et al (2019) Polylactide (PLA) and its blends with poly (butylene succinate)(PBS): A brief review. Polymers 11:1193.
[17] Christopher K. et al (2017), Chapter 19 - Polymer Design and Development, Biology and Engineering of Stem Cell Niches, Academic Press, Pages 295-314, ISBN 9780128027349,
[18] Sundar, N.; et al (2020). Dual purpose, bio-based polylactic acid (PLA)-polycaprolactone (PCL) blends for coated abrasive and packaging industrial coating applications. Journal of Polymer Research, 27(12), 386
[19] Chengcheng et al (2012). Progress in polymer science, Vol. 37, 6, P 767-780

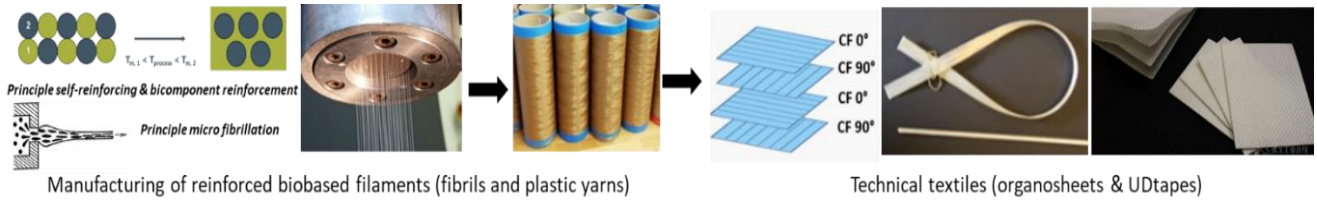


Fig. 5: Technologies that conforms the thermoplastic production process for high customization products: intermediate formats stage

The **final product:** the intermediates will be used to make insole prosthetics by press-moulding (heat-moldable - thermoforminglike). The sheets will be cut-to-size and shape for the creation of multi-ply laminates selectively reinforcement where the simulation designs consider best.

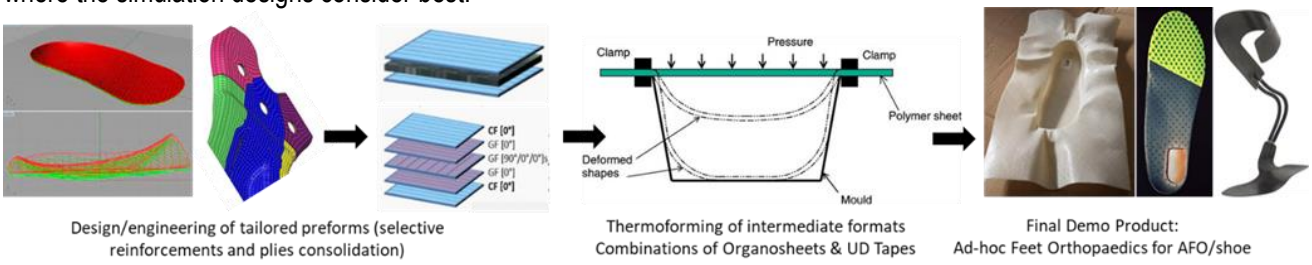


Fig. 6: Technologies that conforms the thermoplastic production process for high customization products: end-product stage

Main Validation tests according to identified Standards: **EN ISO 2062** to determine the single-end breaking force and the elongation at break. For the final product: matching the performance of the current orthosis done by FILLAUER client and have to pass their inhouse (mechanical & thermal) testing & when approved it will be sent for patient testing which takes 12 months.

End-of-life scenario pursued: Insoles may have 3 different end-of-life approaches, Bio-Uptake assesses on the best choice:

a) Monocomponent PLA: recycling (milled and directly used for injection-moulding purposes, re-melted into pellets and converted back to microfibrillar reinforced polymer composites).
b) Bicompponent PLA/PCL: mechanically reprocess the PLA/PCL materials trying to recreate microfibrilles for the same purpose. Otherwise, it may be chemically recyclable to recover the lactic acid and the caprolactone building blocks. Assessed via coordinated activities with ongoing EU projects: MULTICYCLE / VIBES /DECOAT
c) Both polymers are compostable under industrial conditions (hydrolytic & enzymatic degradation - *ISO EN 14995: 2007)*

Digitalization pursued: In-line monitoring technologies: The distribution of the fibrils in the matrix will be monitored to control the quality of the produced materials. Information will be obtained by (1) **machine vision** applied to the to the produced samples; and (2) **spectroscopic techniques** to characterize the composition of the materials.
Simulation support: Artificial Intelligence (AI) analysis will be used to improve the chemometrics discrimination and machine learning algorithms applied to determine the distribution of the microfibrils inside the matrix. A library for this material will be created and simulation trials for thermoforming will be done (adapting the TSIM software) using the eco-designs for the foot insoles. Updates on the libraries when using recycled PLA will be set-up too, and deviations from neat PLA (if any) obtained and corrected for improving the quality of future regenerated products.

Packaging Value Chain: Demo Case → Lid for garbage container (99% biobased)

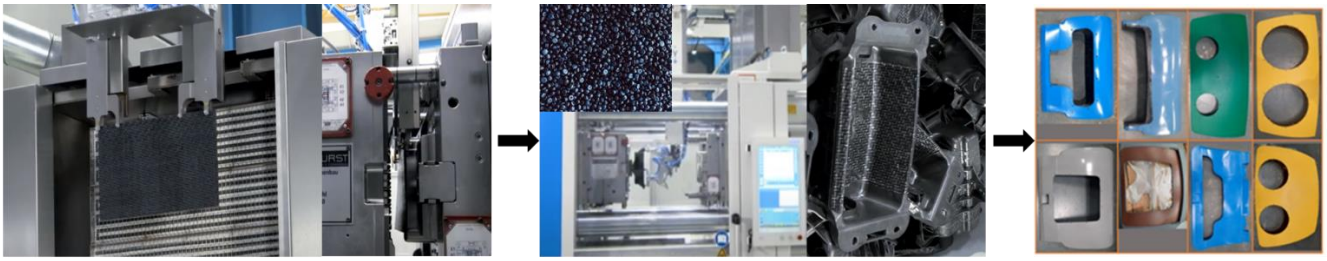
Partners involved in the value chain: SP (CAN-adhesive designed for bioPA/PLA thermoplastic), AIT (Pellets development, end-of-life validation and circularity); CTB & COM (organosheets production), MOS (garbage lid over-injection moulding and product validation), SIM (over-injection simulation), AIM (data collection & interoperability), IRI (in-line monitoring)

Materials used for the demo-case: This value chain will use commercial and recycled PLA[20] (from organosheet reused) and bioPA (Natureplast - NP BioPA1010-201, 100% biobased). This PLA/bioPA will be reinforced with lignin derivates: biobased short Carbon Fiber[21] (TPU-lignin) and wood dust coming from discarded stages of the recycling of furniture (FINSA stakeholder). Pellets for injection-moulding will be developed (these natural fibres will act as impact absorbers, noise mitigation, enhance mechanical performance and provide nicer aesthetical finishing)[22-24]. In addition, a biobased reversible adhesive will be used for debonding overinjected parts of bioPA from structural PLA organosheet, allowing to disassemble targeted prototypes and enable the remanufacture and recycling of each independent components.

[20] Peinado V, et al. Effect of Extrusion on the Mechanical and Rheological Properties of a Reinforced Poly(Lactic Acid): Reprocessing and Recycling of Biobased Materials. Materials (Basel). 2015;8(10):7106-7117.
[21] M Culebras, et al. Biobased structurally compatible polymer blends based on lignin and thermoplastic elastomer polyurethane as carbon fiber precursors. ACS Sustainable Chemistry & Engineering 6 (7), 8816-8825
[22] Aparna,S., et al Effect of short CF content and water absorption on tensile and impact properties of PA_6/PP blend based composites. Polym Compos. 2020.
[23] Ahmad, H., et al. A review of carbon fiber materials in automotive industry. IOP Conference Series: Materials Science and Engineering, 2020,971, 032011.
[24] Peltola, H., et al. Wood based PLA and PP composites: Effect of fibre type and matrix polymer on fibre morphology, dispersion and composite properties. Composites Part A: Applied Science and Manufacturing, 2014, 61, 13–22.

Manufacturing technologies: intermediate formats:1) optimising the extrusion-compounding of pellets (wt% loading, fibre length and shape, adapt shear profile[25] configuration and dosage position[26]….) and 2) continuous pressing to consolidate the desired PLA self-reinforced organosheets.

The final product: injection-moulding process by using the PLA organosheet as a structural component. By warming-up the organosheet over the material Tg (55ºC) with an Infrared light lamp system, spray the adhesive at ≈same temperature and immediately afterwards locate the soft organosheet inside the mould. The mould clamps and by pressure confers the final preform-shape to the laminate while the adhesive cures. In a second step via over-injection of melted bioPA blends will confer the final functional and aesthetical features. Supercritical fluids will be introduced (MuCell® technology) to foam the internal layer. In this way a lightweight but robust structure will be achieved.



Biobased Synthetic Organosheet + Biobased pellets (with short natural fibres)
Over-injection Moulding process (softening with IR lamp + forming with mould clamping force + Over-injection of bioplastic pellets)

Fig. 7: Technologies that conforms the thermoplastic production process for large volume products: intermediate and end-product stage

Main Validation tests according to identified Standards: European Directive 2000/14/EC. 92,0 db: noise emission outdoors; EN 840:2021: Installations and equipment for waste disposal and treatment (In particular: 840-1: Dimensions & Design and 840-5: Performance requirements and test methods). ISO 14006 Ecodesign.

End-of-life scenario pursued: Separation of multimaterial components by inducing a control warm-up to demonstrate CANs reversibility (adhesive) and be able to separate the over-injected plastic from the organosheet structure (peeling-like). The separated materials can be either recycled for the same purpose (re-melted and extruded into pellets) but also other remanufacturing purposes will be seek (e.g. for urban furniture products and insulating facades). Wether the properties are too low, bioPA monomers could be recovered via solvolysis: *cooperating with POLYNSPIRE - PA solvolysis recycling process. Caprolactone obtained from recycled lids and packaging could be also used for polymerizing PCL for the medial value chain.*

Digitalisation: In-line monitoring: organosheets monitored as explained in the medical value chain; while high mature technologies such as extrusion-compounding of pellets and over-injection moulding are already considered smart technologies with enough self-monitoring pro forject purposes. However, a **spectral temperature camera** for continuously control the temperature of the organosheet during its over Tg heating will be used to ensure achieving high conforming and good wetting with the adhesive layer and the melted bioPA. Simulation support: New libraries for injection-moulding grades of the biopolymers developed will be inserted in CADMOULD software. The PLA organosheets materials will be included too for the overinjection module as well as the adhesive layer.

- Innovation activities feeding the project

The main national and international research and innovation activities whose results directly feeds Bio-Uptake and that can be considered as the **unique state of the art of the consortium** has been collected in the following table. The established link is highlighted in blue:

Table 5: research and innovation activities whose results will feed into Bio-Uptake

Project [Date & Partners]	*Key contribution to Bio-Uptake*
VIBES: Improving recyclability of thermoset composite materials through a greener recycling technology based on reversible biobased bonding materials. (H2020-BBI-JTI-2020; GA101023190). *[2021-2025; AIT (coord), SP, LIM]*	VIBES aims at developing biobased covalent adaptable network materials (D-A, vitrimers and supramolecular architecture) of different nature for industrial applications (construction, naval, aeronautical) to separate thermoset resins from fibres. Within Bio-Uptake project, this concept is going to be adapted to be used in the interphase between thermosets and thermoplastics, providing a sustainable end-of-life in the separation and circularity of the components (intermediate formats) conforming the final biobased products.
MANDALA: The transition of Multilayer/multipolymer packaging into more sustainable	MANDALA aims to develop new adhesives with dual functionality (easy to split and barrier properties) with increased biobased and recycled content for multilayer packaging (PUR adhesives) that allows the separation of multilayer flexible packaging. The

[25] Hopmann,C, et al. Method for the Validation of Simulated Mixing Characteristics of Two Dynamic Mixers in Single-Screw Extrusion. Polymers 2020,12,2234.
[26] Imre B, et al. Reactive compatibilization of plant polysaccharides and biobased polymers: Review on current strategies, expectations and reality. Carbohydr Polym. 2019;209:20-37.

multilayer/single polymer products for the food and pharma sectors through the development of innovative functional adhesives. (H2020-BBI-JTI-2018-GA837715) *[2019- 2023; AIT (coord)]*	knowledge developed has demonstrated the feasibility to separate thermoplastic materials of different nature to increase plastic recyclability. As it is in the upscaling phase for the delamination technology, Bio-Uptake will take advantage of the technologies used to activate the debonding mechanisms that induces delamination (IR, UV, convection…), to be a starting point for piloting and upscaling the separation of the final product components.
ECOXY: Biobased, recyclable, reshapable & repairable (3R) fibre reinforced thermoset composites**.** **(**H2020-BBI-JTI-2016-GA744311) [2017-2020; SP, AIT, CID (coord)]	ECOXY develops biobased epoxy resins and fibre reinforcements to produce new sustainable thermoset composites (based on 3R-vitrimer technology). In Bio-Uptake project the acquired knowledge and the optimized bio-based 3R resin formulation will be adapted and upscaled to the new applications and manufacturing technologies. This manufacturing line will allow to get enduring prepreg sheets at different curing stages. These sheets will be layed-up and thermoformed for obtaining a multi-layered composite.
BIO4SELF: Biobased self-functionalised self-reinforced composite materials based on high performance nanofibrillar PLA fibres. (H2020-NMP-GA 685614), [2016-2019, CTB, COM]	BIO4SELF aimed at fully biobased self-reinforced polymer composites (SRPC) for non-clothing applications (automotive & home appliances). The expertise on SRPC will be used to develop biobased thermoplastic self-reinforced organosheets (mono and bicomponent): 1) combinations of high melting PLA and low melting PLA; 2) the combination of PLA and PCL microfribrils will be envisaged. The aid of in-line monitoring (IRI) extrusion parameters will suppose a quality gain in the control of the manufacturing process which can be adjusted to achieve the best properties for these preforms.
BIONANOPOLYS: Open Innovation Test Bed For Developing Safe Nano-enabled Bio-based Materials And Polymer Bionanocomposites For Multifunctional And New Advanced Applications (H2020- NMBP- GA953206) [2021-2024; IRI]	BIONANOPOLYS creates a network of pilot plants and complementary services to speed up the introduction of biobased nano-enabled materials into the market through a Single-Entry Point. Within Bio-Uptake project, advanced monitoring systems for the manufacturing of biobased intermediate products (crystallinity of microfibrils that composed the fibres for organosheets and curing grade of endured pre-pregs) will be designed and implemented in order to provide accurate and reliable compositional information. Machine vision applied to the produced samples and spectroscopic techniques to characterize the composition of the materials will be tailored to the manufacturing stages. AI analysis will be used to improve the chemometrics discrimination and machine learning algorithms. In addition, surveillance on derived novel standards & certification schemes in biobased materials will be transferred to Bio-Uptake.
INTEGRADDE*:* Intelligent data-driven pipeline for the manufacturing of certified metal parts through Direct Energy Deposition processes (H2020-FoF-GA820776): [2018-2023; AIM]	This project develops a new manufacturing methodology capable of ensuring the manufacturability, reliability, flexibility and quality of a target metal component from initial product design. The main input of INTEGRADDE to Bio-Uptake will be the use of AutomationML as an interoperable and neutral format, providing data exchange in the CAx (CAD/CAE/CAPP/CAM) dataflow. The concept of INTEGRADDE aiming at the combination of design and multi-scale simulation and modelling for an optimal manufacturing strategy, will be replicated in Bio-Uptake to demonstrate the interoperability and digitalization of the manufacturing value chains, ensuring a smooth and trustful transition for biobased materials.
ACAPULCO : fAbriCation rApide d'emPreinte d'oUtiLlage d'injeCtiOn French national fund Fonds Unique Interminesteriel) [2017-2020; SIM]	ACAPULCO has enabled to work on innovative moulds and to adapt SIM simulation software to the needs of new processes and materials. In this sense, the methodology developed will be applied in Bio-Uptake project, in order to adapt the simulation programmes of thermoforming and injection-moulding manufacturing technologies to the biobased materials and intermediate formats particularities. The behaviour laws suitable for these types of materials will be integrated into the software tools (CADMOULD and TSIM), building these competences in the simulation of bio-based materials.

In addition to the previous list, there are other national and international projects that are also relevant for Bio-Uptake State-of-the-art, from which results will also feed this project. These have been aligned to the key disciplines involved in Bio-Uptake:

- Regarding further developments of biobased materials to be used in Bio-Uptake: LIBRE - Lignin based Carbon Fibres for composites (LIM; H2020-BBI-720707); BAMCO - BAmboo long fibre reinforced biobased epoxy Matrix Composites (SP, French national fund Fonds Unique Interminesteriel, 2018); PRESERVE - High performance sustainable bio-based packaging with tailored end of life and upcycled secondary use: Development of biobased microfibrillar structures in packaging applications (CTB; H2020-NMBP-952983); SEABIOCOMP - Development and demonstrators of durable biobased composites for a marine environment (CTB, Interreg2Seas); NXGENWOOD developing polymers, chemicals and nutrients from waste forestry materials (LIM, science foundation Ireland 2019PROG704).
- Regarding the development of CANs and other approaches to makes the materials and products more circular and sustainable: CARBO4POWER - New generation of offshore turbine blades with intelligent architectures of hybrid, nano-

enabled multi-materials via advanced manufacturing (CID, AIM: H2020-NMBP-953192); AIRPOXY - ThermoformAble, repaIrable and bondable smaRt ePOXY based composites for aero structures (CID; H2020-MG-769274);

- Regarding product end-of-life, materials circularity and LCA assessment: MULTICYCLE (IRI, CTB; H2020-NMBP-SPIRE-820695); CIRCPACK - Towards circular economy in the plastic packaging value chain (AIT, POL, UNE; H2020-CIRC- 730423); POLYNSPIRE - Demonstration of Innovative Technologies towards a more efficient and sustainable plastic recycling (AIT, H2020-NMBP-ST-IND- 210496601), Development of advanced, prospective LCA for cascading bio-based value chains (LIM, ENVISION doctoral training partnership on sustainable wood value chains, 2017)[27]
- Regarding advancements in selected manufacturing technologies including monitoring, simulation and AI to build a virtual interoperable twin: ACROBA - AI-Driven Cognitive Robotic Platform for Agile Production environments (MOS, AIT; H2020 – ICT - GA101017284);); PRESERVE - High performance sustainable bio-based packaging with tailored end of life and upcycled secondary use: Advanced monitoring solutions applied to bio-based material focused on the new microfibril formation (IRI; H2020-NMBP-952983); SILICONE[28] - simulation of elastomers – thermoset resins (SIM, French national fund Fonds Unique Interminesteriel); DIGIPRIME - Digital Platform for Circular Economy in Cross-sectorial Sustainable Value Network: pilots for de-and-re manufacturing on composite and polymers (POL, H2020-ICT-873111).

- Inter-disciplinary approach

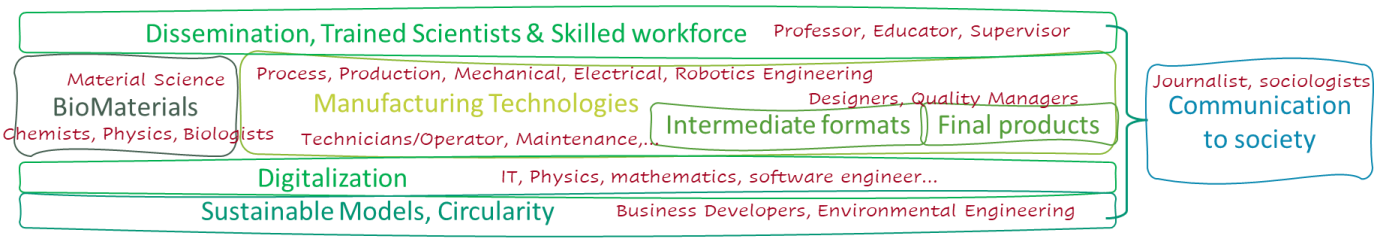


Fig. 8: Bio-Uptake Interdisciplinarity along the Value Chain

Bio-Uptake integrates partners from different disciplines to provide coverage to all the required developments and specifications. The entities involved are highly interdisciplinary and complementary. The Bio-Uptake Work Programme is defined in such a way that the interaction and collaboration among partners is a must in each WP. In this sense, the manufacturing of products with biomaterials is quite a new challenge… *a new frontier of science is a multidisciplinary research field combining engineering, physics, chemistry, biology, material science, which allow the production of biobased products*… that requires to combine knowledge from the following fields (*more details in Part A of the project proposal, this is just a representative group of PIs*):

- For Operational/Manufacturing work: chemists (Alain Graillot, Alaitz Rekondo, Henning Bak) regarding the plastics discipline, material science (Maurice Collins; Pere Castell, Isabel Schrijver) and even biological science (Laura Rodriguez Turiezno) to appropriate combine and produce composites and run characterisation protocols (Davy Van Cauwenberghe) and measures to enhance the material itself, manufacturing engineers (Lien Van der Schueren, Hans Knudsen, Anne Beaucamp, Iván Monzón, José Antonio Dieste) to modify and optimise the manufacturing processes, adapt machines, systems and peripherals (Baltasar Lodeiro Señaris), and process operators (Mikael Söderberg, Therese Berggren) to carry out the production, optimisation of the machine and guide, receive and multiply the training. In addition, the understanding and combination of physics, chemists and engineers for the ad-hoc development and integration of in-line monitoring technologies (Sara Sanchez, Victor Olmos, Nicola Palombo Blascetta). These scientific and technological profiles also cover scientific publications, patents and contributions to new standards as committee managers (Steffen Jenkel, Fernando Utrilla).
- For Digitalisation: informatics (Ludovic Potier), mathematics, software engineers (Romain Chaière), I4.0 system, machine learning and artificial intelligence engineers (Lucía Alonso, Carlos Gonzalez Val, Lara Baria Besada, David Nettleton) are required to understand how to collect the vast amounts of data, design the mathematical algorithms and make the data interoperable as well as understandable for the technicians involved in the process.
- For Circularity and Environmental Science: chemists (Samuel Malburet), experts in logistics (David Ponce), business and product developer (Birgitha Nyström, Peter Lundmark, Raphaël Mosnier) and environmental engineering (David Styles, Ana María López Sabirón).
- For Social Implications: sociologists (Eva Sanchís), journalists (Lara Escudero), supervisors (from industrial partners), professors (from academia and RTOs associate professors) and business developers (Berta Gonzalvo, Elena Jubete, Annabelle Sion) to ensure a proper communication to the different targeted audience (including to reach policy makers and the society in general), as well as be able to properly develop a training programme and finally guide the exploitation and have the business vision.

[27] http://www.envision-dtp.org/2017/developing-environmentally-sustainable-forestry-value-chains/

[28] H. Ou, et al. Experimental characterisation and modelling of rheokinetic properties of different silicone elastomers. Int J Adv Manuf Technol (2017) 92:4199-4211 (Publication from SILICONE project)

- Social Sciences and Humanities

Integrating SSH research across European collaborative projects is essential to maximise the returns to society from investment in science and technology. Under the list of SSH disciplines, Bio-Uptake project has identified three of them under the category of social sciences, education, business and law. In particular, these are related to: i) Social and behavioural sciences; ii) Business and administration; and iii) Education science.

The creation of an effective after-use plastic economy as a main strategy to promote increased recycling rates of polymers directly clashes and influences societal behaviour patterns as well as economics. To contribute creating the after-use plastic economy it is important to create connections among all stakeholders in the value chain as well as involve society. This target is achieved with the combination of the consortium partners and project stakeholder´s. Specific activities with the targeted stakeholders (WP6), workshops and round-tables will take place to share different perspectives and get feedback to boost the concept and its implementation in a real economy. Moreover, the creation of novel sustainable business models for the manufacturing industry and adapted to each of the selected products and sectors in WP5/6 will boost circular loop business, industrial symbiosis, for the real deployment of circular solutions (source and end-of-life), even creating new de-manufacturing businesses and job profiles.

One key relevant point in Bio-Uptake project goals is the development of a training programme to provide knowledge and skills to the European industry workforce about the circular economy and digitalisation good practices as well as to European students in the materials science, engineering and scientists (mainly STEM fields) teaching them new skills for new arising demand in technical jobs (to boost the breeding ground of new talents). The training activities will be orchestrated by ULI at the academia level (researchers and students), with the support of AIT, CTB, AIM and CID at vocational level (industry) inside WP1. Thus, it will be coordinated by Professor Maurice Collins, from a more scientific perspective and as expert in biobased composite materials, in cooperation with Ms. Eva Sanchís (AIT - graduate in sociology and training & master in emerging methodologies and technologies applied to education, e.g. gamification) who will orchestrate the industrial workforce upskilling training.

- Gender Dimension

The plastic industry is a considerable employer in Europe. The plastics industry gives direct employment to more than 1.6 million people in Europe[29]. Because of the ubiquitous nature of plastic, we are intrinsically linked with plastic in many ways due to our assigned gender roles, affecting us through usage, disposal, and through to how it is recycled, depending on cultural and social aspects, economic circumstances, and availability[30]. There is little documented research and knowledge on the specific roles of women and men as agents of change in the management, production and usage of chemicals and plastics. Although workers can be exposed to a toxic soup of chemicals found in most plastics industry environments, women are the ones most likely to be most concerned about environmental issues or impacts on health from exposures to toxins and therefore are more prepared to take action. This is probably due to its relation to biological factors, like effects on the hormonal system, and on menstruation as well as pregnancy and breast-feeding.

Bio-Uptake consortium is aware of the relevant gender and cultural biases and are familiar with gender analysis frameworks that address it, being informed about the EU regulation on achieving gender equality across the project, as stipulated in European Union Polices Com (96)67, Com (97)497, Com (98)131, and Com (98)112. Bio-Uptake follows the EU objectives for gender equality in research as expressed by the European Institute for Gender Equality (EIGE) and has a strong commitment to promote gender equality in the execution of the project focusing from two angles: 1) human resources: ensuring gender balance in decision-making processes and bodies; and 2) content: guaranteeing gender equity in the research and innovation activities of the project and at academic level, since women are sometimes underrepresented in research. In this way we aim at narrowing gender gaps as a result of the implementation of measures to improve gender equality in Europe, and in particular in the coming bioplastic manufacturing industry. One of the main outputs of Bio-Uptake project in this matter is related to the communication, dissemination and training activities in order to boost more women graduating in science degrees, technology, engineering and mathematics (STEM) and more woven actively participating in the plastics labour market.

The Human resources perspective will be tackled in Bio-Uptake by means of a balanced participation, as close as possible to 50/50, of both men and women in the teams and among the leading roles during the project. This will be followed up throughout the entire duration of the Action and reported to the EC at the end of each period and as requested. Continuing with this, Bio-Uptake ensures that at least 40% of the underrepresented sex will participate in the decision-making bodies of the project by nominating them in steering roles. The consortium shall guarantee the absence of gender bias in the performance of the decision makers. Any bias will be reported, discussed, annotated and eliminated. Not only that, the integration of the gender dimension will be promoted as well in the management procedures during the project.

The Content perspective shall include the design and activation of appropriate indicators and monitoring, and revisions of the gender perspective for all relevant activities during the General Assembly meetings and in the Periodic Reports. Such indicators and their monitoring will be taken into account with special care in T6.1, which deals with the communication, dissemination and training strategic plan. Such plan will include a chapter on gender equity for the three activities, built with the inputs of all the partners involved. Stakeholders will also be invited to participate. The integration of women in research and innovative activities

[29] https://www.plasticseurope.org/application/files/9715/7129/9584/FINAL_web_version_Plastics_the_facts2019_14102019.pdf
[30] https://www.wecf.org/wp-content/uploads/2018/11/PlasticsgenderandtheenvironmentHighRes-min.pdf

will be also attained, in part following the European Charter for Researchers – Code of Conduct for the Recruitment of Researchers – and participating in the 'International Day of Women and Girls in Science'.

In all, the consortium agrees (1) to guarantee the total absence of inequalities, either maliciously or unintentionally, created during or as a result of this research project; and (2) to follow the prescriptions of the GEAR Tool, as published in 2016 by the EIGE, including institutional change inside their organizations. Finally, and as a result of the project KER exploitation, Bio-Uptake will generate businesses owned and led by women as well as well-balanced gender jobs. In this way, it is expected to reduce the risk of dropping-out women from their professional careers.

All the actions are in compliance with current and under implementation partners´ gender equality plans (public bodies, higher education and research organisations): AIT, CTB, CID, AIM, LIM, POL, UNE

- Open Science

BIO-UPTAKE will conduct Open Science, including infrastructure, technical, legal, professional and implementational support from institutions by introducing in its research actions the following practices and activities along with outcome dissemination and data management, promoting the main characteristics described according to the Open Science Wheel[31] and helping to set ip appropriate indicators to monitor further the impact of the project:

- **Actions to promote Open research data**: Bio-Uptake aims at train and guide researchers (and in particular young researchers) attitude towards data sharing by organising along the project at least two sessions. These sessions will be flipped classrooms that will boost the "learning by doing" method. They will be based on open courses following FOSTER[32] learning paths (first watching and studying the videos/lessons by themselves and afterwards having a plenary session to debate and put into practice the learned concepts, and putting them into practice with the project selected outcomes). Senior Bio-Uptake researchers will participate from these sessions too, so that it serves as retrain and as a discussion/debate sessions to update about good practices, new repositories and arising funder policies about data-sharing, that may redirect the method used. The research data and bibliographic metadata needed to validate the results presented within the research article publications will be stored in the respective area of the repository *(e.g. INTEGRADDE: Metal Additive Manufacturing Open Repository | Zenodo)* and will be follow FAIR principles as stated in the DMP (D7.2). It will serve as a common exercise to train the project partners in Open Science good practices.
- **Actions to promote Open Access publications:** A dedicated budget for open access scientific publications has been reserved, which will be used for "green" or "gold" publications in reputable journals (identified in *Table 10*). Alternative repositories such as the Open Access Infrastructure for Research in Europe (OpenAIRE), the Registry of Open Access Repositories (ROAR) and the Directory of Open Access Repositories (OpenDOAR) or other institutional repositories with no cost related to their use such as EOSC, ZENODO (supported by the EC) and alternative publishing platforms e.g. ResearchGate (directly managed by each researcher) … etc will be also exploited. In all cases, within six months after publication, LIM as dissemination task (6.3) leader will be responsible to collect a machine-readable electronic copy of the published manuscript in order to be deposited in the project website either as a reproduction or as direct link to the publication. In addition, LIM university library will also act as open repository of Bio-Uptake selected data, metadata and publications. Moreover, all the scientific articles disseminated within the project will be access accelerated by using Pre-Prints mode (the peer review draft will be uploaded afterwards by the journal).
- **Actions to promote Open scholarly communications:** Bio-Uptake most relevant publications will use almetric[33] badges and platforms to capture diverse flavours of the impact that our research outputs have in the community and society. In this way we´ll help the project to have a major impact and promote discussions at scholar and professional level. In this way, Bio-Uptake aims at covering a gap that only by using citations, researchers cannot reach. It will complement with new indicators which are "instantly measurable" (most like in Social Media) that can be applied to other data in general (e.g. software, books, metadata…) and will supplement further monitor the real impact of the dissemination efforts carried out by Bio-Uptake partners. In this way a project output will be identified (DOI) and monitored through traditional and non-traditional sources. Definition and monitoring of project almetrics will be defined in the communication and dissemination plan and will be suirvellanced by LIM. In addition, when possible, the papers to be disseminated will participate from Open Research Reviews, in order to increase transparency, make the outcomes more inclusive and also train the young researchers to see how the constructive critics are received, how these have to be tackled and answer etc, and in other hand research leaders from academic and RTO institutions (e.g. LIM, AIT, CTB, CID, AIM…) will also participate in at least 3 Open Peer Reviews when they are considered the experts of a topic. In this way, young researchers will be also trained in how a Peer Review is done.

In this way, Bio-Uptake will educate and skill researchers (helping to upgrade current senior scientific researchers and training the young ones) in incorporating open science skills as part of career development and also recognise and reward these skills in recruitment and career progression. Moreover, by putting into practice these practices, Bio-Uptake researchers will be able to

[31] Providing researchers with the skills and competencies they need to practise Open Science. Open Science Skills Working Group Report, 2017. DOI 10.2777/121253

[32] https://www.fosteropenscience.eu/

[33] https://www.altmetric.com/about-altmetrics/what-are-altmetrics/#prettyPhoto

train the science speech and popularise science by learning how to collaborate and build networks for the general public, peers and fundraising and investment institutions.

- Data management

Bio-Uptake is developing a DMP within task 7.3. Confidential data related to industrial processes and products will be kept restricted to the consortium partners and not published until it is protected by partners for exploitation. Only non-sensitive publishable data will be eventually shared and made accessible to the research community for verification. Specific procedures for publication clearance will be included in the Consortium Agreement. Contractors will have to inform the consortium of their intention to publish their results that may be relevant intellectual property. Dissemination of intellectual property is granted only with the approval of the consortium, making sure that the period of the GA will also address these aspects, to guarantee that no conflicts arise between the partners on IPR. Publishable results proposed by partners must be accepted by the innovation board and the project coordinator. The CA will also address these aspects. Bio-Uptake will follow a proactive strategy for internal data and knowledge management. A preliminary draft of the DMP based on the specific elements of the proposal:

1. Data summary: Six database (primary data) are envisaged for Bio-Uptake project
1. Simulation and digital threat: code for modifying the simulation programme, material libraries (metadata on biobased materials), algorithms, product mess (FEA) modifications, numerical data from running simulations, … 2. Biobased Materials: original & modified/new datasheets, numerical data coming from characterisation lab & workshop tests (mechanical, rheological, thermal, microscopy etc), application-integration protocols… (numerical, curves, images) 3. Manufacturing processes: Data/parameters coming from experimental work recorded from running the manufacturing processes (by in-line monitoring prototypes and from probes and sensors coming from smart processes), set up of the pilot lines… (numerical data, signals, process curves, metadata, observational, machine drawings and images,…in HDF5) 4. Product: drawings and modifications over original designs (CAD: STEP, IGES…), numerical data from product quality check and functional & environmental validations (internal & ISO applied standards, numerical data, observational outcome) 5. Training exercises, lectures, practices (text, ppt, videos): Interaction between staff, machines & equipment's, and materials, anonymous staff survey… 6. Data coming from communication and dissemination actions: surveys, stakeholder's feedback, scientific publications… (text, observational, numerical data, drawings, images, curves, metadata…)
2. FAIR Data
2.1. Making data findable, including provisions for metadata
A data repository will be managed by the Project Coordinator to store common data, generated metadata files, results, reports and deliverables. Partners will have access to the repository for uploading and downloading documents. This repository will be based on Microsoft Teams or Google Drive (folder restricted to the project consortium). For open Data Repository re3data and/or EOSC (European Open Science Cloud). This double copy will serve as security storage. Following, scientific publications and articles, the dataset deliverables and the final demonstrator research results will be shared through EOSC and other database (such as ZENODO) to promote the data making FAIR. Files and folders at data repositories will be versioned and structured by using a name convention consisting as follow: **BIO-UPTAKE_Dx.y_YYYYMMDD_Vzz.doc**
2.2 Making data openly accessible (related to the Open Science community)
Selected data and results will be shared with the scientific community and other stakeholders through publications in scientific journals and presentations at conferences, as well as through open access data repositories. Identify existing e-Infrastructure and resources: including advice on copyright law, best practices on intellectual property right and exploitation (using persistent identifier such as a DOI or following OpenAIRE+ and ZENODO e-infrastructures to establish the right protocol, or GitHub (e.g. EUDAT, for software). The Bio-Uptake project datasets are first stored and organized in a database by the data owners (institutional secure server) and on the project database. All data are made available for verification and re-use, unless the task leader can justify why data cannot be made openly accessible. All data deposited on EOSC or ZENODO are accessible without restriction for public. Embargo periods of certain publications (green open access) will be solved by sharing pre-prints version. For other data, potential users must contact the IPR team or the data owner in order to gain access (e.g. via Researchgate). If necessary, appropriate IPR procedure (such as non- disclosure agreement) will be used.
2.3. Making data interoperable
AIM with the support of IRI and SIM will be in charge of making data interoperable: machine-to-machine interoperability of the metadata generated in the project. Making files opened, readable and understandable by different software will ease the reproducibility of the new files for replication purposes. All datasets will use the same standards for data and metadata capture/creation. Information on data & metadata vocabularies, standards or methodology to follow facilitate interoperability and whether the project uses standard vocabulary for all data types present to allow interdisciplinary interoperability.
2.4. Increase data re-use
Creative Common Licensing will be used to protect the ownership of the datasets. Both Share-Alike and NonCommercial-ShareAlike licenses will be considered, for which the decision of making that part public has been made by the Consortium.
2.5 Curation and Storage

An internal peer review is performed for the main project deliverables (among WP leaders and through coordinator) to guarantee the deliverable is developed with a high level of quality. The project data will remain re-usable for at least 5 years.
The European Open Science Cloud will enable researchers across disciplines and countries to store, curate and share data. Also, public partners such as public European universities & libraries (e.g. LIM) will provide this service too.

2. IMPACT

2.1. Project's pathways towards impact

- Contribution to the outcomes and the wider impacts

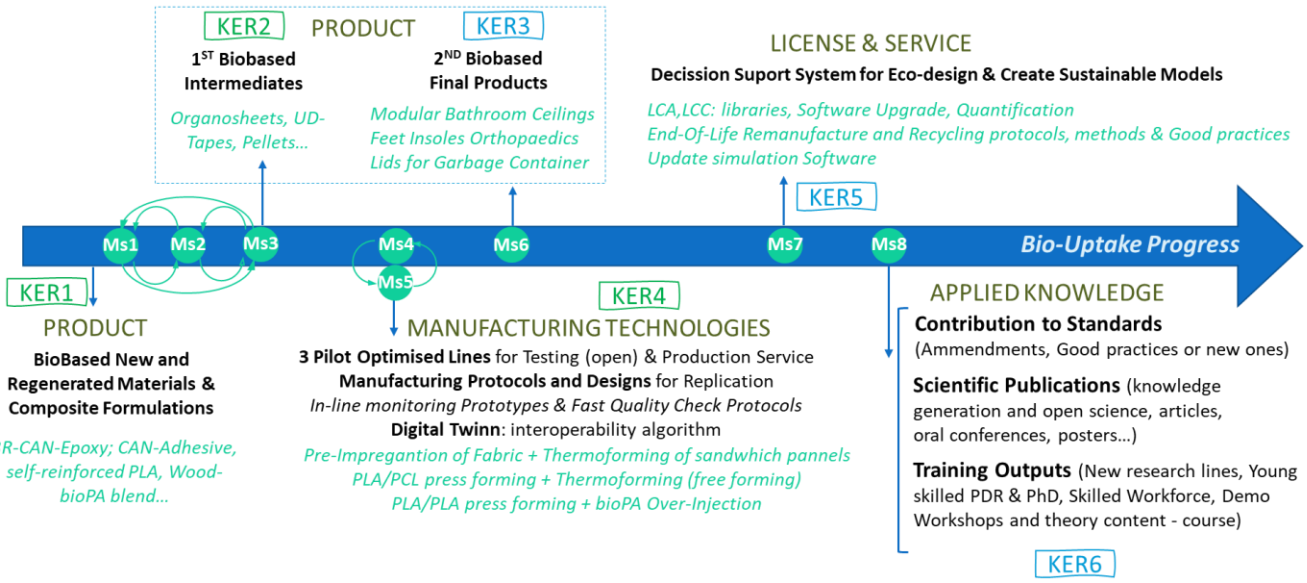


Fig. 9: Definition of Project Outcomes and KERs identification (temporal line according to the achievement of project milestones)

Bio-Uptake contribution to the **Outcomes** specified in this topic:

Demonstrate relevant scale production of innovative bio-based products to substitute traditional materials with high environmental footprint
The manufacturing sector and more in particular the **plastics manufacturing sector is on the centrality of a variety of value chains.** The potential scale production of Bio-Uptake concept and the developed **manufacturing processes** (KER4) goes far beyond the selected value chains. It is based on **modularity:** production of **intermediate formats** (KER2), which are used as components of the final or finished products (industry-to-industry trade – KER3). This is **a European dominated market** with a share$_{2020}$ of 37.75%. The potential to grow is huge, as an example, the Global Organosheet and Semi-Finished Thermoplastic UD-Tape Laminate Market Anticipated to Reach $1,671.2 Million by 2030 at a CAGR$_{2020-2030}$ of 10.03%[34], which can be considered the SAM (Served Addressable Market) of Bio-Uptake developed intermediates. Owing to increasing environmental concerns, this concept is gaining momentum and **Bio-Uptake aims to be the first of its kind introducing and promoting biobased composites** (biobased plastics and fibres/reinforcement) with such a model. Up to know, the commercial materials that dominate these types of intermediates are mainly Glass and Carbon Fabrics, epoxy thermosets, high-tech thermoplastics such as PEEK, PPS and PA, and Polyolefins - PP. In this sense, Bio-Uptake aims at demonstrating that at least for the selected value chains, biobased epoxy and flax fabrics, PLA and bioPA products can be as competitive as petroleum-based counterparts following this modularity approach and with a proper eco-design. The selected demo cases for each value chain (construction, medical and packaging), opens the door for biobased materials to develop and adapt a complete manufacturing process that could be afterwards exploited in these sectors or others. In any case, it permits to carry out the experimental and digital modifications, obtaining physical products to trade-show and opening three direct penetration markets (Bio-Uptake Serviceable Obtainable Market – SOM). The scale production within the selected sectors is therefore: 1) **Prefabricated Bathroom Pods** market is valued at USD 1.41 billion in 2020 and is expected to reach USD 2.95 billion by the end of 2027, growing at a CAGR$_{2021-2027}$ of 10.53%[35]; 2) **foot orthotic insoles** market size was 2.98 billion in 2020 and is projected to grow at a CAGR$_{2020-2027}$ of 6.1% reaching 4.50 billion in 2027[36]; and 3) the global **waste management market** will reach 2.3 trillion U.S. dollars in 2027, based on a CAGR$_{2020-2027}$ of 5.5% from 2020[37].

[34] Business Wire. Research & Markets 2021. Global Organosheet and Semi-Finished Thermoplastic UD-Tape Laminate Market Report 2020-2030
[35] CISION – Valuates Reports (2021): Prefabricated Bathroom Pods Market Size
[36] FORTUNE BUSINESS INSIGTHS – Medical Device / Foot Orthotic Market. Forecast 2021-2027
[37] STATISTA (2021): Market size of waste management worldwide in 2019 and a forecast for 2027

Develop products with similar or better mechanical, physical and chemical properties, while having a substantially lower environmental footprint and being sustainable, non-toxic and recyclable when compared to non-biobased materials

The conception of the intermediate formats was originally developed in the aerospace and defence industry, which dominated the market until 2019 (quite a new trend). In this way, lightweight products with outstanding strength and design flexibility were conceived. Each intermediate format is designed to provide a particular feature/function and once these individual components are joined, the resultant product presents a **synergistic technical behaviour** that meets target requirements for an application. This concept is applied in Bio-Uptake to overcome the well-known technical barriers of biopolymers[1].

Introduction & comparison in properties of Bio-Uptake products and non-biobased counterparts:

The **garbage containers** are designed for society (for councils in urban areas, industries and home-residential areas) adapting them to ease trash deposition in the most hygienic way. These are made of PE (oil-based polyethylene) + EPDM (oil based elastomeric material) conceived to present good mechanical properties and not to be affected by environmental harshness's or vandalism actions. Up to now, there has been some approaches to make this product more sustainable, but none based on Bio-Uptake concept: CONTENUR is manufacturing containers with 100% recycled PE (CIRCLE ®) and PLASTIC OMNIUM (EU World Leader) 100% sugarcane-based polyethylene. Bio-Uptake aims at migrating this material for a combination of self-reinforced PLA organosheet (which presents mechanical properties[38] compared and even higher than current PA, while reducing the product overall cost and whose industrial production its being strategically foster in EU[39]) overinjected with European bioPA (to provide UV, water, chemical and mechanical resistance specially toughness at impacts[40,41]). This bioPA will be reinforced with different derivates of lignin microfibers (European feedstocks): from bioCF to r-wood. Lignin will provide natural UV- blocker protection[42], noise reduction (it is used in bio-asphalt[43] for this purpose), aesthetical properties (e.g. for the organic container) and antimicrobial[44] effect. A clear reinforcing effect in the case of the selected bioCF coming from LIBRE and VIBES developments is expected: > 2 GPa tensile strength and > 200GPa modulus[45]).

The **modular bathroom pods** are simple to lift, place and connect (e.g. hotels, caravans, offices…). These flat panels can be also used as walls and floors of the whole bathroom pod solution (modular plug). Currently, these panels as produced via resin vacuum infusion of conventional epoxy resin[46] and glass fibre and within Bio-Uptake it will be substituted by a thermoforming process of flax-epoxy layers and a wood basal core. For this product it is of outmost importance avoiding water uptake (accurate control of the curing stage of the bioepoxy), be light while mechanically resistant and absorb noise too.

Foot pain affects more than 30% of the general population. **Feet orthotic insoles** are shoe inserts designed: a) to cure the foot deformities (e.g. medical for plantar fasciitis, bursitis and arthritis…), b) stabilize the biomechanical misalignment of the foot (e.g. sports & athletics) and c) provide personal comfort in daily life. Bio-Uptake aims at demonstrating a solution to "adaptable hard" insoles to substitute *Thermoformable Composite Insole* rigid insoles 2.5 mm thick and made to be used as a rolling insole for diabetic patients & of Dynamic Walk AFO designed to allow people with dorsiflexor paresis or paralysis to confidently regain their stride. These two models are now made of Carbon Fibre and medical PEEK grades. In this case self-reinforced PLA and PCL/PLA microfibrillar bicomponent organosheets will confer mechanical properties comparable to current hard insoles (rigidity for the plantar arc, impact absorption).

Sustainability and environmental comparison, contributing to emissions reduction: In terms of environmental footprint, LCA studies have demonstrated the benefits of introducing these biobased materials to achieve decarbonisation and Green Deal GHG goals. Three categories are foreseen:

- Regarding the **thermoplastics** used for Bio-Uptake products: PA is made of adipic acid (AA) building block. Conventional AA account for about 5% of anthropogenic N_2O emissions which is a powerful greenhouse gas (265 time the CO_2 reference) and have a global warming 7.84 kg CO_{2eq}. Biobased AA (coming from softwood - celluloses either via levulinic acid or enzymatic route) contains biogenic carbon (carbon positive) and the total global warming is of 4.28 kg CO_{2eq}, this is about 45% less GHG emissions[47]. Moreover, if a circular solution is encouraged such as in Bio-Uptake, reusability and/or recyclability is contemplated and in a second loop this biobased recycled material greenhouse emission is cut down by 50% (about 2.3 kg $CO_{2\,eq}$)[48].
- **Reinforcements** such as the high added value Carbon Fibre (PAN - oil-based polyacrylonitrile) has a global consumption or 170,000Tn and its production generates 38,9 kg CO_2eq/kg CF[49]. According to LIBRE project LCA results, lignin-based

[38] L. Van der Schueren et al. BIO4SELF: High stiffness PLA yarns for biobased self-reinforced composites – Journal Chemical Fibres International, April 2018
[39] European Biotechnology 2021. Progress for PLA plant in EU (Total Corbion & NextChem, *new PLA operational plant in France 2024*)
[40] NaturePlast bioPA grades (TDS). 100% biobased. Comparable mechanical and chemical properties to oil-PA. Non-toxic, even Food Safe.
[41] J. Reolid, et al (2019). Functionalization of Partially Bio-Based Poly(Ethylene Terephthalate) by Blending with Fully Bio-Based Poly(Amide) 10,10 and a Glycidyl Methacrylate-Based Compatibilizer. Polymers. 11. 1331.
[42] H. Sadeghifar, et al (2020). Lignin as a UV Light Blocker—A Review. Polymers. 12. 1134.
[43] Gosselink R.J.A., Landa P., Lignin-based bio-asphalt, WO2019/092278 (PATENT)
[44] Lobo, F.C.M.; et al (2021). An Overview of the Antimicrobial Properties of Lignocellulosic Materials. Molecules, 26, 1749.
[45] A.Beaucamp (2019).Carbon Fibre From Renewable Resources:The Role of Lignin Molecular Structure on its Blendability with Biobased PET.Green Chem21
[46] A. La Rosa, et al (2016). Recycling treatment of carbon fibre/epoxy composites: Materials recovery and characterization and environmental impacts through life cycle assessment. Composites Part B: Engineering 104.
[47] Achille-B, et al (2017). A Prospective Cradle-to-gate LCA of a Bio-based Adipic Acid. Maastritch University & Chemlot InSciTe. FNRS LCA Contact group
[48] B. Cushman-Roisin et al (2021). Data, Statistics, and Useful Numbers for Environmental Sustainability. Bringing the Numbers to Life. Chapter 1 - Materials. Elsevier Books. ISBN 0128230118, 9780128230114
[49] Carbon Fibre LCA data in EcoCalculator, An essential extension. Composites Europe - 2018. EUCIA – European Composites Industry Association

CF can cut in half the CO_2 footprint of CF production. The impact of Glass Fibre production is 1.5 kg eq. CO_2 per Kg of virgin GF produced[50]. The production of E-glass (technical fibre) also emits SO_x and NO_x from batch material decomposition. In Bio-Uptake Glass Fibre will be substituted for flax fabrics in the construction applications. The CO_2 impact of using flax for thermoset applications varies depending on the origin of the biomass (e.g. European versus Chinese fibre due to more sustainable agricultural practices). Bio-Uptake will promote and use exclusively flax coming from EU (promoting EU local resources *Be-Fr-Nl*). In a comparison study carried out by Elin Jacobson[51] (2018), results showed best values for global warming potential of -1.4 kg CO2 /kg fibre considering carbon dioxide sequestration (biogenic). These data have been also corroborated by Nova-Institut report[52].

- The use of toxic and environmentally hazards chemicals in conventional epoxy **thermoset** (Bisphenol-A & Epichlorohydrin are highly toxic and are categorized as carcinogenic, mutagenic, and reprotoxic) has triggered efforts among researchers on developing epoxies from various bio-sources such as vegetable oil (as in Bio-Uptake with formulations developed under ECOXY project and own SP research). Studies comparing oil-based epoxy with vegetable derived epoxy have shown about a 38% reduction in Global Warming Potential (GWP)[53].

Demonstrate disruptive innovation of biobased materials production in at least three different manufacturing value chains

Bio-Uptake disruptive innovation has been brought by combining **material and process development** new concepts together with their **digital twin** (digital thread and digital manufacturing creating a virtual process that enables to accurately simulate, monitor and optimise the final product). The disruptive innovations which are specific for the peculiarities of each Manufacturing Process (KER4) and are highlighted below:

- In the **manufacturing process for thermoset composites**, (construction) pre-fabricated panels for modular bathroom pods will be developed thanks to: 1) developing biobased endured organosheets based on bioepoxy +3R-CANs + flax fabrics, which are reshapeable; can be stored at room temperature without refrigeration; if certain curation appears it does not mean that is has to be discarded permitting its reuse (this will terminate with the unintentional generation of scrap); 2) set up a semi-automated process to work with these endured-pre-pregs by monitoring the curing level of the resin will allow to introduce and faster and less energy consumption; 3) migrate the current infusion of pod panels (slow process) for a quicker thermoset process; 4) new TSIM simulation module to enable designing products under this concept.
- In the **manufacturing process for highly customised thermoplastics**, (medical) orthotic insoles are developed thanks to 1) biobased monocomponent and bicomponent reinforced fibres and the creation of microfibrillar structures monitored by machine vision (AI analysis) and spectroscopic techniques to ensure the quality, 2) free forming of thermoplastic organosheets layers allows a selective reinforcing for customised configuration.
- In the **manufacturing process for one-shot large volume thermoplastic production**, (packaging) a lighter garbage container lid will be developed thanks to: 1) introducing a self-reinforced core structure (organosheet intermediate monitored via spectral temperature camera) and making an overinjection to provide selective reinforcing and a protective layer for environmental harsh; 2) spraying a bonding-debonding adhesive on the organosheet surface prior injection to create an interphase which will favour materials separation and recycling under controllable stimuli (de-manufacturing); 3) a new CADMOULD simulation module to enable designing products under this biobased pre-fabricated concept.

Develop sustainable business models for materials sourcing and recycling

A circular business model is a means of implementing one or more circular goals in an economically feasible way. Indeed, shifting to a circular way of producing and consuming requires fundamental changes in the way products and services are conceived; new types of relationships between producers, retailers, consumers and service providers; alternative notions of material and product value and associated revenue models. Bio-Uptake will develop one circular sustainable business model for the European manufacturing sector. This model is built on three basic pillars:

1. Agility by means of manufacturing with intermediate formats. This is based on the concept of **pre-fabricated** parts (coming from the construction sector).
2. **Decentralized** Manufacturing (boosting start-ups and SMEs with expertise in a specific process) to produce with flexibility and faster, using **local** resources and labour workforce and costs from location area (closer to final customers and permitting more interaction and co-creation).
3. **Manufacturing-as-a-Service** (MaaS)[54] manufacturing thanks to the implementation of the process digital twin.
4. Boosting new **Value Capture revenues due to product circularity** opportunities to ensure that the value remains in the value chain: the product comes back to industry for being de-manufactured, reused, recycled... as regenerated material (permitting the creation of jobs & growth by launching new industries and/or opening business lines for manufacturers)

[50] Life cycle assessment of CFGF. By PwC – Sustainable Performance and Strategy (Report for GlassFibreEurope) 2016
[51] Master´s Thesis – Mid Sweden University. Environmental Impact Analysis of Flax Fibre Cultivation for Composite Reinforcement. Elin Jacobson, 2018
[52] N. de Beus et al (2019). carbon footprint and sustainability of different natural fibres for biocomposites and insulation material - study providing data for the automotive and insulation industry. comprehensively revised 2nd edition.
[53] A.D. La Rosa, et al (2014). Bio-based versus traditional polymer composites. A LCA perspective, Journal of Cleaner Production, Vol. 74, 135-144
[54] Pezzotta, G., et al (2017). Collaborative product-service systems engineering: Towards an active role of customers and stakeholders in value co-creation. International Conference on Engineering, Technology and Innovation (ICE/ITMC).

Business models are defined as a combination of value proposition (VP), value creation (VCr) and value capture strategies (VCp)[2]: The **VP** of Bio-Uptake is **KER4** (the Bio-Uptake manufacturing processes). This means, that the model can only be implemented if KER4 reaches exploitation (which needs to be complemented by the other KERs, specially KER1 and KER2, as explained in *Fig. 10*). The **VCr** is demonstrated within Bio-Uptake is double: first by developing **products as intermediates (KER2)** and second **end-products (KER3) in three different sectors**: construction (pre-fabricated biobased panels for modular bathroom pods), medical (feet insoles - AFO oriented) and packaging (garbage container lids with biobased and recyclable materials). Moreover, the particularisation of this model to the three manufacturing value chains will imply new **VCp** (a current gap that still exists due to the current linear model). The implication of citizens and policy makers will be key to enable new access-based modes, that will suppose new opportunities to create further value capture strategies via:
a) **material** (source) **cost reduction** either due to the use of CANs (endured thermosets and adhesives) which will avoiding factory scrap, permitting material rework, or due to the substitution of costly materials such as PEEK or Carbon Fibre;
b) revenues will come from **selling the product** itself and,
c) new opportunities may arise by working in the **access-based model** of each sector and end-user product. Specific circular access-based proposals for Bio-Uptake KER3 products are: i) after an in-house renovation, reforms´ SMEs and waste managers will ensure that pre-fabricated composite panels come back to the re-manufacturing site; ii) providing incentives to end-users when getting rid of the insoles which can be handled through the orthopaedic shops (physical or virtual) were are redirected to a waste/de-manufacturer manager; and iii) making some rental of the containers to institutions (councils, hospitals, residence-communities, shopping malls…) so the return of the product is guaranteed to the de-manufacturer.
For the three products, whenever reusing and remanufacturing (including industrial symbiosis) is not feasible/possible due too critical damage, recycling alternatives will be sought with new network stakeholders (creation of network value). These recycling alternatives will be cooperative actions with current EU projects (task 5.4) based in chemical and enzymatic recycling. In this sense, both the monomer building-blocks of the polymer used and the reinforcing fibres, will be recovered so as to be re-introduced by chemical companies into the manufacturing value chains as biobased-recycled material sources.

A successful shift to a circular economy could help meet the Sustainable Development Goals, the Paris Climate Agreement targets, the ambitions of the European Green Deal and reach carbon neutrality in Europe by 2050.
Bio-Uptake contribution to **wider impacts** specified in the destination: ***Global leadership in clean and climate-neutral industrial value chains, circular economy and climate-neutral digital systems and infrastructures (networks, data centres),*** through innovative production and manufacturing processes and their digitisation, new business models, sustainable-by-design advanced materials and technologies enabling the switch to decarbonisation in all major emitting industrial sectors, including green digital technologies. In addition to the explanations previously given to the specific project outcomes, and according to Destination 4 wider impacts, it should be highlighted:

Bio-Uptake human dimension: Bio-Uptake has conceived a training plan (Task 1.4) with defined impact indicators to train young researchers (including at RIA stage also engineers) and industrial workforce with the skills required to set up the new processes and models, in which digitalisation and low-carbon circular economy principles are embraced (technological change + master digital components). Moreover, the training model applied in the project is based on the "learning-by-doing" method, with the aim of providing sufficient levels of flexibility and learning autonomy, preparing for a more resilient labour jobs (preparing future workers for a short-term and a long-term learning (proactive attitudes, flexibility, teamwork and communication skills). Gamification will be also applied so as to create workers/scientist engagement, behaviour changes and create culture (in favour of new attitudes at work). This approach aims at making more attractive and safer the future jobs for humans.

Bio-Uptake business dimension: Bio-Uptake project will act as an example to introduce circular practices and new approaches and technical models which will be needed in European future factories. These have been designed following the Made in Europe Partnership SIRA[3] Key technologies and enablers, in order to motivate a change from a mass production model to a mass customisation model. Bio-Uptake processes and sustainable models have been designed to set up the future decentralised, cleaner and greener, more flexible and smaller factories of the future. Thanks to the modularity concept of the disruptive technologies, Bio-Uptake processes will be easily to robotise (advanced robotics). Bio-Uptake project has been designed so provide a global solution that answers the 4 R&I specific objectives of the manufacturing strategic agenda: *1) Excellence, Responsive & Smart Factories & supply chains (WP1,WP5); 2) Circular Products & Climate Neutral Manufacturing (WP2,WP3,WP4), 3) New integrated business, product-service and production approaches; new use models (WP6), and 4) Human-centered & human-driven manufacturing innovation (WP1).* In addition, this project structure is linked to the following Sustainable Development Goals: *SDG4-Quality Education, SDG8-Decent work and economic growth, SDG9-Industry, Innovation and Infrastructure, SDG12-Responsible Production and Consumption, SDG13-Climate action.*

Bio-Uptake serving as cradle for the New European Bauhaus [55]: Bio-Uptake defined conformal processes allows the manufacturing not only of final products for construction applications but also for other products such as tertiary packaging or even urban furniture, in a greener and more customised way (mimetising the particular needs of a designed urban environment, opening the designs to be more artistically-oriented and ultimately *"shaping more beautiful, sustainable and inclusive forms of living together"*). In the particular demo case of the garbage lid, a message to society is sent: "your waste is a valuable resource",

[55] https://europa.eu/new-european-bauhaus/index_en

where impact on society behaviour is sought by manufacturing the garbage container lids with biobased and recyclable materials (VCr), making this product, greener, lighter, ergonomic, aesthetically to integrate itself in the urban environment.

Bio-Uptake considerations on the "Do Not Significant Harm Principles": According to the latest report of the technical working group of the Platform on Sustainable Finance (PSF)[56], manufacturing, waste management and construction (sectors covered within Bio-Uptake project), appears on the operational list of economic activities to steer green investments that will make a substantial contribution to a/some of the 6 environmental objectives. Bio-Uptake project is considered taxonomy-aligned to:

Table 6: Bio-Uptake contribution to DNSH Objectives

DNSH Objectives	Bio-Uptake contribution
OB4: the transition to a circular economy	One of the major threats to our ecosystem is the large increase in the accumulation of plastic waste. Thanks to the CANs innovation and the de-centralised manufacturing concept based on intermediate formats, Bio-Uptake will allow the remanufacture and recyclability of thermoset materials (which nowadays ends at landfill or burnt) and boosts the circularity and recyclability of biobased thermoplastics. Bio-Uptake will lead to new and regenerated products that will boost the emergence of a disruptive value chain that can be perfectly integrated as part of local systemic solutions to augment the circular economy and also the industrial symbiosis across different sectors: e.g reuse and recycle PLA, recover caprolactone monomers from PCL and bioPA, reshape and remanufacture bioepoxy thermosets etc. Bio-Uptake will cooperate with other projects providing solutions at different stages of the value chain with regard to the circular economy (e.g. BIZENTE, VIBES, DECOAT, MULTICYCLE…). It will also deploy the usage of high-value applications for sustainable materials such as lignin which has currently no application - *with 2% burned as low value fuel*
OB3: the sustainable use and protection of water and marine resources	It is not expected either to generate significant inefficiencies or pollution to air, water or soil and the project directly or indirectly. LCA and safety validation assessment (WP1) to evaluate the impact will be performed and in case of any inefficiency or pollutant is found, propose alternatives to minimise its impact. Overall, we expect to lower emissions, pollution and energy consumption utilising advanced low energy manufacturing techniques developed in previous EU projects (e.g. LIBRE). If any unexpected pollutant particle appears in the project results, corrective measures will be proposed. The re-manufacturability and reshapeability of plastic composites will open an important second economy activity for a multitude of products whilst providing extra value to the materials. This will avoid negative practices such as sunk & abandoned vessels or goods, reducing also the amount of plastics and fibres in the oceans, waterways and soil. The project will in fact avoid the incineration or landfill of valuable bioplastics (zero pollution, zero landfill).
OB5: pollution prevention and control	

Bio-Uptake makes a substantial contribution to those objectives while there is not foreseen any intentional significant harm to any of other objectives. Environmental and safety activities gathered in WP1 will monitor the developed activities and in case of any significant harm appears as potentially affecting any of the objectives a mitigation measure and potential contingency activities will be defined.

- Requirements, potential barriers and mitigating measures

The dynamics necessary to transform current business models into circular ones, involves certain requirements such as innovation of business models, technology or social practice, as well as enabling factors such as policy measures and education. Apart from identified technical barriers (which are specific for the manufacturing processes selected), there exists other non-technical systemic pre-conditions and barriers that need to be addressed in favour of a large-scale implementation of new circular business models. The main identified ones are:

***Technical barriers**:* Bio-Uptake project needs to overcome multi-faceted technological barriers, which are basically related to the manufacturing of biobased materials (those which are not commercial with higher risk) and with the Digital twinning and fabrication. These are related to the *speed, stability and robustness* of the new green and smart conformal manufacturing processes (between different series or batches) that are adopting CAN innovative materials (basically 2 in the thermoset process & 1 in the packaging process). The concept of decentralised factory based on intermediate products (prefabricated) will help to overcome this barrier, as process can be divided and improved easier without affecting others (localised).
Social & Cultural barriers - Resistance to change and lack of awareness and sense of urgency: Powerful stakeholders have vested interests in keeping production processes linear, such as waste incineration businesses. Circular Economy requires a **systemic perspective** and **high cooperation among actors** at all levels of the manufacturing and de-manufacturing supply chain and **a systemic change needs to be gradual** to allow such businesses to recoup their investments. For consumers, buying recycled or bio-based products can be seen as assuming unnecessary risks than buying fossil-based products. For businesses, limited knowledge about the circular economy leads to inertia and aversion to taking risks that disrupt current processes. And while nearly all European regions governments have matured urban waste systems based on generic collection systems, it can be challenging to **focus more on quite selective fractions** and specific selective. *WP1 (training) and WP6 (dissemination and communication) are a combined approach to provide skills to the stakeholders to better understand and work in Circular Economy and to allow motivated communities to reach a higher audience.* In case of industrial

[56] Platform on Sustainable Finance: Technical Working Group. Taxonomy pack for feedback - August 2021

symbiosis, where waste generated by an industrial ecosystem can be used as source of materials and energy for other, there is also the need for cooperation among different supply chains and avoid uncertainties about the data generated by providing data security. Establishing **trustful conditions among all supply chain actors** is indeed one of the major barriers. *WP1 (LCA, LCC assessments) and WP4 (digital twin, building a virtual factory) will develop multilateral information systems to manage the whole product life-cycle from supplier to end-user, while adding value and maximizing the resource utilization.*
Legislation barriers constitutes an additional barrier to the implementation of Circular Economy businesses. Policies, rules and regulations were likewise designed with linear processes in mind and may restrict innovation in circularity. This issue affects to many of the current regulations, directives and standards; and activities related to de-manufacturing and remanufacturing are still legally undefined in most countries, even though ongoing initiatives are addressing end-of-waste regulation (at EU level Bio-Uptake will benefit from the Circular Economy Package: new circular economy action plan, Directives 850/2018 and 851/2018…). *In WP1, UNE will surveillance current standards, helping the project in contributing to new ones. In addition, WP6 targets Policy Makers too, and ensures actions to update and provide feedback with them.*
High Economic costs: Transitioning to a Circular economy incurs costs to businesses in terms of management, planning, research and development, physical and digital infrastructure. While government can provide incentives or subsidies, these costs nonetheless bring uncertainty and risk. New manufacturing plants will be initially supported by dedicated policies, but they must be gradually turned into **self-sustaining** sites by adopting appropriate business models. *Sustainable business models have been foreseen and will be further elaborated within WP5/6 in Bio-Uptake.*

- Scale and Significance of Bio-Uptake´s contributions → KPIs

Table 7: Scale and Significance of Bio-Uptake´s contribution

Contribution	Scale & Significance	Justification
OUT1: Relevant Production	*Scale*: market share 34% (SOM - construction); 28% (SOM - medical), 38% (SOM – packaging); *Significance:* thermoset thermoforming: 30% faster, 17% cheaper; free-forming thermoplastics: 34% more efficient, 12% lower cost, 50% shorter process; over-injection: 22% cheaper.	As justified in excellence section and OUT1: Due to the new prefabricated – modular concept using the flexible intermediates and conformal technologies, a reduction in CAPEX will be achieved. Also, in OPEX due to saving in materials, zero waste achievements, reusing & recyclable rates. Digitalisation permits a more accurate control of the overall process and of the product quality.
OUT2:Similar properties, lower Env. Foot.	*Scale:* biobased content of end-products: sandwich panel (87%); feet insoles (75%); garbage lid (99%); *Significance:* average CO_2 reduction [12%-33%]	As justified in excellence and OUT2: Due to the type of materials used (and feedstocks – EU local), the efficiency of the manufacturing processes, the reduction in chemicals (including end-of-life due to re-manufacturability).
OUT3: Disruptive innovation in 3MVC	*Scale:* Increment the uptake of biobased products by 39%. *(Average of SOM x %biobased x MF). Significance:* Increase portfolio of biobased products in at least 3 new products per targeted sector and at least 1 new product in 3 other new sectors.	Thanks to the incorporation of these technologies (KER4) to the market a **multiplier factor**[57] (MF≈1.3) is rapidly achieved as not further CAPEX are required (none leverage factor = need of new investments), what permits a quick adaptability for **new products and markets**. Thanks to the intermediate formats, re-design & MaaS production will be straightforward.
OUT4: Sustainable Business Models	*Scale:* 1 new sustainable business model; *Significance:* 3 direct business cases	As explained in OUT4, this model is based on pre-fabricated intermediate products & de-centralised manufacturing, permits flexible processes adapted to territorial industry & society needs. It is more competitive than rigid solutions, being easier to adapt & replicate with minimum expenses.
WIDER IMPACTS	*Scale:* 24 new direct jobs derived from the implementation of the project (partners) and about 1000 indirect jobs (short term) & 1M jobs at long term. *Significance:* stable and versatile jobs (adaptable)	This manufacturing model based on flexible processes, digital jobs, allows to keep the talent within EU. De-localized manufacturing permits also delocalised workers, more home-office, and more versatile jobs, that can be realised by disadvantaged and vulnerable social groups in rural and territorial areas and that permits a fully gender equality. Long-term job creation is estimated to reach 3% of the total EU manufacturing industry (30M jobs)[58].

[57] Schanz, H, et al. (2019). Markets as leverage points for transformations of economic systems: The example of the German bioeconomy. Environmental Innovation and Societal Transitions.
[58] Destination 4 – Climate Neutral, Circular and Digitised Production. European Commission (2021)

2.2. Measures to maximise impact – Dissemination, exploitation and communication

- Plan for the dissemination and exploitation including communication activities

The **Plan of exploitation and dissemination of results (PEDR),** will be designed in order to fully achieve the expected impact of the Bio-Uptake project and to prepare the transition towards industrial and commercial uptake. The PEDR will describe the strategy and the activities to be undertaken (how and by whom) in order to ensure the exploitation beyond the project itself. The exploitation strategy will reflect and will be built-up as a result of sound analysis of the **market trends, potential users, and financial sustainability**. In order to optimize dissemination and exploitation of the results, one specific WP have been described: **WP6 - Exploitation, dissemination and communication**, including activities aimed at engaging stakeholders, business strategy and results deployment, showing to our envisaged target audience (*Table 8*) the benefits of the project and therefore facilitate the transfer of experience and knowledge needed to replicate the Bio-Uptake approach. This strategy is also completed with **WP1 - Manufacturing plan, sustainability assessment and good practices distillation**, in particular with Task 1.3 – Standardization and Task 1.4 – Training (which have to be taken under consideration for exploitation and dissemination purposes too). The communication, dissemination and exploitation strategy will be elaborated at the very beginning of the project (M6) in D6.1 and updated periodically during the project.

This PEDR plan will rely on three pillars: exploitation, dissemination and communication therefore it will be orchestrated by the Exploitation leader (AIT) and the Dissemination leader (LIM) in cooperation with the Communication leader (POL) with the collaboration of all partners. It will be defined under T6.1 and as a "lived" plan, will be updated as project results arises (*Fig. 9*) and the best strategy to maximise its impact is decided and executed as define in the following tables (*Table 9* - Communication &*Table 10* - Dissemination).

The **business model** of Bio-Uptake is conceived on the exploitation of the newly developed manufacturing processes (KER 4), these industrial processes permit the transformation of new materials (KER 1) and production of pre-fabricated intermediate products (KER 2) which can also be exploited individually outside the manufacturing process. Other related services such as software development (KER 5) and training & consultancy services (KER 6) cannot be exploited individually but are services that provided added value to the exploitation of KER 4 (additional packages). Finally, the combination of these KERs as a commercial offer will mean the possibility of manufacturing new final biobased industrial products (KER 3).

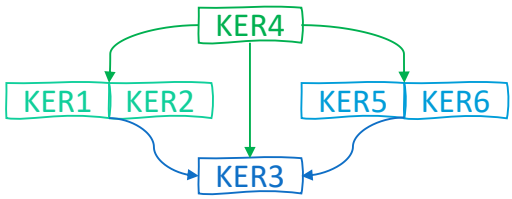


Fig. 10: Bio-Uptake KERs relation & prioritization for exploitation

Table 8: Bio-Uptake target audience and alignment to potential KERs of interest

Target Audience	Examples of Stakeholders	KER
(1) Policy makers (EU, National, local level)	European decision-makers, Governmental, Environmental EU agencies, The European Bioeconomy Panel, Directorate-General Enterprise and Industries (DG ENTR), Environment Directorate-General (DG ENV), Regional and Local organizations for implementing bioeconomy, Regional and Local organizations for implementing circular economy…Public authorities at local, regional, national and EU levels (from partners countries and regions to EU replication level)	4,6
(2) Working Groups of influence, platforms, partnerships & Standardisation bodies	Sectorial groups and sectorial associations – Made in Europe, Circular Bio-based Europe (CBE – previous BBI), Vanguard Initiative, Processes4Planet (SPIRE), CleandSky, Shift-to-Rail, EGVI, European-Biotechnology Network, European Technology Platforms e.g. SusChem, EuropaBio,… European Innovation Partnerships - EIPs: EIP Raw Materials & EIP Agri, EITs (Manufacturing and Digital KIC, Raw Materials and Climate), Digital Europe Programme – DEP, Network of National Contact Points (NCPs) for the Climate & Environment Programme; REACH, AMFEP, CEN, ISO, SETAC, AETOX, TÜV, ISC.	6
(3) Industrial Stakeholders)	Industries (Large industries, SMEs, start-ups and industrial clusters) along the entire Value Chain related to: (a) Biobased Materials producers; (b) Technology Uptakers; (c) Plastic Product developers (for same: medical, construction & packaging; or others for replication: automotive, electronics, sports, naval, railway, leisure, furniture, etc >; (d) Recyclers & Waste Managers	1,2, 3,4, 5,6
(4) Research & Scientific communities	Universities, Research Institutes, PhD candidates, Technological centres…at EU level. Scientific forum organizing conferences of interest for the project such as: FMF (Applied polymer research), European Bioplastics, European Biomass Industry association, Different chemical societies from Europe (RSC, GDCh…) and outside (ACS…)	6
(5) Investors and financial actors	Horizon Results Platform (HRP), European Bank for Reconstruction and Development (EBRD), the European Circular Bioeconomy Fund (ECBF), European Investment Bank (EIB)	4
(6) Wider audience (public at large)	NGOs, Consumer's Associations… European citizens, customers, civil organisations	4

Table 9: Communication measures. These have been designed to inform and reach specific target groups/audience at different levels and show the activities performed, project outputs and the benefits these will have for the European society. The communication activities pave the way for future exploitation activities by engaging interested target groups.

Tool/Channel	Audience	Objective & Message	Related KPI	Target
Website / Website	General Public. **(1 - 6)**	Generate awareness on project and serve as central contact point for external queries and information request Update information about: the project (general overview); news about developments; publishable activities and results; list of relevant events, meetings and conferences; meet consortium partners & provide contacts.	Nº website users	3,500
			≈ session duration	00:01:00
			Countries reached	35
Social Media / *Twitter & Youtube*	**(6) (1)**	Building a community that shares the same interests in bioeconomy, circular economy, manufacturing technologies…Increasing visibility to stakeholders active in social media, raising awareness and redirecting to website, articles published, related news or debate forum. Multiplier effect & motivate citizens to take action. Inform about the labour of R&D EU cooperative research. Videos of the project (also short pills), topics relevant and related to the project in terms of novelties in legislation, standards regarding circular and plastics bioeconomy, twinning/connections with other project, news about live-ongoing events, spots of intermediates and products, links to project results and publications…	Nº of followers	2,000
LinkedIn	**(2) (3) (5)**		Nº of posts	450
ResearchGate	**(4)**		Nº of reactions	1,000
Press Kit released: TV, Radio, Newspapers & Magazines*	To general public, particular emphasis to **(6) (3)***	Bio-Uptake benefits known to potential end-users and consumers of biobased products. Catch-up attention on circular bioeconomy, influence on society habits and show how close are biomaterials in their daily routines. Press releases will be covered at EU in English and at national and local level in consortium partner´s local language. *Few Examples: TV: futuris-euronews, Horizon Results Platform (HRP)-TV, TVE,..), AragónTV, ETB, Radio: France Bleu, DR, Radio Sysvast… Newspapers: Cordis, Heraldo de Aragón, L´Echo…; Industry Magazines: Chemical & Engineering news, European Plastics News, Composite World Magazine, JEC Magazine, Eco de l'Ain, Empresas XXI ……*	Nº of journalism impacts	80
			Nº of press kit distributed	30
			Participation in TV/Radio	10
Newsletters / emailing	**(2) (3) (4)**	Updates on project developments, promotion of reached milestones and give visibility to the participation in relevant events. Communication in and understandable manner to the project's subscribers. Catch attention of industry, platforms, clusters to have a direct multiplier effect to other sectors and boost replication (potential adopters). Inform scientific community about research lines carried out in project institutions (to promote collaboration and career positions).	Nº of subscribers	200
			Nº Newsletters	8
Promotional materials for social media (e.g Youtube); exhibitions, open-days, congresses summits, infodays, Brokerage events, Fairs, webinars…	General Public. **(1 - 6)**	Communication of the project's activities and Unique Selling Proposition in a captivating and engaging manner. Illustrate the work. 2 general **videos** (M4-framing the problem, solution and presenting the consortium & M40 showing the results – project wrap-up); other intermediate videos-pills explaining specific technical actions e.g. digitalization for interoperability, benefits of CANs, eco-design with biotextiles? Raise citizens' about cooperative R&D projects.	Nº of videos	5
			Nº of views	700
	(2) (3) (4)	Catch up Public attention and interest in events, inviting public to come across the stands and ask partner representant: **1 infographic (poster/roll up)** explaining the Bio-Uptake concept & solutions. **2 project leaflets**: presenting the project's main elements (beginning) and another summarising project achievement at the mid-end	Nº of events (in a one way comm.)	30
			Nº people asking for information	500
	Adapted to **(1-7)** audience	**Project general and specific presentations** (PPT/Prezi). To spread the word in Bioeconomy, Bio-Uptake products and Circular Economy practices. Impact in end-user consumption habits to boost biomaterials´ demand. Create a debate. Bring bioeconomy closer to daily routines (close examples). *Examples of events: The European Biopolymer Summit, ECOMONDO, K-Trade-Fair, Fakuma Show, Annual JEC World Fair, ACHEMA 2021, Composites for Europe, Expoquimia, The Greener Manufacturing Show, TECMA,* Participation in webinars and events organised by CSA, clusters and PPP: pitching, matchmaking, networking. Starting point to collaborate with other EU initiatives. Receive feedback and engage in a dialogue. Open a door to share results and avoid duplicate mistakes. Multiplier effect to reach other sectors for replication purposes. *Examples: Infoday CBE, BIOECONOMY VENTURES, VIBES, POLYNSPIRE, MANDALA, …*	Nº of events presented (in a two-way comm)	20
			Nº of initiatives liaised with	15

Dissemination activities within the project are conceived to have an impact in **(4)** audience group. However other target audiences such as **(2), (1)** and **(3)** with multidisciplinary teams, are also considered.

Table 10: Dissemination activities

Tool & Examples of identified Channels	Objective	KPI	Target
Scientific publications in: • **Identified Journals**: Polym. Sci., J. Reinf. Plast. Compos., J. Mater. Process. Technol., J. Adv. Manuf. Syst., J. Manuf. Mater., Int. J. Precis. Eng.Manuf., J. Intell.Manuf., Digit. Disc. (RSC), Adv.Model.Simul.Eng. Sci • **Identified scientific congress/conference:** Composites Recycling Conference–JEC; BIOPOL International Conference on Biodegradable Polymers & Sustainable Composites, COMPOSIFORUM Composite forum for industry and researchers, Composites Europe, MECHCOMP International Conference on Mechanics of Composites, …	Scientific peer validation of project's approach and findings. Sharing of knowledge at European & International level. Put into practice Open Science & teach Young Researchers (oriented to STEM and environmental multidisciplinary researchers)	Nº papers published	8
		Organization of congress	1
		Oral Contribution	20
		Poster Contribution	8
Scientific internal transfer of knowledge: • **Research and innovation programmes**: Tutoring PDR & PhD by senior IPs & researchers to start project's innovation lines. • **Training transfer:** Demo Workshops on pilot lines, classrooms and short conceptual lessons. Promote *Learning-by-doing*. Inhouse & Open training sessions on manufacturing technologies: experimental and digital uses.	Creating talent, integrating multidisciplinary profiles Training researchers and workforce in circular bioeconomy & digitalisation	Nº research lines initiated	5
		Nº young researchers	20
		Nºworkshop	3
		Nº trained workers	40
Standardization Activities: Channel: UNE. UNE will introduce the project to the appropriate standardisation groups, organising working sessions… *To start during project implementation and **to continue after the end of the project** (Influencing biobased materials & processing technologies market)*	Contribute to & complement current standards as technical specifications-*TS* or CEN Workshop Agreements-*CWA*	sessions with standariz. groups	3
		Nº TS/CWA	3
Scientific influence on Policy Makers – Preparation of Public final Layman Report. *To start during the project implementation and **to continue after the end of the project** (contribute to Pathways to sustainable industries[59] and the 4th Industrial Revolution - digitalization & human dimension[60])*	Become an Important Project of Common European Interest (IPCEI) & provide feedback to Projects for Policy P4P initiative	Nº distributed Layman Reports	200

** In favour of boosting Open Science practices, from envisaged scientific journals, books… Bio-Uptake will promote scientific publications in open access journals were open peer reviews and pre-pints modes are in use: Elsevier, MDPI, F1000Research, Sci-Post, PeerJ, etc. Some, such as Elsevier and MDPI journals also permits the inclusion of almetrics. Examples taken in Table 10, belongs mainly to these platforms.*

Strategy for the management of intellectual property

Bio-Uptake will continuously contribute to generate new knowledge that will be instrumental for shaping the expected project outputs, several of which may qualify for **Intellectual Property (IP) protection**. The consortium will comply with the obligation to disseminate results as well as open access rules and obligations, whilst safeguarding at the same time the rights of the consortium partners to protect their IP thus **enhancing the chances of an effective commercial exploitation of the project results**.

Based on the **background** of the consortium and once beneficiaries have agreed about the **access rights** on a royalty-free basis, (which will be identified and agreed in the Consortium Agreement), Bio-Uptake will define a policy for the protection of the project's results whenever are expected to become commercially exploitable. An internal **IP protection plan will be elaborated within WP6** (T6.1 & T6.2). Partners will continuously evaluate the potential contribution to new IP generation, whenever project outputs are suspected to be commercially exploitable and protecting them is possible, reasonable and justified. IP protection measures (such as patents, copyrights, trademarks, trade-secrets confidentiality, and other forms of protection) shall follow the Grant and Consortium Agreements. Any issue that might arise from the IP protection initiative during the project will be dealt with **by the General Assembly and coordinated by the exploitation leader** (AIT). For protectable IP, the following steps will be taken as provisional IP protection **strategy**:

1. The concerned partner **notifies** the Exploitation leader about the technical **content** of any protectable IP. The **ownership** rights will be defined: i) Bio-Uptake results are owned by the beneficiary that generates them; ii) whenever results have been jointly generated by two or more beneficiaries, the joint owners must agree on the allocation & terms (joint ownership agreement).

[59] PATHWAYS TO SUSTAINABLE INDUSTRIES – Energy efficiency and CO2 utilisation. Directorate-General for Research and Innovation, 2018.
[60] CAPITALISING ON THE BENEFITS OF THE 4TH INDUSTRIAL REVOLUTION - Directorate-General for Research and Innovation, 2018.

2. Preliminary copyright, trademark, or patent **analysis** are conducted by the interested partner supported by the Exploitation leader **to check "freedom-to-operate**".

3. Filing of the related IP protection application is then followed directly by the relevant partner(s), in accordance with the perimeter agreed with the other partners and the perimeter of innovation to maximize the potential for protection of the result.

Once IP has been managed according to Bio-Uptake strategy and the targeted KER is clearly stated, consortium partners will put efforts in reaching target audience (investors **(5)**, Policy makers **(1)**, Stakeholders, business partners & customers **(2) (3)**) through its conventional exploitation channels. In addition, the exploitation leader (AIT) will seek for further booster tools to maximise output´s impact, such as request guidance to the Horizon Results Booster and generate (with the support of KER owners) a KER profile to be published in the Horizon Results Platform (HRP). In this sense, a combination of partners own resources plus potential **capital from investors** and/or cooperation with other business third parties, may **speed up the upgrade in TRL** (from 6 to 9) and reach before market opportunities. In addition, **policy makers** may pave the way for a quicker up-take of the project KERs by amending or launching new legislation within the Circular Action Plan and The European Green Deal policies: *Biobased, biodegradable and compostable plastics, global action on plastics, plastics packaging, microplastics…* **creating demand** on biobased circular products. **Industrial partners** (POD, CON, MOS) will put efforts in looking for potential clients and business parties interested both in the biobased end-products an also in replication purposes: enlarging business in new products for the same sector and for other sectors. Industrial partners (MOS, SIM, IRI) and **technological partners** (RTOs) will look for exploiting the manufacturing technologies themselves: complete process including operational prototypes & optimised equipment's (multiplier effect). The first strategy will allow industrial partners to **grow** in production volume and turnover (one-off revenue model). In the second strategy the **creation of spin-offs and joint-ventures** will be promoted as well as the creation of other incomes return based on **license**-royalties payment (combined revenues).

Table 11: Envisaged Ownership of KERs, Protection measures and Exploitation routes

KER	Ownership	Protection & Exploitatio
KER 4 Manufacturing Processes: *- Reprocessable Thermosets - RT* *- Highly Customised Thermoplastics - HCT* *- Large Volume Thermoplastics - LVT*	AIT, CTB, CID, SP, AIM, IRI, SIM, POD, COM, MOS, LIM,	In T6.2 the potential to protect these KERs will be envisaged. Initially, patentability will be sought with particular interest for KER4, KER1 & KER2. Whether a patent cannot be applied; the Utility Model will be studied too. Otherwise, Industrial Secret will be the best option. This protection methodology will be evaluated too for the rest of KERs, including Copyright & Trademarks options. Direct Selling – KER2 & KER3 (industrial partners directly exploit KER4 via an *Exploitation Agreement*) Service Selling – KER6 & KER5 (training, LCA service etc) Licence Selling – KER1, KER2, KER4, KER5 (software, materials formulae, products, processes)
KER 1 Bio-Uptake New Biobased and Regenerated materials (formulae)	SP (bioepoxy + biobased dynamic hardener, biobased 3R thermoset formulation +biobased and reversible adhesive formulation) LIM (bioCF & lignin derivates) CID (3R-CANs for thermosets)	
KER 2 Intermediate formats	CTB+CON (thermoplastic organosheets and UDTapes) SP+CID (thermoset organosheet) AIT, LIM Pellets formulae	
KER5 Software – DSS & environmental	SIM, AIM, IRI, LIM, AIT	
KER6 Training and consultancy	UNE, AIT, AIM, CID, LIM, CTB, POL, SIM	
KER3 Final biobased industrial products	POD, CON, MOS	

2.3. Summary Table 12: Key elements of the impact section

SPECIFIC NEEDS

- Making sustainable products the norm in EU
- Building a sustainable manufacturing industry
- Biobased & Circular by design products
- Reducing environmental footprint of waste streams
- Strengthen European manufacturing industry
- Leader in MaaS (pre-fabrication, customization, de-centralization factories)
- Adapt/contribute standards to manufacture with biobased materials
- Modify workers attitude at job (autonomous, attractive, safer)
- Multidisciplinary teams & Skilled scientists & workforce (in biobased-materials manufacturing and digitalization)

EXPECTED RESULTS

KER 4 Manufacturing Processes

*- Reprocessable Thermosets - **RT***
*- Highly Customised Thermoplastics - **HCT***
*- Large Volume Thermoplastics - **LVT***

- KER 1 New Biobased & Regenerated materials
- KER 2 **Intermediate** formats
- KER6 Training & consultancy
- KER3 Final biobased industrial **products**
- KER5 Software – DSS & environmental

D & E & C MEASURES

- Plan for the Exploitation, Dissemination and Communication
- IPRs and exploitation strategy: business potential

8 Papers
20 Conferences
20 Y. researchers, 40 workers

- Open Science practices
- Scientific Publications
- Training programmes for upskill scientists &workers
- Scientific contribution to standards & influence on Policy Makers

- Communication tools: website, social media, press kit, newsletter & promotional materials

2000 f_w
450 p_t
1000 r_e
8 N_{wL}

TARGET GROUPS

- Policy makers (EU, National, local level)
- Industrial Stakeholders)

MF ≈ 1.3

200 L_yR_p

- Investors and financial actors
- Research & Scientific communities

35 Countries
- Wider audience (public at large)
- Working Groups of influence, platforms, partnerships & Standardization bodies

3 working sessions & 3 TS/CWA

OUTCOMES

- Intermediate formats: functional behaviour
- Biobased products by 39% in EU
- Lower CO_2 emissions [12%-33%]

Disruptive innovations :

RT: bioepoxy endured pre-impregnated organosheets, semi-automated process (thermoforming+TSIM simulation+inline resin monitorization)

HCT: self-reinforced fibres, machine visión, spectroscopic techniques, free forming, customisation & TSIM simulation

LVT: self-reinforced core, over-injection & bonding-debonding adhesive, CADMOULD simulation.

- **New circular sustainable business model** for the EU manufacturing & **3 business cases**

IMPACTS

- Create stable & versatile jobs: 24 direct + 1000 indirect in the short term & 1M jobs long term
- From mass production model to mass customisation model: decentralized, cleaner & greener, more flexible, smaller factories
- Products serving as cradle for the New European Bauhaus
- Contributions to "Do not significant Harm Principles" OB4, OB3, OB5

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

3.1. Work plan and resources

Bio-Uptake aims to accelerate the market introduction of sustainable products by means of developing new bio-composite formats with similar or better properties when compared to non-biobased materials and demonstrating the green manufacturing of sustainable products in the construction, medical and packaging value chains. To do this, **WP1** will define the testing protocols, design the pilot lines and adapt the assessment methods to monitor properly the impact of the biobased materials, boosting the standardisation recommendations and preparing the workforce to implement biomaterial-based manufacturing activities. **WP2** will deal with the development of the new bio composite intermediate formats based on thermoplastic and thermoset matrix respectively, while improving and adapting the in-line characterisation methods to the bio-based material requirements. **WP2** will also explore self-reinforcement and dynamic chemistry technologies to allow the combination of different formats in the same manufacturing process and their subsequent separation for boosting circularity. **WP3** will tackle the main factors affecting the current manufacturing models by testing the new bio-based formats in free-forming, over-injection and thermoforming conformal manufacturing processes as well as to adapt and enhance the quality control for the new bio-based end-products. In parallel, **WP4** will enable the bio-based product design and quality control as well as the process optimisation by preparing the bio-based manufacturing value chains for an effective digitalisation.

The data generated during the bio-based format characterisation, the demonstrations and the process parameterization will be used to improve the models of SoA simulation software. The smart systems will be integrated and the pilot lines will be optimised in **WP5** to demonstrate the enhanced mechanical, physical and/or chemical properties of the new bio-composite formats as well as their circular functionalities for three uses cases: i) ceilings for modular baths, ii) orthopaedics feet insoles and iii) lids for tertiary packaging - garbage collection. In addition, **WP6** will boost the fast uptake of the new bio-based material formats and bio-based products by implementing an updated and adapted plan for the exploitation, dissemination and communication of the results. Finally, the activities to be developed during the project execution will be supported by coherent and smooth coordination as well as data management for interoperability following FAIR principles and ethics, which is articulated under **WP7**.

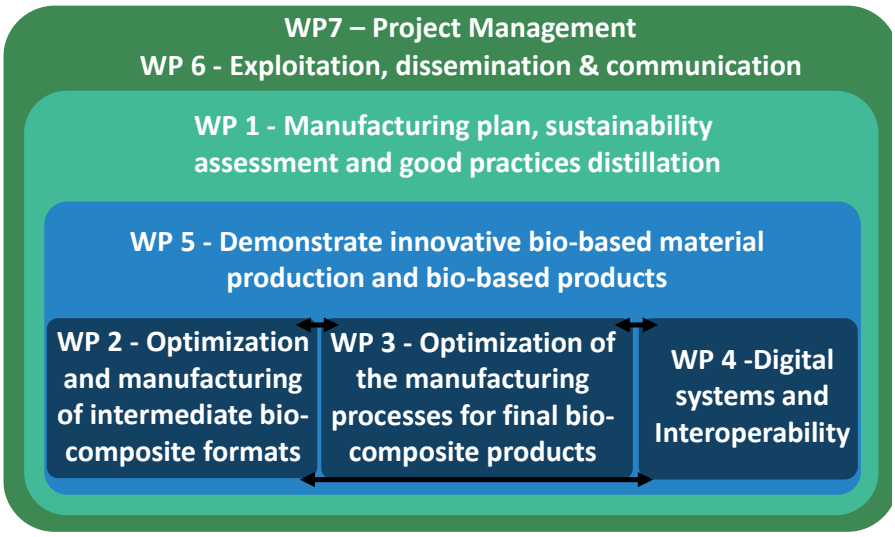


Fig. 11: Bio-Uptake PERT

Table 13: List of work packages

No	Title	No	Name	PM	Start	End
1	Manufacturing plan, sustainability assessment and good practices distillation	11	LIM	105	1	41
2	Optimization and manufacturing of intermediate bio-composite formats	2	CTB	252.5	1	30
3	Optimization of the manufacturing processes for final bio-composite products	7	SIM	133.5	13	30
4	Digital systems and Interoperability	5	AIM	85.5	16	38
5	Demonstrate innovative bio-based material production and bio-based products	1	AIT	102	30	41
6	Exploitation, dissemination & communication	8	POL	76.5	1	42
7	Project Management	1	AIT	57.5	1	42
			Total	812.5		

BIO-UPTAKE Project: 42 Months	Leader	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42
WP1 Manufacturing plan, sustainability assessment and good practices distillation	**LIM**	
T1.1 User cases specifications and pilot lines configuration	AIT	D1.1
T1.2 Sustainability and safe assessment	LIM	D1.2 D1.2 D1.2
T1.3 Standardization activities	UNE	D1.3 D1.3 D1.3
T1.4 Training for skilled workforce	LIM	D1.4 D1.4
WP2 Optimization and manufacturing of intermediate bio-composite formats	**CTB**	Ms1 Ms2 Ms3
T2.1 Extrusion of self-reinforced bio-based fibres	CTB	D2.1 D2.1
T2.2 Synthesis of bio-based dynamic chemicals	SP	D2.2 D2.2
T2.3 Preforming of thermoplastic bio-sheets	COM	
T2.4 Compounding of fibre reinforced bio-pellets	AIT	
T2.5 Production of thermoset bio-composites	CID	D2.3 D2.3
T2.6 In-line characterisation methods	IRI	D2.4
WP3 Optimization of the manufacturing processes for final bio-composite products	**SIM**	Ms4
T3.1 Process modelling	SIM	D3.1
T3.2 Free-forming of orthopaedics	COM	
T3.3 Over-injection of tertiary packaging	MOS	
T3.4 Thermoforming of bath ceillings	POD	
T3.5 In-line quality control of products	IRI	
WP4 Digital systems and Interoperability	**AIM**	Ms5
T4.1 Interoperable embedded platform	AIM	D4.1
T4.2 Data analytics for adaptive manufacturing systems	AIM	D4.2
T4.3 Data analytics for product design	SIM	
T4.4 Hybrid-digital twin	SIM	D4.3
WP5 Demonstrate innovative bio-based material production and bio-based products	**AIT**	Ms6-7
T5.1 Smart systems integration	AIM	D5.1
T5.2 Pilot line optimisation and testing	AIT	
T5.3 Validation og user cases in the medical, packaging and construction value chains	AIT	D5.2
T5.4 Sustainable end-of-life models for new circular and regenerated products	MOS	
WP6 Exploitation, dissemination and communication	**POL**	Ms8
T6.1 Exploitation, dissemination & communication plan	POL	D6.1/D6.2 D6.1 D6.1 D6.1
T6.2 Exploitation activities	AIT	D6.2 D6.2
T6.3 Dissemination activities	LIM	
T6.4 Communication and cooperation activities	POL	
T6.5 Clustering activities	AIT	
WP7 Project Management	**AIT**	
T7.1 Coordination Processes, Quality Assurance and Risk Management	AIT	D7.1
T7.2 Administrative, financial and legal management	AIT	
T7.3 Data Management Plan	AIT	D7.2 D7.2 D7.2 D7.2
T7.4 Ethics	AIT	D7.3

DX: Deliverable; MsX: Milestone

Fig. 12: Bio-Uptake Gantt

Table 14: Work package description

WP No	1		Start	1		End	41		Lead beneficiary		LIM		
Title	Manufacturing plan, sustainability assessment and good practices distillation												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	12	7	3	2	4	1	1.5	4.5	6	2.5	48	1	12.5

Objectives: The main objective of WP1 is to put in a common framework all the needs to define the manufacturing pilot lines: product specifications, designs, technologies and equipment's, staff skills, current benchmarks for sustainability and economical assessment, identify current standards etc. In this way the pilot lines for each manufacturing value chain can be set up with guarantees. In WP1, standards monitoring and contribution, LCA and LCC assessments on final Bio-Uptake products and specific training actions will be carried out with the aim of facilitating biomaterials up-take and market acceptance.

T1.1 User case specifications, pilot lines configuration and manufacturing development (Leader: **AIT**, CTB, CID, SP, AIM, IRI, SIM, POD, COM, MOS, M1-M6): End Users will define the requirements that their use cases should accomplish and based on the analysis of this information, a list of functional specifications for the industrial demonstrators will be obtained. Analysis and specifications will include: i) detailed description of the current manufacturing workflows, technologies and baseline values for each use case (AS-IS analysis), ii) detailed description of the manufacturing processes and technologies to be supported during the project (TO-BE analysis), iii) detailed description of the requirements for technologies to be developed during the project and their integration in the pilot lines and iv) full validation protocols, for product, process and systems will be produced, considering the product requirements, the expected process control capacities and the overall production targets. These protocols and criteria defined in this task will be employed for the demonstration in WP5 for assessing pilot tests after integration of systems developed along the project.

T1.2 Sustainability and safe assessment (Leader: **LIM**, All, M1-M40)

Based on production data for biocomposite intermediate formats in WP2 and bioproducts demo application scenarios in WP3, this task will build value chain scenarios representing plausible bioproducts production, use and end-of life (EoL) management (e.g. remanufacture or recycling) for the 3 assessed products. These value chains will then be analysed through life cycle assessment (**LCA**) including the social perspective (**s-LCA**). System boundaries will be established for entire production-use-EoL value chains for each product, accounting for multiple remanufacture and recycling loops where relevant (impact on new and regenerated products). Equivalent system boundaries will be established for reference product value chains based on conventional materials and manufacturing processes (benchmark). In the first instance, the functional unit for attributional LCA foot-printing will be a single use of each product. Inventories of relevant processes and inputs along the value chain will be developed using detailed primary data on inputs and processes involved in the thermoplastic and thermosetting manufacturing steps. These foreground activity data will be combined with background data from the Ecoinvent v3.7.1 (or higher) database to represent, e.g. environmental burdens associated with bio-based feedstock production and electricity generation. Life cycle impact assessment will be based on the Environmental Footprint 2.0 methodology, generating indicators of damage potential across 16 environmental categories (from climate change, through abiotic resource depletion, to human toxicity). Sensitivity analyses around e.g. type of bio-feedstock used and energy sources will inform optimisation of product design and manufacturing (WP5). Insight gained from product foot-printing will be translated into a set of best practise guidelines for bio-composite product foot-printing (D5.2), accompanied by peer-reviewed publications (T6.3). Input and process inventory compilation for the LCA will be combined with estimates of e.g. labour requirements and capital investment to undertake the parallel life cycle costing (**LCC**). Finally, to more holistically quantify the scalable contribution of bio-composite material substitutions to wider circular (bio)economy and climate neutrality targets, value chain analyses will be out-scaled to deployment scenarios that will be evaluated using prospective, consequential LCA. This will identify total environmental savings, and potential trade-offs, associated with what-if deployment scenarios at European scale. New evidence generated in this task will be synthesised in a report containing recommendations for bio-composite deployment to achieve circular economy and environmental sustainability objectives (D1.2).

T1.3 Standardisation (Leader: **UNE**, All, M1-M41). The relevant standardization landscape together with applicable standards will be identified (M6). The possibility and the extent to which a contribution to standardization can be made will be investigated through interacting with the relevant technical standardization committees (monitoring their information, attending meetings, establishing formal liaisons, organizing joint events, etc.), disseminating the project and getting feedback from their members. The main outcome is the contribution of Bio-Uptake to the development of new standards in topics related to the project's objectives and based on its results (new and regenerated biobased products). Depending on the ongoing works of the standardization committees around the midpoint of the project, this may involve providing information, participating in ongoing works, submitting proposals for the modification of existing standards or promoting the elaboration of new standards like technical specifications (TS). The development of fast-track standards like CEN Workshop Agreements (CWA) is usually the best-suited option in the frame of a Horizon Europe project. Standardization is made in technical committees external to the consortium through the consensus of various stakeholders. Thus, project decisions regarding the subject, development track

and start of new standardization works are unpredictable now and shall be studied and coordinated during the project life, according to the specific standardization environment and under the advice of UNE.

T1.4 Training for skilled workforce (Leader: **LIM**, CTB, AIT, AIM, CID, POD, COM, MOS, POL, SP, IRI, M1-M41)

This task aims at making transfer of knowledge at different levels to current and future European workers and specialists in Circular Economy product development and manufacturing practices:

- From the academic point of view Bio-Uptake will **train young generations of scientists and engineers**, by training emerging and upskilling established researchers (at least 20 people in new technologies in the field of biocomposites: materials, products, processing, characterisation, digitalization). Across all WPs, approximately 10 postdoctoral researchers (PDR) will be involved in the research who will in turn help mentor 10 PhD students (attracting talent that may be uptake by the industry and public institutions as new workforce for the jobs to be created). PhD students and PDRs will **visit partner institutions and companies**, not only to develop skills required for composite manufacture, but also to **broaden and enhance their career options**. This will deliver different experiences across knowledge-driven public laboratories and customer-centric industrial environments. Details about all secondments or visits will be agreed by partners by M12.
- From the industrial point of view, **in-house training of workforce** and open training to new staff/workers will be carried out. RTOs are closer to the industry needs in terms of developing new products and technologies. An internal training programme for transfer of knowledge will be done hand-by-hand between the lead RTO and the industrial manufacturer for each value chain to provide new skills so that staff is able to evolve and adapt to new circumstances. At least 4 intensive sessions will be organised: AIT-MOS (injection moulding), CID-POD (thermoset thermoforming), CTB-COM (thermoplastic thermoforming) and a common one dedicated to digitalisation performed by AIM-SIM-IRI to the three end-user companies and interested partners of the project. In addition, at least other 4 **open training sessions** will be given by RTOs to identified stakeholders oriented to vocational training of professionals (e.g. Fundación San Valero within their *Centre for Continuing and Occupational Training)*, and also to other potential end-users that can replicate and multiply the applications of the technology. Products and end-of-life practices. In this way, Bio-Uptake will upskill engineers and technical staff across partner RTOs and industry parties to strengthening their innovation potential and competitiveness.

Deliverables:

D1.1 Bio-Uptake detailed baseline and expectations (R;SEN;M6;**AIT**)
D1.2 LCA and LCC assessment report and recommendations to achieve circularity (R;SEN; M6; M18; M40; **LIM**)
D1.3 Report on the contribution to standardization (landscape and new contributions) (R; PU; M12, M24, M41; **UNE**)
D1.4 Training Programme: intellectual outputs and session summary (R; PU; M18; M41; **LIM**)

WP No	2		Start	1		End		30		Lead beneficiary		CTB	
Title	Optimization and manufacturing of intermediate bio-composite formats												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	12	20	33.5	43	0	69	1	3	29	0	42	0	0

Objectives: The main objective of this WP2 is to research, develop and optimise the technologies and manufacturing processes required for the development of each of the intermediate biobased formats. In this way the first stage of each demo pilot plant is adapted and optimized to be operated with the selected biobased materials.

T2.1 Extrusion of reinforced bio-based fibres (Leader: **CTB**, IRI, M1-M24): Mono- and multifilaments of PLA materials, both monocomponent and bicomponent (core/sheath), will be developed using low melting grades as matrix and high melting grades as reinforcement. Besides low melting PLA grades, PCL grades will also be investigated to develop mono- and bicomponent mono- and multifilaments which can be used as matrix material in combination with high melting PLA filaments as reinforcement. An evaluation of the potential for extrusion / melt spinning will be done using a pilot scale extrusion process. The materials will be analyzed according to their processability in textile filaments extrusion (e.g. high draw ratios and extruder temperatures) by evaluating the flow behavior via structural and rheological measurements and a full characterization of the produced yarns will be performed (tensile strength, modulus, elongation at break, shrinkage triggered by heat and resulting shift in mechanical properties and crystalline structure, melting temperature). Different methods of cooling (using a cooling water bath or using a cooling drum, air cooling or a combination of both) will be investigated to cool the PCL filaments down below their melting temperature of 60°C before reaching the uptake rolls. Unidirectional composite sheets will be manufactured by compression moulding to perform a first screening and quality control of the produced filaments.

Polymer pairs showing appropriate rheological and thermal properties (characterized via capillary and rotational rheometry, MFI, DSC, MDSC, TGA and SEM) will be selected for **microfibril formation** on textile fibre extrusion lines. Filament extrusion of **fibrils** will be obtained via an extrusion process starting from a compounded or dry blend of both polymers (PLA:PCL 5:95 up to 50:50). The produced filaments will be further stretched on the extrusion line in order to induce the fibril formation. Maximizing elongation and orientation of the fibrils will be pursued. Filaments drawing without breaking the microfibrils or disturbing the internal matrix (voids) will be evaluated. Following parameters are important in the optimization process:

temperature profile, shear rate, mass flow, draw ratio etc. Based on the process denied previously, preliminary specifications and samples will be set as input for the development of the monitoring system. First approach for the model development will be carried out aligned with T2.6.

T2.2 Modelling and Synthesis of bio-based dynamic chemicals (Leader: **SP**, CID, M1-M24). This task will include the modelling (to accelerate the stages of preselection guiding the experimental research) and experimental synthesis of reversible bonding agents for a) thermoset bioepoxy resins (so called bio-based 3R epoxy resin) and b) to develop a reversible adhesive for thermoplastic bioPA/PLA separation. At any case, it will be necessary to define the main descriptor (most representative physical parameter of a given property) as well as the adaptability of this approach to other physical quantities by the creation of new modelling chains. Correlations with experimental work will allow improving the relevance between modelled and experimental data: *(This task will also feed T3.1)*

a) In a first stage the bioepoxy matrix formulation will be fine-tuned and upscaled based on Diglycidyl ether of Vanillyl alcohol (DGEVA with customisable Tg 70º-120ºC), adequate to meet the bath ceiling requirements. DGEVA will be pre-polymerized and formulated with other biobased amorphous oligomeric building-blocks for tunning the rheological behaviour and crosslinking density: Monoglycidyl ether (clove oil), Diglycidyl ether (Lignin) and Triglycidyl ether (algae or sugar bioconversion) or resveratrol (grape). In parallel, SP and CID will identify and select the dynamic hardeners which will combine covalent adaptable bonds (disulfide moieties, imines, esters…) with epoxy reactive chemical moieties such as amine or carboxylic acids. Once optimized the chemical composition, SP will upscale it at kg level (≈20kg).
b) In a second stage, a biobased reversible adhesive will be tuned to satisfy the main specifications considered for the lid demo: (i) implementation via spray coating; (ii) curing cycle consistent with garbage lid manufacturing process, (iii) the reversibility commutation temperature must be consistent with the PLA/bioPA thermal behaviour to allow proper creep properties. Pre-polymerization of DGEVA is foreseen to allow to increase pendant hydroxyl groups that enhance the adhesion. The viscosity (sprayability) will be enhanced with biobased building-blocks (reactive diluents, plasticizer, solvents) and reactivity will be adjusted too. (50-100g for lab characterisation and best formulation upscaled 1Kg)

T2.3 Preforming of thermoplastic bio-sheets (Leader: COM; CTB, IRI, M6-M30)

After characterization and selection of suitable PLA and PCL grades in task T2.1, monocomponent low and high melting multifilaments will be produced on an industrial extrusion line at COM. In addition, focus will be given to the tailoring of filaments facilitating their further processing and improving the resulting properties. This will include filament sizing and optimal titer for processing to hybrid yarns and weaves; proper spinfinish not disturbing the consolidation process; optimal temperature resistance for processing window during consolidation. A selection of the most optimal filaments will be made based on mechanical properties, processability to intermediates and performance in the final composite parts. A close interaction with WP3 will be established to allow constant feedback. The selected filaments will be combined in different structures and transformed into self-reinforced and microfibrillar reinforced organosheets and unidirectional tapes that can be used as preforms for the three selected cases that will be developed in WP3. Consolidation temperatures, pressure and time according to sheet thickness and weaving pattern will be optimized for each combination of materials (PLA/PLA and PLA/PCL) and samples will be characterized in terms of microstructure and mechanical properties.

Organosheet: thickness - *0.5 mm to 5mm* - varying the number of plies and fibre orientation and size (≈600mm length); and UD tapes thickness *-0.3mm to 2mm-* and with 6-50mm, will be adapted according to final product specifications. Based on the process denied previously, preliminary specifications and samples will be set as input for the development of the monitoring system. First approach for the model development will be carried out aligned with T2.6

T2.4 Compounding of fibre reinforced bio-pellets (Leader: AIT; LIM, IRI, CTB, M6-M30)

BioPolyamide will be used as matrix for the development of biobased pellets aiming at injection of garbage lids. BioPA will be reinforced with short carbon fibres coming as inkind from VIBES -BBI project lignin-TPU upscaled fibre. LIM will condition the fibre to be used for this application. In addition, wood fibre from recycled furniture coming from FINSA industrial process will be also in-kind provided, pre-treated if necessary (by AIT) and tested. The fibres, apart from a mechanical reinforcement, are expected to act as impact absorbers, noise mitigator and also permits achieving different aesthetical finishing (e.g. organic garbage). Extrusion compounding will be used to develop and optimize the biopellet formulations. In case that other additives such as impact modifiers or UV absorbers (Sukano). Particular design of the process set-up will be developed: drying station for PA and fibres, screw configuration, feeder configuration, adaptation of the cooling system (to avoid moisture uptake) and pellet chopping (to respect fibre size and structure). Characterisation tests will be carried out by AIT and with the support of (LIM/SP): mechanical tests, rheological behaviour, notch impact, UV resistance, thermal stability (DSC, TGA) and SEM microstructure. Lab trials will be carried out to define and characterize the best formulation (1kg each) and the best candidates (3-5 selected formulations) will be upscaled for product validation at WP3 (200-300 Kg). Based on the process denied previously, preliminary specifications and samples will be set as input for the development of the monitoring system. First approach for the model development will be carried out aligned with T2.6

T2.5 Production of thermoset bio-composites (Leader: CID, LIM; M3-M30)

This task is aimed at the realization of experimental campaign for manufacturing of thermoset pre-pregs to define the process boundary conditions required to produce these formats based on the different CANs formulations obtained in T2.2.a). The prepreg manufacturing process will be optimized (resin viscosity, process temperature, process speed, etc) to obtain the most suitable enduring prepregs for their final application: prepregs with different curing degrees will be produced and their

thermomechanical properties and processability via thermoforming will be assessed by studying the pressure, temperature and time needed for the thermoforming of the final product. Due to the dynamic character of the here described bio-based 3R epoxy resin, this new class of enduring prepregs can be prepared as totally cured prepregs (they will not be on a B-stage) and are able to be stored at room temperature (either in rolls, sheets or tapes) and still be appropriate to produce multilayered composite parts in WP3. This task will also provide the samples that will be evaluated/analysed in T2.4 for developing the in-line characterization method (aimed at measuring curing degree).

T2.6 Development of in-line characterisation methods (Leader: IRI, CID, CTB, COM, POD, SP, LIM; M1-M30)

Several photonic techniques will be assessed, and monitoring prototypes developed to provide real time data on relevant parameters to improve the efficiency and quality of the manufacturing processes. The prototypes will be delivered with an ad-hoc software to implement the models, acquire and process the data, and show the results using a friendly Interface. The systems performance will be validated in the users' facilities. FAT procedure will be implemented according to ISO 9001:2008. The following monitorization prototypes will be developed:

Inline monitorization of the curing process for the thermoset pre-impregnated sheets considering the mixing ratio of the multicomponents (flax fibres, resin and additives). HSI and Spectral reflectance in probe mode will be tested by IRI. CID will provide representative samples of different curing levels to establish a reference method. Calibration models for the most representative formulations performing the required mathematical treatment of the images/spectra to predict the degree of curing will be developed. If necessary, the temperature or other relevant parameters will be included to improve the accuracy of the results. The most promising and cost-effective approach will be selected and a prototype will be designed, built and integrated inline, and validated at CID facilities.

Inline monitorization of the manufactured thermoplastic filaments providing information on the distribution and orientation of the polymeric fibres. IRI will evaluate several spectroscopic techniques, such as NIR, RAMAN, LIBS and XRF. The chemometric discrimination will be enhanced by machine vision to provide information on the distribution and orientation of the fibres. CTB will provide characterized samples and mathematical processing of the developed models and the images/spectra will be carried out. Depending on the results of the offline trials, the most suitable spectroscopic technique will be selected and the architecture of the system hardware will be conceptualized, designed and built for the inline integration at CTB facilities including the optical systems, as well as the enclosure and other optomechanical parts.

Monitorization of the thermoplastic spinning process yarn providing information on the crystallinity of the resulting polymer. The crystallinity of the polymer extruded will be monitored applying an at line approach, using the Visum Palm system available at IRI (portable infrared technology, provide real time quality control, characterization and chemical composition analysis of the incoming material at critical points along the line). In this task, tailor made predictive models will be developed for the handheld analyzer. Representative sets of samples in terms of crystallinity will be provided by CTB to establish a reference method, develop, and implement the chemometrics-based software. The system performance will be validated at CTB´s facilities and transferability to industrial production (COM) evaluated.

Deliverables:

D2.1 Report about the development of reinforced fibres (R, SEN, M12, M24, **CTB**)
D2.2 Report on biobased and dynamic thermoset resins and adhesives (R, SEN, M12, M24, **SP**)
D2.3 Report on manufacturing thermoplastic & thermoset intermediates (R, SEN, M18, M30, **CID**)
D2.4 In-line characterisation prototypes (DEM, SEN, M30, **IRI**)

WP No	3		Start	13		End	30		Lead beneficiary		SIM		
Title	Optimization of the manufacturing processes for final bio-composite products												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	10	9	20	3	7	4	9	11	15	29	16	0.5	0

Objectives: The objective of WP3 is to set up the manufacturing lines of the selected products for each of the value chains. In particular, model, simulate, up-grade and optimisation of a semi-automated thermoforming line for thermosets (bath ceilings), a press forming process for thermoplastic free forming (feet insoles) and over-injection + mould forming for thermoplastic large parts manufacturing (lids for garbage/refuse containers).

T3.1 Process modelling (Leader: **SIM**, M13-M30):

Simulation of thermoforming and injection moulding in order to establish the boundary conditions for each manufacturing process. The goal is to define range of process conditions for manufacturing of products, including desing-of-experiments based on characteristics of intermediate materials formats developed in WP2. This might include to enhance/adapt current mathematical models for simulating bio-based materials behaviour during thermoforming/injection together with their experimental validation. FEM work based on mechanical and rheological analysis and process simulation will be followed, in a concurrent approach to provide an upgraded configuration of the product geometry, which considers both mechanical and processing aspects. This might include to redefine the meshing strategy, redesign of parts geometry, taking into account fiber orientation, materials combinations (and results and libraries coming from material modelling WP2) and sequence (if

applicable), thickness, size, etc. Indications for end-users for the design of moulds will be given. At the end of this task a redesigned CAD model will be available.

T3.2 Free-forming of orthopaedics (Leader: **COM**, CTB, M13-M30):

In this task, an orthopedic AFO/shoe will be developed and manufactured by using combinations of the self-reinforced and multicomponent composite organosheets and UD tapes developed in task 2.3. The product will be designed to fulfill the demand from market. This includes preliminary evaluation of the dynamic and flexural behavior of the part and good processing properties, but also to reach the thermo-mechanical requirements. The manufacturing process will include a layup station of the blanks (sheet and tape layers) according to design and modeling indications, followed by a heating stage and then pressing forming stage (with adjusted temperature, time, pressure, need for ramping, mould temperature…) that will need to be optimized with the required tooling which will be developed. Finally, a trimming station will be required to make the finishing of the targeted product. Adapted cutting technologies will be tested to evaluate and avoid any induced/undesired damage. Quality criteria and data-sets will define to feed task 3.5 and task 3.1, respectively.

T3.3 Over-injection of tertiary packaging (Leader: **MOS**, AIT, LIM, CTB, IRI, M13-M30):

In this action MOS will define, adapt and demonstrate the manufacturing process of lids for street garbage containers to collect municipal waste. The targeted product will combine intermediate formats coming from WP2. The whole manufacturing process is the combination of 4 main stations: first the PLA organosheets (task 2.3) will be warmed-up over Tg in an IR lamp to make it soft and mouldable. Then the organosheet surface will be sprayed with the CAN adhesive developed (task 2.2) and located in the mould cavity. It will be essential to ensure a quick and homogeneous pulverization (tempered the adhesive at PLA-Tg will be evaluated). Afterwards, the mould clamps and stamps the structural-preform. In this stage the formality of the organosheets will be evaluated. Then, it will be overinjected with bioPA reinforced pellets (task 2.4). Parameters such as melting, noozle and mould temperature, injection pressure, time and cooling will be optimized in order to obtain a high quality part without injection defects (e.g. flow lines, vacuum voids, burn marks, weld lines, warping etc) and define the quality criteria (to feed task 3.5). Once the process is robust enough, foaming the core structure of the lid will be tested by injecting the bioPA using Mucell® (supercritical fluids). The aim is to make the part lighter, use less material and fulfil other requirements such as noise absorbance. The last station will be dedicated to par trimming and finishing the lids. Process data-sets will be generated to improve simulation models in T3.1. Characterisation tests to verify reaching basic properties will be carried out. IRI will support the selection of IR camara to monitor the surface temperature of organosheets before the injection process.

T3.4 Thermoforming of bath ceilings (Leader: **POD**, M13-M30):

POD aims in this task at the realization of the experimental program define in T3.1, in order to: 1) evaluate the processability of bio-based thermoset sheets/tapes produce in T2.2 in a thermoforming process combined with flax fabrics to produce the ceiling structures of modular bathrooms. This will involve the evaluation of different process parameters and conditions of bio-based sheets, such as plies orientation, number of plies, ply thickness, fibre orientation, heating profile, forming pressure, etc., with respect to product characteristics (through-thickness consolidation, mechanical performance, geometry accuracy, etc.). In addition, in a second stage the integration of the wood basal core will be carried out by applying a CAN reversible adhesive applied to glue the interphase to the thermoformed structure in order to ensure the EoL separation of composite components. In this task, process data-sets will be generated to improve simulation models in T3.1 and feed the development of the process data-model in WP4. Quality assessment criteria will be defined within this task to feed the in-line inspection solutions develop in T3.5 for quality control of the final product.

T3.5 Fast quality control measures of the final products (Leader: **IRI**, AIM, AIT, POD, MOS, COM, M13-M30):

This task will combine different destructive (establishing a systemic study method of randomly selected parts) and non-destructive testing approaches (MW, US) to identify the most critical defects in order to provide feedback to the manufacturing line and make in-line corrections (teaching both the algorithms and staff). POD will prioritize to avoid voids formation and also to selectively integrate some moisture sensor with RFID that can measure water absorption at different stages of the process (which would be critical for the quality of the bathroom parts). COM will look into the full adhesion of conforming layers (e.g.peeling tests) and MOS will require a system to rapidly detect surface defects (blotches, stains, flow lines…), evaluate if there is also any lack of adhesion between the components, finding any internal crack and in foamed samples avoid large bubbles or coalescence (US and microscopy observance will be developed).

Deliverables:

D3.1 Establishing the manufacturing line for biobased products (R, EU-C, M30; **COM**)

WP No	4		Start	16		End	38		Lead beneficiary			AIM	
Title	Digital systems and Interoperability												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	2	0	3	0	53	4	23	0	0	0	0	0.5	0

Objectives: i) Definition of communication protocols and data exchange between process modelling software, process monitoring and quality monitoring systems; ii) To develop data analytics and Decision Support System to support the design

optimization and process parameters optimization; iii) To develop a hybrid digital twin merging virtual and real data for a continuous updating model.

T4.1 Interoperable embedded platform (Leader: **AIM**, SIM, IRI, M16-M30):
This task will be focused on the development of a common embedded platform for process monitoring ensuring interoperability across the different machines and processes employed to manufacture intermediate formats and the final products. This embedded platform will allow multimodal monitoring combining imaged-based sensor for the inline characterization methods and in-line quality control systems developed in T2.6 and T3.5 and analogue signal from the machines/equipment monitoring in a synchronous way. The embedded platform developed by AIM will be interoperable with different fieldbuses -e.g. Profinet, EtherCAT,…- using different gateways and connectors and the communication layer will be based on a OPC UA server. The goal is to provide a data format to ensure data exchange between process parameters and characterization data extracted from devices/systems employed for the manufacturing of intermediate formats, and process and monitoring data together with part geometry and numerical simulation of final bio-based products. The data gathered during the manufacturing process by the platform will be used for, first, process optimization based on data analytics algorithms in T4.2 and, second, for product design optimization in T4.3.

T4.2 Data analytics for adaptive manufacturing systems (Leader: **AIM**, SIM, IRI, M26-M36):
This is to develop and implement an adaptative system for manufacturing process optimization based on a cognitive approach that uses the information from the characterization of intermediate formats jointly with process parameters' data acquired from WP2 experimental trials and data extracted from the manufacturing of final products (WP3). Together with information data provided by the quality inspection systems (T3.5). Data-driven algorithms and AI-based pattern recognition solutions will be proposed to reveal significant details and dependencies among all acquired data. The final goal is to develop an empirical data-driven model of the entire manufacturing value chain allowing to recalibrate/modify processes for manufacturing of intermediate formats and the final products, increasing robustness against environment and production process variations and enabling operators to get a deeper understanding of process parameter relationships crossing the boundaries of each processing step.

T4.3 Data analytics for product design (Leader: **SIM**, IRI, M28-M38):
A Decision Support System (DSS) will be developed to aid the Product Engineer in the design of new bio-based products, combining different bio-based materials and ensuring not only manufacturing of the actual product but further usability onto new or regenerated products. This will involve the development of specific data mining algorithms to extract the relationships between process parameters, product quality and material characteristics for a specific product design and with respect to future recirculation of bio-composite products (reuse and/or remanufacturing). A Graphical User Interface will be created to ensure adequate output presentation to the Product Engineer, revealing significant details, interdependencies and effects of the design parameters and materials combinations in the entire manufacturing flow.

T4.4 Hybrid-digital twin (Leader: **SIM**, M20-M38):
The goal of this task is the implementation of a Hybrid Digital Twin along the entire manufacturing value chain and aimed at improving the smart manufacturing of bio-composite products (WP3). The objective is to create the architecture that will merge the product geometry and the numerical simulation design-based information to the data acquired along the manufacturing chain (process parameters, materials characteristics and quality monitoring), in order to generate new knowledge for a continuous updating model of the entire manufacturing process (including intermediate formats and the final products).

Deliverables:
D4.1 Interoperable embedded platform (OTHER, SEN, M30 **AIM**)
D4.2 Data Analytics and knowledge-based system for adaptive manufacturing systems (OTHER, SEN, M36; **AIM**)
D4.3 Decision Support System for Product engineering optimization (R, SEN, M38; **SIM**)

WP No	5		Start	30		End	41		Lead beneficiary		AIT		
Title	Demonstrate innovative bio-based material production and bio-based products												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	30	6	6	2	18	4	3	11	4	15	2	1	0

Objectives: The aim of this WP5 is to demonstrate the smooth run of the designed pilot lines at operational and digital level (now smart), manufacturing a short batch of demonstration prototypes and validating their functional and environmental performance according to industrial standards and circular premises.

T5.1 Smart systems integration (Leader: **AIM**, M30-M38):
This task concerned with the integration and implementation of the different sub-systems developed along the project, from in-line characterization methods (T2.6), fast quality control systems of the final products (T3.5), and communication protocols enabling data flow between the different manufacturing stages (intermediate formats and the final products). It involves the sensing, mechanical, and electronics integration of the different quality systems as well as the implementation and integration of the embedded platform enabling the synchronous process parameters monitoring. Data gathering from the process (T5.2)

together with the product and process modelling data (T3.1) will be the input for the development of the data analytics algorithms (T4.2 and T4.3). This task will serve as demonstration of the complete dataflow, merging data from the shopfloor and data from process simulation in the three pilot lines.

T5.2 Pilot line optimisation and testing (Leader: **AIT**, IRI, POD, COM, MOS, AIM, SIM, M30-M38):
Bio-Uptake pilot tests will be tested in detail, analysed, and validated against the requirements defined in T1.1 and following the validation protocol defined in WP4. A test plan will be defined including iterative trials towards full operation availability and conformity of all key process and quality parameters with the process windows defined in WP3 and with respect to the characteristics of intermediate materials formats in WP2. Once optimized the pilot lines for each value chain, batches of demonstration prototypes with best candidate materials and optimised designs will be produced for their validation in task 5.3.

T5.3 Validation of user cases in the medical, packaging and construction value chains (Leader: **MOS,** M30-M41)
This task will be focused on the validation of the three demo products according to market and end-user specifications as well as ISO standards. POD directly represents the final construction end-user. MOS is a Tier 1 for lids of the large company CONTENUR (and will follow established CONTENUR validation requirements) and COM is provider and product developer based on organosheets of the medical company FILLAUER (FILLAUER will establish the main validation tests for the AFO/shoe insoles). Each industrial leader will be responsible to carry out the necessary validation tests:

Construction: Factory Acceptance Test (FAT) procedure will be implemented following ISO 9001:2008 processes. For the final product: The composite sandwich has to match the mechanical properties of current POD ceilings according to ISO 178 three point bending tests; the fire properties of the ceiling has to match fireclass D-s2,d0, ISO 5660-1 Cone Calorimeter and EN ISO 11925-2 small flame test and EN 13823 Single Burning Item; emission tested according to ISO 16000-3:2011(Indoor air - Part 3).

Medical: EN ISO 2062 to determine the single-end breaking force and the elongation at break. For the final product: matching the performance of the current orthosis done by FILLAUER client and have to pass their inhouse testing and when approved it will be send for patient testing which takes 12 months.

Packaging: For the garbage lid, it is essential that it is UV-stabilised, weather-resistant, impact-resistant at low temperatures and resistant to heat and chemicals: European Directive 2000/14/EC. 92,0 db: noise emission in the environment by equipment for use outdoors (sound power level); European Standard EN 840:2021: Installations and equipment for waste disposal & treatment (840-1: Dimensions & Design and 840-5: Performance requirements & test methods). ISO 14006 Ecodesign.

T5.4 Sustainable end-of-life models for new circular and regenerated products (Leader: **AIT**, CTB, MOS, COM, POD, SP, M30-M41)
The goal of this task is to validate that the final prototypes will comply with the conceived end-of-life: i) demonstrate the separation of components thanks to the bonding-debonding adhesives (wood basal core from 3R-organosheet structure & bioPA from PLA organosheet) and set-up guidelines for a proper decommissioning and separation; and ii) propose alternative uses of the biobased components to obtain regenerated products (for the same sector in similar uses or even for potential applications in other sectors); iii) whether reuse and remanufacture is not possible, propose and corroborate recycling alternatives for the materials (e.g. recycling tests of the PLA/PCL to reuse it as matrix material). This task will also feed LCA/LCC assessment and results will impact in the eco-design principles of the selected products too.

Deliverables:
D5.1 Process dataflows and systems integration (DEM; SEN; M38; **AIM)**
D5.2 Operational guide of the Manufacturing processes, prototypes production and validation (R; EU-C; M41; **AIT**)

WP No	6		Start	1		End	42		Lead beneficiary		POL		
Title	Exploitation, dissemination and communication												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	13	4	3	3	6	3	3	4	4	3.5	8	21	1

Objectives: The aim of WP6 is to widely communicate and disseminate the project's activities and results for strengthening Bio-Uptake's impacts as well as to maximize the outreach and impact of innovations of the project enhancing market and business opportunities, growth and jobs in Europe.

T6.1 Exploitation, dissemination & communication plan (Leader: POL; All, M01-M42)
This activity is devoted to defining the overall dissemination and communication strategy. As a result of this task a complete an interlinked strategy of the three pillars will be issued on M6 and periodically updated if needed in the respective deliverables due to M18, M30 and M42 (dynamic and flexible tool). The Plan will constitute the core document outlining the activities at the basis of the project's dissemination and communication activities as well as the guidelines for developing business plan for market deployment. More specifically it will identify the target groups and define the communication tools and distribution channels. The Plan will be prepared as a collaboration of the Communication, Dissemination and Exploitation leaders with the support of the coordinator and project partners/beneficiaries, and will address the following aspects:

- Overall communication and dissemination strategy and its expected results;
- Project identity, logo and graphical layouts guidelines;
- Identification and segmentation of communication and dissemination targets;
- Tools needed to implement successful communication activities, such as communication materials;
- Events relevant to the different sectors involved in the Bio-Uptake project (where partners can represent the project);
- Calendar of dissemination and communication activities;
- The communication levels (EU/National/Regional) and the responsibilities/roles attributed to each partner;
- Guidelines for implementation of communication & dissemination actions: messages to convey, internal reporting rules…
- Database of stakeholders' contacts: list of all stakeholders and potential end-users. The database will be implemented and allocated in the platform throughout the project lifetime with the contribution of the partners;
- Pathways for exploitation covering calendar of exploitation activities including exploitation workshops for the consortium planning of activities to create links with parallel European initiatives related possibly with different sectors and guidelines for the implementation of the long-term market deployment and replication plan of manufacturing for bio-based materials.

T6.2 Exploitation activities (Leader: AIT; All, M6-M42)

Following the initial structure of Key Exploitable Results (KERs) defined under *Table 11* & *Fig. 10*, partners will work firstly in the definition and sensitivity of knowledge and data, so that critical decision making on exploitation, dissemination, data and knowledge protection can be taken. Under this task the following actions will be covered:

- Identification of own background and third parties background (aligned with the Consortium Agreement)
- Determine the ownership of results: who owns what?, definition of transfers and conditions for exploitation.
- Evaluate the best procedure to protect new knowledge and data (Patent fillings, trademarks, utility models, etc)
- Finally, once the alternatives for protection have been considered, exploitation models will be evaluated at different level for the defined KERs: creation of spin-offs, launch of a PhD thesis, further new side research projects etc. The exploitation of research knowledge will cover as well the replicability of the training activities/content contemplated in WP1.

T6.3 Dissemination activities (Leader: LIM; All, M6-M42)

This Task will manage the dissemination activity plan to maximise impact while controlling confidentiality to protect the IP interests of the partners (promoting open access whenever is possible and supporting scientists training and skills – task1.4). Scientific results will be disseminated through publications in peer-reviewed international journals and presentations at international conferences and workshops, as indicated in *Table 10* and targeting the defined audiences. Particular emphasis will be on the collaborative technological advances achieved with EU support, and on the wider applications in other sectors.

T6.4 Communication and cooperation activities (Leader: POL, All, M1-M42)

This task will be led by POL, with the help of all the partners to develop a stakeholders' engagement strategy to clearly identify Bio-Uptake target audience groups and propose actions to involve and keep them engaged in the project. The aim of the task is to establish communications needs, the best communication channels and the development of suitable communication material for each target group in sectors (website, press-kit, leaflets, e-newsletter, infographic, social networks, videos…). See *Table 9*. This will help to prepare and update the suitable Strategic Dissemination Plan section (task 6.1). The list of identified stakeholders will be updated as need arises throughout the project, in accordance with any new project results and findings.

T 6.5 Clustering activities (Leader: AIT, All, M7-M42)

This task is oriented to interact with projects and networks on similar topics oriented to introduce biopolymer in the market. Alliances with ongoing projects (such as collaborative workshops with BIOECONOMY VENTURES – CSA, and whenever is beneficial to close Bio-Uptake business cases to run proof-of-concept test trials with POLYNSPIRE, MULTICYCLE, MANDALA, DECOAT and VIBES projects dedicated to sustainable and green chemical recycling) from BBI/CBE, Made in Europe and other platforms such as Renewable Carbon Initiative, https://renewable-carbon-initiative.com/, European-bioplastics, etc. This clustering will be oriented towards opening cooperation, sharing results and experiences in the processing of biopolymers, sharing dissemination forums, and enabling the easier uptake in the market of the solutions developed in Bio-Uptake.

Deliverables:

D6.1 Communication & Dissemination Plan & Achievements (including clustering activities) (R;PU; M6/M18/M36/M42; **POL**)
D6.2 Exploitation and IPRs strategy (R;SEN; M6/M30/M42; **AIT**)

WP No	7	Start	1	End	42	Lead beneficiary	AIT						
Title	Project Management												
No	1	2	3	4	5	6	7	8	9	10	11	12	13
Short name of Participant	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
PM	45.5	1	1	1	1	1	1	1	1	1	1	1	1

Objective: To structure, foster and daily maintain a high level of commitment from all partners; and to provide the consortium with adequate tools for guaranteeing timely delivery of all results in the project. To facilitate and follow up the fulfilment of the project's key goals, through periodical and detailed control of all tasks/sub-tasks and potential risks; To set up a framework for

efficient communication, collaboration, exchange of documents and fast decision-making, The correct management of all legal, financial & administrative duties of the project, pondering the special characteristics which the Horizon Europe provides, Aiding WP6 towards the transferability of project's results after its conclusion and its further use by partners.

T7.1 Coordination Processes, Quality Assurance and Risk Management (Leader: AIT; All, M1-M42)

This task sets the processes for the overall strategic project management of the project. T7.1 will ensure due decision-making, clear and operational internal communication, metrics and timely risk control. T7.1 processes are fundamental for T7.2.

- Set up a quality assessment mechanism based on project metrics that will be introduced in D7.1, then elaborated as the project evolves, and reported in the Periodic Reports
- Monitor the project for foreseeable potential risks. Promptly initiate contingency plans to lessen or eliminate negative effects. An initial list of risks will be introduced-D7.1, then as the project evolves updated & reported in Periodic Reports
- Establish a bottom-up and top-down communication flow among all the responsible partners that will include decision-making not only in terms of discussion of internal milestones, risks and deliveries, but also in terms of external positioning, key alliances and sustainability of the project's results
- Set up management structures assigning clear responsibilities to all relevant partners

Establish, coordinate and oversee all the project management processes including, but no limited to, transversal coordination of all partners and WPs according to the work plan and the project's objectives, external reviews, internal project meetings, compilation and circulation of meeting minutes and action points, internal reporting procedures and communication channels.

T7.2 Administrative, financial and legal management (Leader: AIT; M12-M42)

Daily management of all legal, financial and administrative issues; within the consortium and formal communication to the EC:

- Monitor full compliance with the current regulations of all activities, agreements, amendments, expenditures, IPR management and any other relevant issue at stake.
- Set up a secure, organized and useful collaborative space for the efficient communication, collaboration and exchange of documents among all partners (M3). Maintain and deliver templates for all financial, administrative & reporting duties.
- Ensure periodic reporting and fluent communication with the EC services and Project Officers for purposes of coordination, progress monitoring and reporting; including the preparation and submission of Periodic Reports.
- Request, collect and review inputs from all partners with regards to any report, official deliverable, document and/or information as required by the EC or by the daily routine of the project. This will be an internal reporting, in the form of Activity Reports (periodic technical reports plus a follow-up of the Use of Resources and expenses) due every six months

Implementing any potential change in the Grant Agreement as per decision of the General Assembly and/or any other legally enforced reason by ensuring the fulfilment of the contractual obligations in the Grant Agreement.

T7.3 Data management (Leader: AIT; All M01-M42)

Data Management Plan (DMP) will be generated, taking into account data for implementation purposes and end-users private data to be handled during demonstrations, apart from collaboration in external dissemination and communication activities covering internal and external perspectives of the project. This DMP will be conceived as a "living" document outlining how data collected, stored or generated will be handled during and after this project. All data generated during the project will follow FAIR and secure principles in order to ensuring the interoperability for data sharing.

T7.4 Ethics (Leader: AIT; All, M1-M3)

The objective is to ensure compliance with the 'ethics requirements' that the project must comply with. The general assembly will appoint a Data Protection Officer (DPO) and the contact details will be made available to all data subjects involved in the research. Templates of the informed consent forms and information sheets will be kept on file and the informed consent procedures in regard to data processing will be collected accordingly.

Deliverables:

D7.1 Project Handbook and Quality Assurance Plan (R;SEN; M3;**AIT**)
D7.2 Data Management Plan (DMP; PU; M6/M18/M28/M42;**AIM**)
D7.3 Ethics Report (R, PU,M3**;AIT)**

Table 15: List of Deliverables

No	Name	WP No	Lead	Type	Diss.Level	Date
D1.1	Bio-Uptake detailed baseline and expectations	1	AIT	R	SEN	M6
D1.2	LCA and LCC assessment report	1	LIM	R	SEN	M6; M18;M40
D1.3	Report on the contribution to standardization	1	LIM	R	PU	M12,M24,M42;
D1.4	Training Programme: intellectual outputs and session summary	1	LIM	R	PU	M18;M40
D2.1	Report about the development of reinforced fibres	2	CTB	R	SEN	M12, M24
D2.2	Report on biobased and dynamic thermoset resins and adhesives	2	SP	R	SEN	M12, M24
D2.3	Report on manufacturing thermoplastic & thermoset intermediates	2	CID	R	SEN	M18, M30

D2.4	In-line characterisation prototypes	2	IRI	DEM	SEN	M30
D3.1	Establishing the manufacturing line for biobased products	3	COM	R	EU-C	M30
D4.1	Interoperable embedded platform	4	AIM	OTHER	SEN	M30
D4.2	Data Analytics and knowledge-based system for adaptive manufacturing systems	4	AIM	OTHER	SEN	M36
D4.3	Decision Support System for Product engineering optimization	4	SIM	R	SEN	M38
D5.1	Process dataflows and systems integration	5	AIM	DEM	SEN	M38
D5.2	Operational guide of the Manufacturing processes, prototypes production and validation	5	AIT	R	EU-C	M41
D6.1	Communication & Dissemination Plan and Achievements	6	POL	R	PU	M6/M18/M36/M42
D6.2	Exploitation and IPRs strategy	6	AIT	R	SEN	M6/M30/M42
D7.1	Project Handbook and Quality Assurance Plan	7	AIT	R	SEN	M3
D7.2	Data Management Plan	7	AIT	DMP	PU	M6/M18M28/M42
D7.3	Ethics Report	7	AIT	R	PU	M3

Table 16: List of Milestones

No	Name	WP(s)	Date	Means of verification
Ms1	First batch of intermediate formats ready for validation	2	M12	Delivery of physical formats to end users and report of preliminary results from D2.1 & D2.2
Ms2	Second batch of intermediate formats with improved performance according to manufacturing feedback	2	M20	Delivery of physical formats to end users and report of preliminary results from D2.3
Ms3	Final batch of fully optimized intermediate formats	2	M30	Delivery of physical formats to end users and final results from D2.1, D2.2, D2.3 & D2.4
Ms4	Pilot manufacturing lines ready to start its validation at operational level	3	M30	Final set up and optimization of the three pilot lines as reported in D3.1 by industrial users.
Ms5	Pilot manufacturing lines ready to start its validation at digital level	4	M38	Digital threat ready to interoperate as described in D4.1, D4.2 & D4.3
Ms6	At least one OK product demo prototype per sector (OK ceiling, OK insole, OK lid)	5	M41	Final manufactured Prototype with good quality and functional validation results from D5.1 & D5.2
Ms7	Demonstrate the remanufacture and recyclability of the developed smart composites under the proposed EoL	5	M41	Results coming from D5.2 and EoL guidelines and improving actions (if deviations appear)
Ms8	Achieving training, standardization communication and dissemination KPIs.	1/6	M42	Monitoring training & standardization WP1, communication & dissemination activities WP6, results included under D1.3, D1.4 & D6.1, continuous reporting in participant portal

Table 17: Critical risks for implementation

Description of risk: **L**ikelihood (Probable, Remote, Improbable) & **S**everity: Low/Medium/high			WP	Mitigation measures
Description of risk	**L**	**S**		
1) Insufficient data on bioplastic production, use end-of-life provided by partners to complete the life cycle inventory	R	M	1	Early & ongoing dialogue with partners to ensure that the necessary data are recorded, along with use of relevant default process data from latest Ecoinvent database (v3.7.1 or higher).
2) Not reaching a sufficient number of professionals and students for the training activities planned	I	L	1	The training activities will be planned well in advanced in the beginning of the project, by establishing the content and steps required with universities and RTOs Promotion in newsletter, social media and open portals. Increase the number of training course, workshops and events.
3) Low PCL thermal stability, degradation during extrusion	R	M	2	Extra stabilized grades will be used and compatiblizers will be added to improve the adhesion between PCL and PLA
4) Microfibrillar and self-reinforcing structure produced for the PLA/PLA organosheets	R	H	2	In depth study (simulation & experimental) and optimisation of overinjection times (shortest as possible). Low melting bioPA. The effect of the adhesive as intermediate layer will dissipate (while curing) heat. Mould refrigeration will be

is affected (melted) during the overinjection step of bioPA due to the difference in Tm				optimised to ensure fast-cooling. The thickness of the organosheet will be increased (more heat needed, more manoeuvre). If previous measures do not work, the biopolymer grades will be changed (PLA with higher Tm, test bioPA to produce a organosheet – based on current CTB expertise).
5) Not being able to produce bioPA pellets with short CF from VIBES project due to lack of supply	R	L		AIT (VIBES coordinator) and LIM (VIBES developer of the CF) will ensure to construct this interaction between projects by anticipating this batch in both projects planning. Otherwise, a batch may be purchased (depending on kg required) from DIFT pilot plant. Trials with oil-based CF will be carried out and with other lignin-based developments from LIM to find sustainable alternatives
6) The obtained construction prototype (final product) does not meet functional requirements	I	H	3/5	The sandwich structure will be re-designed in order to achieve mechanical performance. Different curing grades of resin/hardener will be design and tested too (to meet water uptake ratio and comply with indoors normative). Further additives to enhance specific performance will be included (e.g. introduce a layer with basalt fibre in the bath ceiling to reach fire properties)
7) The obtained medical prototype (final product) does not meet functional requirements	I	H	3/5	Redesign and insertion of additional reinforcement zones. If redesign is still insufficient the incorporation of stiffer fibres such as biodegradable or biobased carbon fibres would be considered. (biobased CF or alternative GF)
8) The obtained packaging prototypes (final product) does not meet functional requirements	I	H	3/5	The organosheets will be modified in terms of plies, fibre orientation, thickness etc, to ensure obtaining enough rigidity. BioPA grade might be changed to ensure toughness. The adhesive will be compatible and provide stability. Modifications in the lid design to reinforce the final part will be also carried out and implemented in the prototype mould. Additives to reach UV resistance or decrease clamping noise will added to comply with ISO standards.
9) Reshapeability of thermoset sheets &desired separability of PLA/bioPA sheets is not achieved.	R	M	2/5	Material modelling, simulations and correlation with experimental work will allow an ad-hoc design of the formulations. Increase dynamic bond response and wt% if demo prototypes do not reach EoL. Alternative recycling is proposed with ongoing EU funded projects.
10) Computational speed of digital twin too low to be updated every cycle	R	L	4	The computational algorithms developed will be sufficiently performing and will use adequate strategy to optimise the computational time (e.g. parallelisation), and the computational power assigned will be adapted
11) Digital infrastructure incompatible to equipment	R	H	4/5	If it is not compatible with existing partners equipment, the development of additional software/hardware interfaces for integration will be carried out.
12) The performance of AI systems has a slower learning curve rate due to lack of experimental data	I	H	4	If the first iteration of the AI systems doesn't provide the optimal response. Leveraging on bigger experimental datasets will provide the knowledge base to boost system performance. Bio-Uptake proposes several demonstration steps to ensure a growing dataset for system development
13) Not reaching agreement for the exploitation KERs	I	H	6	Organized extraordinary negotiation meetings between the affected partners and activate Consortium Agreement conflict procedure
14) Accumulated delays in tasks/WP cause delays in project expected progress	R	H	7	Strict control of the project management with in-time delivery submission and milestones control. Addition of more personal resources assumed by the partners involved in the delay. Negotiation with EC.
15) Partner withdrawal	I	H	7	Exhaustive control of activities, small consortium with agreed and negotiated tasks, meditated & measured tasks and efforts. In case difficulties to continue appeared, current partners may count on their third parties and collaborators to look for further resources (capacity or human resources) if it is necessary.

Risk management is considered under WP7 as an internal policy for their mitigation (periodically reviewed). The following aspects are in any case to be included in the risk analysis:

- ✓ Identification and categorization of risks (SEVERITY)
- ✓ Probability of risk appearance and its impact (LIKELIHOOD)
- ✓ Roles and responsibilities (who is responsible/s for collecting the required data? Who is responsible/s for implementing the contingency plans? Who is responsible/s for providing inputs for risk management?
- ✓ Documentation of the risks and of the mitigation of their effects

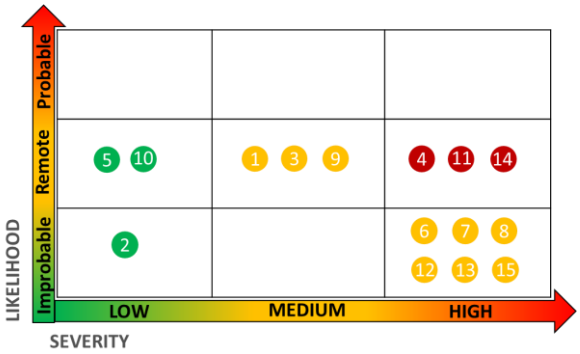


Fig. 13: Bio-Uptake risks representation

Table 18: Summary of staff effort

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total
1/AIT	12	12	10	2	30	13	45.5	124.5
2/CTB	7	20	9	0	6	4	1	47
3/CID	3	33.5	20	3	6	3	1	69.5
4/SP	2	43	3	0	2	3	1	54
5/AIM	4	0	7	53	18	6	1	89
6/IRI	1	69	4	4	4	3	1	86
7/SIM	1.5	1	9	23	3	3	1	41.5
8/POD	4.5	3	11	0	11	4	1	34.5
9/COM	6	29	15	0	4	4	1	59
10/MOS	2.5	0	29	0	15	3.5	1	51
11/LIM	48	42	16	0	2	8	1	117
12/POL	1	0	0.5	0.5	1	21	1	25
13/UNE	12.5	0	0	0	0	1	1	14.5
Total	105	252.5	133.5	85.5	102	76.5	57.5	812.5
%	12.9%	31.1%	16.4%	10.5%	12.6%	9.4%	7.1%	100%

The overall Bio-Uptake budget is **5,994,886 €** with the following distribution: **Personnel [62%], Other Direct Costs [17%], Subcontracting [1%] and indirect costs [20%** being 25% of personnel and other direct costs**]**

Table 19: Subcontracting costs' items

Subcontracting	Cost (€)	Description of tasks and justification
4/SP	30,000€	WP2 Pilot scale production of DGEVA
7/SIM	44,000€	WP4 Development of thermoforming simulation for fibre loaded sheet with anisotropy in bi-axial stretching

The purchase costs for **CTB(2), CID(3), SIM(7), POD(8) and COM(9) does not exceed the 15% of the personnel costs** for each participant. *Table 20* collects the purchase costs for those partners exceeding the 15% of the personnel costs. The record lists cost items in order of costs and starting with the largest cost item, up to the level that the remaining costs are below 15% of personnel costs.

Table 20: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

1/AIT		
Purchase Costs	Cost (€)	Justification
Other goods, works & services	77,000	Raw materials for pellets preparation (*30,000€*); Prototype drying line for lignin based formulations (*25,000€*); Consumables and prototype tools for EoL validation (*22,000€*)
Travel & subsistence	26,000	7 Consortium meetings (2 Person *800€)+ 1Review Meeting + 7 dissemination and technical travels (800€)+ Travel and subsistence costs for invited experts and project coordinators (liaison/cooperation activities)
Remaining Costs	67,715 (12,1%)	
Total	170,715	
4/SP		
Purchase Costs	Cost (€)	Justification
Other goods, works & services	72,000	Consumables: synthetic precursors (lab- and pre-pilot scale) for coupling agents and resins, solvents and catalyst, synthesis and treatment solvents needed for synthesis and purification, safety lab equipment, Consumables for analytical equipment (NMR, DSC, SEC, TGA, Rheometer consumables) (*52,000€*); Patentability study and toxicity/reglamentation studies (*20,000€*)
Equipment	31,083	Laboratory equipment for lab scale synthesis (***19,000€:*** *791.7 €/month*24 month*100% of allocation*); Reactor system for pre-pilot scale synthesis (***12,083€:*** *2014 €/month*12 month*50% of allocation*)
Remaining costs	30,503 (14.8%)	
Total	133,586	
5/AIM		
Purchase Costs	Cost (€)	Justification

Travel and subsistence	44,800	Participation in consortium meeting (8 meetings*700€*2people) + Participation in Technical Meetings (WP4,WP5) (6 meetings*700€*2 people) + Equipment integration and test (6 travel*3 use cases*2 people*700€)
Other goods, works & services	20,000	Embedded platform and electronic components for online process monitoring for 3 use cases (*20,000€*)
Remaining costs	24,500 (11,1%)	
Total	**99,300**	
6/IRI		
Purchase Costs	**Cost (€)**	**Justification**
Other goods, works & services	120,000	*Parts for the 3 monitoring systems (sensors, optomechanics, specific computer to drive and process the data of the developed monitoring systems prototype, machine vision camera …) (120,000€);*
Remaining costs	49,000 (13,6%)	
Total	**169,000**	
10/MOS		
Purchase Costs	**Cost (€)**	**Justification**
Other goods, works &services	98,000	Prototype Mould for lid validation (*48,000€*); Consumables for over-injection process (IR lamp adaptation, sensor integration, fast check systems…) (*21,000€*); Analysis for product validation (*15,000€*); Raw materials for process optimisation and purging (*14,000€*)
Remaining costs	18,163 (10,2%)	
Total	**116,163**	
11/LIM		
Purchase Costs	**Cost (€)**	**Justification**
Equipment	27,000	Oven (***15,000€:*** *357.1 €/month*42 month*100% of allocation*), Mechanical characterization equipment (***12,000€:*** *285.7€/month*42 month*100% of allocation*)
Travel and subsistence	17,500	Dissemination at conferences/Seminars 2 PIs, and 2 PhDs (project meetings) purposes. This also includes consortia meetings and attendance at conference/ workshop meetings for training and dissemination purposes
Remaining costs	43,000 (13,7%)	
Total	**87,500**	
12/POL		
Purchase Costs	**Cost (€)**	**Justification**
Travel and subsistence	21,000€	Participation in project meetings (7 meetings*1 person*1000€) + participation in dissemination events and activities (7 events*2 people*1000€)
Remaining costs	19,000 (9,5%)	
Total	**40,000**	
13/UNE		
Purchase Costs	**Cost (€)**	**Justification**
Travel and subsistence	7,000	Participation in consortium meetings (7 meetings*1 person*700€) + Participation in dissemination events, especially meetings of standardization technical committees (3 meetings*1person*700€)
Remaining costs	4,300 (6,7%)	
Total	**11,300**	

3.2. Capacity of participants and consortium as a whole

Bio-Uptake consortium forms a strong platform of experts along the manufacturing of (bio)plastics value chain for composite materials. It has been intentionally kept to a minimum number of partners to avoid duplication of efforts. It consists of **13 partners** with complementary competencies, **6 industries** (SP, IRI, COM & MOS as SMEs; and POD & SIM as Large companies) plus **2 OTHER** (POL which is the industrial French plastic Cluster & UNE the Spanish Standardisation Body); **4 RTO** partners (AIT, CTB, CID & AIM) and **1 academia** (LIM) to give support to the companies in the consortium and to develop the main scientific and technological activities in the work plan. The undertaking involves **6 European countries** (ES, FR, IE, BE, SE, DK). All partners have large expertise in participating and coordinating EU projects in previous initiatives (FP7, H2020, Life, etc) in activities closely related with Bio-Uptake (see B1 and A sections). Bio-Uptake consortium is totally industrially driven.

Fig. 14, shows how each partner contributes to the value chain and how the interaction among them is built (complementarity). In this sense, *Table 4* in the excellence section already introduces the partner collaboration in each demo case. The consortium is rich in inter-disciplinary Sciences and includes experts with deep knowledge in the technical aeras (material science, chemistry, manufacturing processes, digitalisation, circularity and sustainability, …) as well as Social Sciences and Humanities (business development, training methodologies…) and high communication and dissemination skills (including open science practices, gender aspects etc).

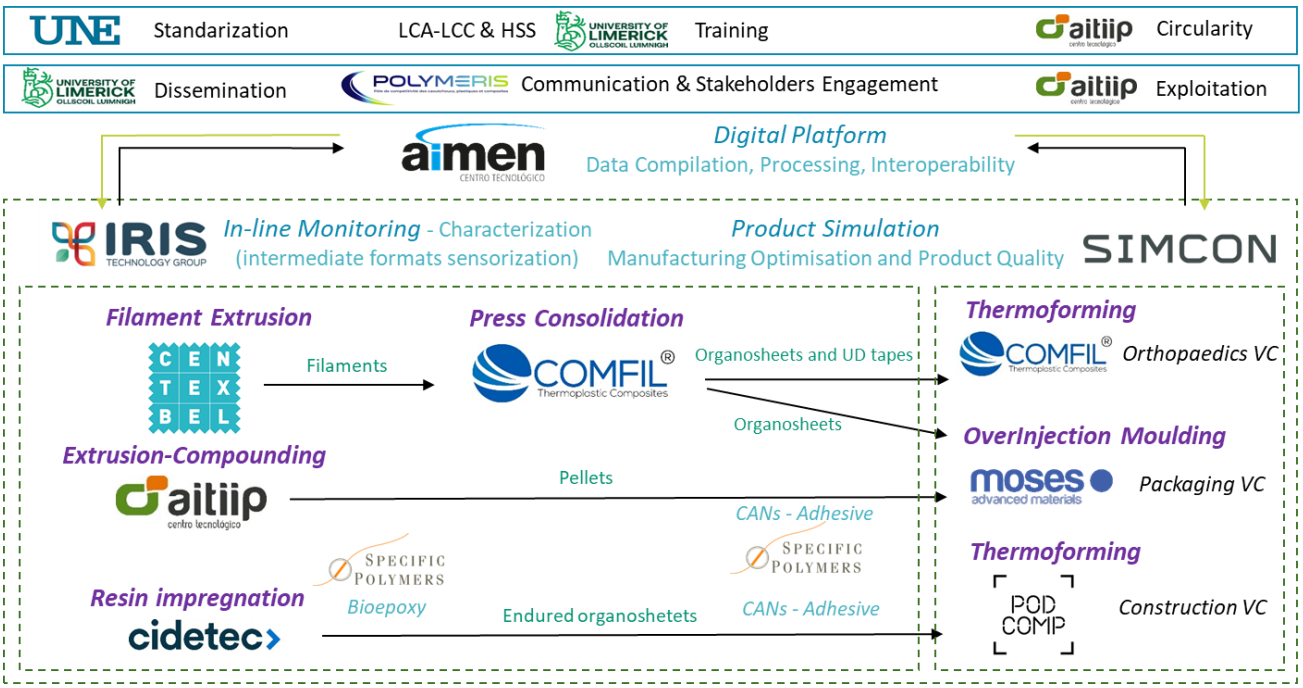


Fig. 14: Bio-Uptake partners cover the three manufacturing VC

Table 21: Access to critical infrastructure

Access to Infraestructure	AIT	CTB	CID	SP	AIM	IRI	SIM	POD	COM	MOS	LIM	POL	UNE
Lab and pilot production of chemical building bloclks and polymeric resins	X		X	X							X		
Characterisation equipment´s for thermoplastics and thermosets	X	X	X	X							X		
Lab and pilot acces for pre-impregnation of fabrics with endured thermosets			X										
Thermoforming pilot line of endured themosets			X										
Extruders for yarn and fibrillar production at lab and pilot level		X							X				
Press-moulding and thermoforming machines for producing thermoplastic organosheets & UDTapes		X							X				
Extrusion-compounding machine for pellets production	X									X			
Injection-Moulding Machines for over-injection	X									X			
Sprayation guns for coatings and adhesives (pilot and industrial)	X				X					X			
Machines and equipments for product standards and internal validation								X	X	X			
Software for product design, material libraries, process simulation (fully Access and license for modification)							X						
Cameras, sensors, spectroscopy equipments… to build lab-on-plant monitoring prototypes (engineering) for real-time data collection					X	X							
Algorithm design, software Access, interoperability					X								
Training tools, Access to course design, physical and virtual rooms & workshops	X	X	X		X						X	X	
Fully Access to standards add working groups													X

Table 22: Partners' contribution to Bio-Uptake, Benefits taken from the project and potential Exploitation

Name	Role	Contribution	Benefits	Exploitation
AIT	Coord, WP5 & Exploit Leader	Expertise on project coordination, extrusion-compounding & injection-moulding of biobased composites. Expert in circular solution and models for plastic products.	Increase knowledge management in Horizon Europe, in working with lignin-based reinforcements & over-injection moulding with organosheet too. Guide the transfer of knowledge and skills to workforce (Transfer of knowledge to industry)	TECHNOLOGY SERVICE
CTB	WP2 leader	Expertise in fabrics transformation technologies, self & bi-component fibre structures with outstanding mechanical properties	Enlarge capabilities in synthetic biobased fibres and fabrics to attract private contracts, design new material formulations and act as knowledge transfer to industries	TECHNOLOGY SERVICE
CID	Partner	To design the endured pre-peg material adequate for industry applications and set-up both pre-peg and thermoforming pilot lines	Bio-Epoxy + 3RCAN thermoset material and their manufacturing technologies (to be patented) and exploited in several value chains. Transfer of knowledge to industry.	PRODUCT TECHNOLOGY SERVICE
SP	Partner	Biobased CANs chemistry, adhesive and bioepoxy developer, characterisation and chemotyping	Develop new thermoset-based products (enlarging current portfolio). New service-based material modelling	PRODUCT SERVICE
AIM	WP4 leader	Algorithm developer, interoperability of data. Visual quick quality check	Provide service for digitalisation manufacturing industries related to plastic transformation (and others)	SERVICE
IRI	Partner	Development in-line monitoring *lab-to-pilot* prototypes for real-time characterisation & correction	Offer the prototypes developed for curing thermosets, controlling microfibre structures and intermediate formats quality to industry.	TECHNOLOGY PRODUCT
SIM	WP3 leader	Adapt current simulation software for injection and thermoforming to include new material libraries and adapted processes	Offer a more complete software to plastic industry to design and work with biobased plastics. More software Licenses.	PRODUCT SERVICE
POD	Partner	Knowledge in the construction sector, regarding composites, designs, validation and feedback. End-user	Increase the knowledge with a solution of great relevance for its products and clients. Modify current production technology for a quicker and more economic one. Promotion of new business in circular economy	TECHNOLOGY PRODUCT
COM	Partner	Knowledge in the orthotic sector, regarding composites, designs, validation and feedback. End-user	Increase the knowledge with a solution of great relevance for its products and clients. Offer biobased intermediates with enhanced thermo-mechanical properties. New business in the circular economy	TECHNOLOGY PRODUCT
MOS	Partner	Knowledge in the packaging sector, regarding composites, designs, validation and feedback. End-user	Increase the knowledge with a solution of great relevance for its products and clients. Modify current technology process for lids production. Offer a lighter and more resistant biobased product. New business in the circular economy	TECHNOLOGY PRODUCT
LIM	WP1 & diss. leader	LCA, LCC; Health & Safety, lignin based composite materials knowledge, open practices in academia and training	Be able to provide specific services to future industrial clients in the field: LCA-LCC, sustainability, ad-hoc training programmes etc	PRODUCT SERVICE
POL	WP6 & Comm. leader	Communication of outputs to industry & society. Engaging stakeholders	Transfer of knowledge of Bio-Uptake solutions to the French plastic industry	SERVICE
UNE	Partner	Standardization knowledge, interaction with working groups, upgrade standards	Broad Services and Commit with its mission of acting as a driver for more sustainable products in EU	SERVICE

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